INCH-POUND
MIL-M-38510/341F
4 May 2004
SUPERSEDING
MIL-M-38510/341E
28 APRIL 1989

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED, SCHOTTKY TTL, FLIP-FLOPS, CASCADABLE, MONOLITHIC SILICON

Reactivated after 4 May, 2004 and may be used for either new or existing design acquisition.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced Schottky TTL, flip flop microcircuits. Two product assurance classes and a choice of case outlines/lead finish are provided and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.3).

1.2 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types are as follows:

Device type	Circuit
01	Dual D-type positive edge-triggered flip-flop
02	Dual J$\overline{K}$ positive edge triggered flip-flop
03	Dual JK negative edge-triggered flip-flop
04	Quad D-type positive edge-triggered flip-flop
05	Octal D-type positive edge-triggered flip-flop with three-state outputs
06	Octal D-type positive edge-triggered flip-flop with three-state inverted outputs
07	Hex D-type positive edge-triggered flip-flop
08	Parallel D-type positive edge-triggered register (with enable)
09	Quad parallel D-type positive edge-triggered register (with enable)
10	Octal D-type positive edge-triggered flip-flop with three-state outputs and bus structured pinouts
11	Octal D-type positive edge-triggered flip-flop with three-state inverted outputs and bus structured pinouts

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, 3990 East Broad St., Columbus, OH 43216-5000, or emailed to bipolar@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
A	GDFP5-F14 or CDFP6-F14	14	Flat pack
B	GDFP4-F14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
D	GDFP1-F14 or CDFP2-F14	14	Flat pack
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack
R	GDIP1-T20 or CDIP2-T20	20	Dual-in-line package
S	GDFP2-F20 or CDFP3-F20	20	Flat pack
2	CQCC1-N20	20	Square leadless chip carrier
X	CQCC2-N20	20	Square leadless chip carrier

1.3 Absolute maximum ratings.

Supply voltage range ..... -0.5 V dc to +7.0 V dc
Input voltage range ..... -1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range ..... -65° to +150°C
Maximum power dissipation, (P_D) 1/ :
- Device type 01 ..... 88 mW
- Device type 02 ..... 93.5 mW
- Device type 03 ..... 105 mW
- Device type 04 ..... 187 mW
- Device types 05, 06, 10, 11 ..... 473 mW
- Device type 07 ..... 247 mW
- Device type 08 ..... 247 mW
- Device type 09 ..... 220 mW

Lead temperature (soldering, 10 seconds) ..... +300°C
Thermal resistance, junction to case (θ_{JC}): (See MIL-STD-1835)
Junction temperature (T_J) 2/ ..... 175°C

1.4 Recommended operating conditions. 2/

Supply voltage (V_{CC}) ..... 4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH}) ..... 2.0 V dc
Maximum low-level input voltage (V_{IL}) ..... 0.8 V dc
Case operating temperature range (T_C) ..... -55 °C to +125 °C
Width of clock pulse high:
- Device type 01 ..... 4.0 ns minimum
- Device type 02 ..... 4.0 ns minimum
- Device type 03 ..... 5.0 ns minimum
- Device type 04 ..... 4.0 ns minimum
- Device type 05 ..... 8.0 ns minimum
- Device type 06, 07 ..... 7.0 ns minimum
- Device type 08 ..... 4.0 ns minimum
- Device type 09 ..... 4.0 ns minimum
- Device types 10, 11 ..... 5.0 ns minimum

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).
2/ Maximum junction temperature should not be exceeded except for allowable short duration burn-in screening conditions in according with MIL-PRF-38535.

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Width of clock pulse low:
- Device type 01 ..... 6.0 ns minimum
- Device type 02 ..... 5.0 ns minimum
- Device type 03 ..... 5.0 ns minimum
- Device type 04 ..... 6.0 ns minimum
- Device type 05 ..... 8.0 ns minimum
- Device type 06 ..... 6.0 ns minimum
- Device type 07 ..... 7.5 ns minimum
- Device type 08 ..... 5.0 ns minimum
- Device type 09 ..... 5.0 ns minimum
- Device types 10, 11..... 5.0 ns minimum

Width of set pulse:
- Device type 01 ..... 4.0 ns minimum
- Device type 02 ..... 4.0 ns minimum
- Device type 03 ..... 5.0 ns minimum

Width of clear pulse:
- Device type 01 ..... 4.0 ns minimum
- Device type 02 ..... 4.0 ns minimum
- Device type 03 ..... 5.0 ns minimum

Width of master reset pulse:
- Device type 04 ..... 5.0 ns minimum
- Device type 07 ..... 6.5 ns minimum

Setup time J, K, or D high to clock pulse:
- Device type 01 ..... 3.0 ns minimum
- Device type 02 ..... 3.0 ns minimum
- Device type 03 ..... 5.0 ns minimum
- Device type 04 ..... 3.0 ns minimum
- Device type 05 ..... 2.5 ns minimum
- Device type 06 ..... 2.0 ns minimum
- Device type 07 ..... 5.0 ns minimum
- Device type 08 ..... 4.0 ns minimum
- Device type 09 ..... 3.0 ns minimum
- Device type 10 ..... 3.0 ns minimum
- Device type 11 ..... 2.5 ns minimum

Setup time J, K, or D low to clock pulse:
- Device type 01 ..... 4.0 ns minimum
- Device type 02 ..... 4.0 ns minimum
- Device type 03 ..... 2.5 ns minimum
- Device type 04 ..... 4.0 ns minimum
- Device type 05 ..... 2.0 ns minimum
- Device type 06 ..... 2.5 ns minimum
- Device type 07 ..... 5.0 ns minimum
- Device type 08 ..... 4.0 ns minimum
- Device type 09 ..... 3.0 ns minimum
- Device type 10 ..... 2.5 ns minimum
- Device type 11 ..... 3.0 ns minimum

Hold time J, K, or D high to clock pulse:
- Device type 01 ..... 2.0 ns minimum
- Device type 02 ..... 1.0 ns minimum
- Device type 03 ..... 2.5 ns minimum
- Device type 04 ..... 1.0 ns minimum
- Device types 05, 06, 07..... 2.0 ns minimum
- Device type 08 ..... 2.0 ns minimum
- Device type 09 ..... 1.0 ns minimum
- Device types 10, 11..... 2.0 ns minimum

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Hold time J, K, or D low to clock pulse:
- Device type 01 ..... 2.0 ns minimum
- Device type 02 ..... 1.0 ns minimum
- Device type 03 ..... 0.0 ns minimum
- Device type 04 ..... 2.0 ns minimum
- Device types 05, 06..... 2.5 ns minimum
- Device type 07, 08 ..... 2.0 ns minimum
- Device type 09 ..... 1.0 ns minimum
- Device types 10, 11..... 2.0 ns minimum

Recovery time $\overline{SD}$, $\overline{CD}$, or $\overline{MR}$ to CP:
- Device type 01 ..... 3.0 ns minimum
- Device type 02 ..... 2.0 ns minimum
- Device type 03 ..... 5.0 ns minimum
- Device type 04 ..... 6.0 ns minimum
- Device type 07 ..... 6.0 ns minimum

Input clock frequency:
- Device type 01 ..... 0-80 MHz
- Device type 02, 07 ..... 0-70 MHz
- Device type 03 ..... 0-90 MHz
- Device type 04 ..... 0-80 MHz
- Device types 05, 06..... 0-60 MHz
- Device type 08 ..... 0-90 MHz
- Device type 09 ..... 0-90 MHz
- Device types 10, 11..... 0-60 MHz

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines

(Copies of these documents are available online at http://assist.daps.dla.mil;quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 2.

3.3.2 Truth tables. The truth tables and logic equations shall be as specified on figure 3.

3.3.3 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity (DSCC-VAS) upon request.

3.3.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 10 (see MIL-PRF-38535, appendix A).

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535, appendix B.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{IH} = -1.0 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	All	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 20 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	All		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	All		-1.2	V
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		20	∝A
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V	All		100	∝A
Low level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	All	0.0	-0.6	mA
	I_{IL2}		01, 02, 03	0.0	-3.0	mA
	I_{IL3}		03	0.0	-3.0	mA
	I_{IL4}		07	-.05	-1.2	mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IL} = 0.0 V	01		16	mA
			02		17	mA
			03		19	mA
			04		34	mA
			07, 08 09		45	mA
Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IL} = 0.0 V	10, 11		86	mA
	I_{CCZ}	V_{CC} = 5.5 V, V_{IL} = 0.0 V	05, 06		86	mA
			10, 11		90	
Off-state output leakage current	I_{OZH}	V_{CC} = 5.5 V, V_{ZH} = 2.7 V	05, 06, 10, 11		50	∝A
	I_{OZL}	V_{CC} = 5.5 V, V_{ZL} = 0.5 V	05, 06, 10, 11		-50	∝A
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{OUT} = 0.0 V 1/	All	-60	-150	mA
Output drive	I_{OD}	V_{CC} = 4.5 V, V_{IN} = 5.5 V, V_{OUT} = 2.5 V	01, 02, 03, 04, 07, 08, 09	60		mA
			05, 06	35		mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
Maximum toggle frequency	f_{MAX}		01	80		MHz
			02, 07	70		MHz
			03	90		MHz
			04	80		MHz
			05, 06	60		MHz
			08	60		MHz
			09	70		MHz
			10, 11	60		MHz
Propagation delay time low to high level		V_{CC} = 5.0 V, C_L = 50 pF ±10% See figure 4				
CP to Q output	t_{PLH1}		01	3.8	8.5	ns
			02	3.8	9.0	ns
			03	2.5	9.5	ns
			04	3.5	8.5	ns
			05	3.0	10.5	ns
			07	1.0	11.0	ns
			08	2.0	9.5	ns
			09	2.0	8.5	ns
			10	2.5	9.5	ns
CP to $\overline{Q}$ output	t_{PLH2}		01	3.8	8.5	ns
			02	3.8	9.0	ns
			03, 11	2.5	9.5	ns
			04	3.5	8.5	ns
			06	4.0	10.5	ns
$\overline{SD}$, $\overline{CD}$, to Q, $\overline{Q}$ output (CP high)	t_{PLH3}		01	3.2	8.0	ns
			02	3.2	9.0	ns
			03	2.0	9.0	ns
$\overline{SD}$, $\overline{CD}$, to Q, $\overline{Q}$ output (CP low)	t_{PLH4}		01	3.2	8.0	ns
			02	3.2	9.0	ns
			03	2.0	9.0	ns
$\overline{MR}$ to $\overline{Q}$ output (CP high)	t_{PLH5}		04	4.0	10.0	ns
$\overline{MR}$ to $\overline{Q}$ output (CP low)	t_{PLH6}		04	4.0	10.0	ns
Propagation delay time, high to low level						

See footnotes at end of table.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
CP to Q output	t_{PHL1}	V_{CC} = 5.0 V, C_L = 50 pF ±10% See figure 4	01	3.8	10.5	ns
			02	3.8	10.5	ns
			03	2.5	9.5	ns
			04	3.5	10.5	ns
			05	3.0	11.5	ns
			07	1.0	13.0	ns
			08	2.5	10.5	ns
			09	2.5	10.5	ns
			10	2.5	9.5	ns
CP to $\overline{Q}$ output	t_{PHL2}		01	3.8	10.5	ns
			02	3.8	10.5	ns
			03, 11	2.5	9.5	ns
			04	3.5	10.5	ns
			06	4.0	11.0	ns
$\overline{SD}$, $\overline{CD}$, to Q, $\overline{Q}$ output (CP high)	t_{PHL3}		01	3.2	11.5	ns
			02	3.2	11.5	ns
			03	2.0	9.5	ns
$\overline{SD}$, $\overline{CD}$, to Q, $\overline{Q}$ output (CP low)	t_{PHL4}		01	3.5	11.5	ns
			02	3.5	11.5	ns
			03	2.5	9.5	ns
$\overline{MR}$ Q, output (CP high)	t_{PHL5}		04	4.5	15.0	ns
			07	1.0	17.0	ns
$\overline{MR}$ to Q output (CP low)	t_{PHL6}		04	4.5	15.0	ns
			07	1.0	17.0	ns
Propagation delay time, low level to off-state						
$\overline{OE}$ to Q output	t_{PLZ1}		05	1.5	7.5	ns
			10	1.0	7.0	ns
$\overline{OE}$ to $\overline{Q}$ output	t_{PLZ2}		06	1.5	7.5	ns
			11	1.5	7.0	ns
Propagation delay time high level to off-state						
$\overline{OE}$ to Q output	t_{PHZ1}		05	1.5	8.0	ns
			10	1.0	7.0	ns
$\overline{OE}$ to $\overline{Q}$ output	t_{PHZ2}		06	1.5	8.0	ns
			11	1.5	7.0	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
Propagation delay time, Off-state to low level		V_{CC} = 5.0 V, C_L = 50 pF ±10% See figure 4				
$\overline{OE}$ to Q output	t_{PZL1}		05	2.0	10.0	ns
			10	2.5	10.5	ns
$\overline{OE}$ to $\overline{Q}$ output	t_{PZL2}		06	2.0	10.0	ns
			11	2.5	10.5	ns
Propagation delay time, off-state to high level						
$\overline{OE}$ to Q output	t_{PZH1}		05	2.0	14.0	ns
			10	2.0	10.5	ns
$\overline{OE}$ to $\overline{Q}$ output	t_{PZH2}		05, 06	2.0	14.0	ns
			11	2.5	10.5	ns

1/ Not more than one output should be shorted at a time.

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TABLE II. Electrical test requirements.

MIL-PRF-38535 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 7, 9
Group A test requirements	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 8, 9, 10, 11
Group B test when using the method 5005 QCI option	1, 2, 3, 9, 10, 11	N/A
Group C end-point electrical parameters	1, 2, 3, 9, 10, 11	1, 2, 3
Group D end-point electrical parameters	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified herein and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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Pin number	Device type 01		Device type 02		Device type 03		Device type 04	
	CASES							
	A, B, C, and D	2, X	E, F	2, X	E, F	2, X	E, F	2, X
1	$\overline{CD}_1$	N/C	$\overline{CD}_1$	N/C	$\overline{CP}_1$	N/C	$\overline{MR}$	N/C
2	D_1	$\overline{CD}_1$	J_1	$\overline{CD}_1$	K_1	$\overline{CP}_1$	Q_0	$\overline{MR}$
3	CP_1	D_1	$\overline{K}_1$	J_1	J_1	K_1	$\overline{Q}_0$	Q_0
4	$\overline{SD}_1$	CP_1	CP_1	$\overline{K}_1$	$\overline{SD}_1$	J_1	D_0	$\overline{Q}_0$
5	Q_1	N/C	$\overline{SD}_1$	CP_1	Q_1	$\overline{SD}_1$	D_1	D_0
6	$\overline{Q}_1$	$\overline{SD}_1$	Q_1	N/C	$\overline{Q}_1$	N/C	$\overline{Q}_1$	N/C
7	GND	N/C	$\overline{Q}_1$	$\overline{SD}_1$	$\overline{Q}_2$	Q_1	Q_1	D_1
8	$\overline{Q}_2$	Q_1	GND	Q_1	GND	$\overline{Q}_1$	GND	$\overline{Q}_1$
9	Q_2	$\overline{Q}_1$	$\overline{Q}_2$	$\overline{Q}_1$	Q_2	$\overline{Q}_2$	CP	Q_1
10	$\overline{SD}_2$	GND	Q_2	GND	$\overline{SD}_2$	GND	Q_2	GND
11	CP_2	N/C	$\overline{SD}_2$	N/C	J_2	N/C	$\overline{Q}_2$	N/C
12	D_2	$\overline{Q}_2$	CP_2	$\overline{Q}_2$	K_2	Q_2	D_2	CP
13	$\overline{CD}_2$	Q_2	$\overline{K}_2$	Q_2	$\overline{CP}_2$	$\overline{SD}_2$	D_3	Q_2
14	V_{CC}	$\overline{SD}_2$	J_2	$\overline{SD}_2$	$\overline{CD}_2$	J_2	$\overline{Q}_3$	$\overline{Q}_2$
15		N/C	$\overline{CD}_2$	CP_2	$\overline{CD}_1$	K_2	Q_3	D_2
16		CP_2	V_{CC}	N/C	V_{CC}	N/C	V_{CC}	N/C
17		N/C		$\overline{K}_2$		$\overline{CP}_2$		D_3
18		D_2		J_2		$\overline{CD}_2$		$\overline{Q}_3$
19		$\overline{CD}_2$		$\overline{CD}_2$		$\overline{CD}_1$		Q_3
20		V_{CC}		V_{CC}		V_{CC}		V_{CC}

FIGURE 1. Terminal connections.

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Pin number	Device type 05		Device type 06		Device type 07		Device type 08	
	CASES							
	R, S	2, X	R, S	2, X	E, F	2, X	E, F	2, X
1	$\overline{OE}$	$\overline{OE}$	$\overline{OE}$	$\overline{OE}$	$\overline{MR}$	N/C	$\overline{E}$	N/C
2	Q_0	Q_0	$\overline{Q}_0$	$\overline{Q}_0$	Q_0	$\overline{MR}$	Q_0	$\overline{E}$
3	D_0	D_0	D_0	D_0	D_0	Q_0	D_0	Q_0
4	D_1	D_1	D_1	D_1	D_1	D_0	D_1	D_0
5	Q_1	Q_1	$\overline{Q}_1$	$\overline{Q}_1$	Q_1	D_1	Q_1	D_1
6	Q_2	Q_2	$\overline{Q}_2$	$\overline{Q}_2$	D_2	N/C	D_2	N/C
7	D_2	D_2	D_2	D_2	Q_2	Q_1	Q_2	Q_1
8	D_3	D_3	D_3	D_3	GND	D_2	GND	D_2
9	Q_3	Q_3	$\overline{Q}_3$	$\overline{Q}_3$	CP	Q_2	CP	Q_2
10	GND	GND	GND	GND	Q_3	GND	Q_3	GND
11	CP	CP	CP	CP	D_3	N/C	D_3	N/C
12	Q_4	Q_4	$\overline{Q}_4$	$\overline{Q}_4$	Q_4	CP	Q_4	CP
13	D_4	D_4	D_4	D_4	D_4	Q_3	D_4	Q_3
14	D_5	D_5	D_5	D_5	D_5	D_3	D_5	D_3
15	Q_5	Q_5	$\overline{Q}_5$	$\overline{Q}_5$	Q_5	Q_4	Q_5	Q_4
16	Q_6	Q_6	$\overline{Q}_6$	$\overline{Q}_6$	V_{CC}	N/C	V_{CC}	N/C
17	D_6	D_6	D_6	D_6		D_4		D_4
18	D_7	D_7	D_7	D_7		D_5		D_5
19	Q_7	Q_7	$\overline{Q}_7$	$\overline{Q}_7$		Q_5		Q_5
20	V_{CC}	V_{CC}	V_{CC}	V_{CC}		V_{CC}		V_{CC}

FIGURE 1. Terminal connections - Continued.

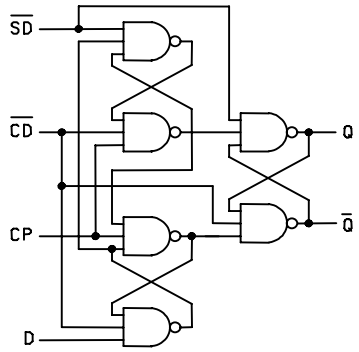
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Pin number	Device type 09		Device type 10		Device type 11	
	CASES					
	E, F	2, X	R, S	2, X	R, S	2, X
1	$\overline{E}$	N/C	$\overline{OE}$	$\overline{OE}$	$\overline{OE}$	$\overline{OE}$
2	Q_0	$\overline{E}$	D_0	D_0	D_0	D_0
3	$\overline{Q}_0$	Q_0	D_1	D_1	D_1	D_1
4	D_0	$\overline{Q}_0$	D_2	D_2	D_2	D_2
5	D_1	D_0	D_3	D_3	D_3	D_3
6	$\overline{Q}_1$	N/C	D_4	D_4	D_4	D_4
7	Q_1	D_1	D_5	D_5	D_5	D_5
8	GND	$\overline{Q}_1$	D_6	D_6	D_6	D_6
9	CP	Q	D_7	D_7	D_7	D_7
10	Q_2	GND	GND	GND	GND	GND
11	$\overline{Q}_2$	N/C	CP	CP	CP	CP
12	D_2	CP	Q_7	Q_7	$\overline{Q}_7$	$\overline{Q}_7$
13	D_3	Q_2	Q_6	Q_6	$\overline{Q}_6$	$\overline{Q}_6$
14	$\overline{Q}_3$	$\overline{Q}_2$	Q_5	Q_5	$\overline{Q}_5$	$\overline{Q}_5$
15	Q_3	D_2	Q_4	Q_4	$\overline{Q}_4$	$\overline{Q}_4$
16	V_{CC}	N/C	Q_3	Q_3	$\overline{Q}_3$	$\overline{Q}_3$
17		D_3	Q_2	Q_2	$\overline{Q}_2$	$\overline{Q}_2$
18		$\overline{Q}_3$	Q_1	Q_1	$\overline{Q}_1$	$\overline{Q}_1$
19		Q_3	Q_0	Q_0	$\overline{Q}_0$	$\overline{Q}_0$
20		V_{CC}	V_{CC}	V_{CC}	V_{CC}	V_{CC}

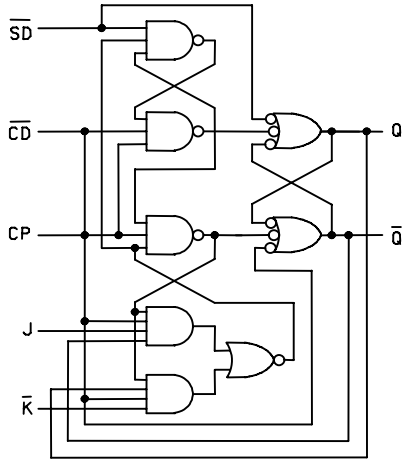
FIGURE 1. Terminal connections - Continued.

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DEVICE TYPE 01 (1/2 SHOWN)



DEVICE TYPE 02 (1/2 SHOWN)



DEVICE TYPE 03 (1/2 SHOWN)

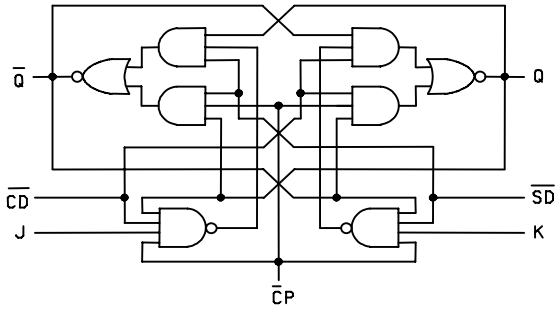
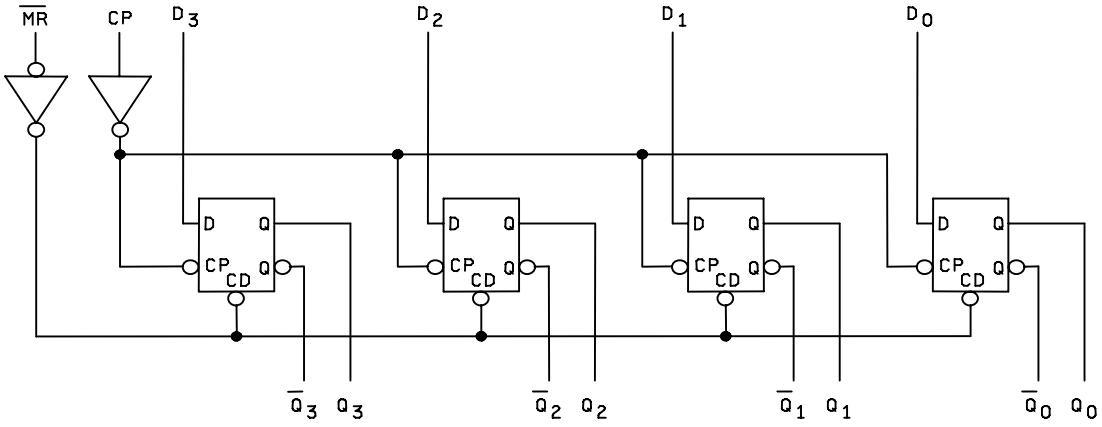


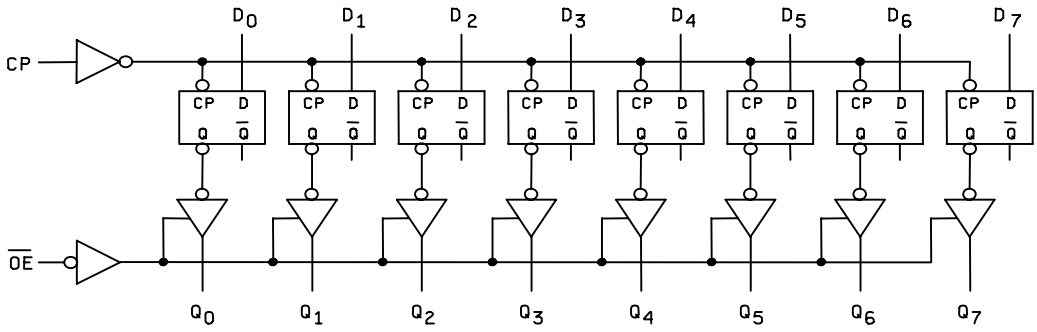
FIGURE 2. Logic diagrams

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DEVICE TYPE 04



DEVICE TYPE 05



DEVICE TYPE 06

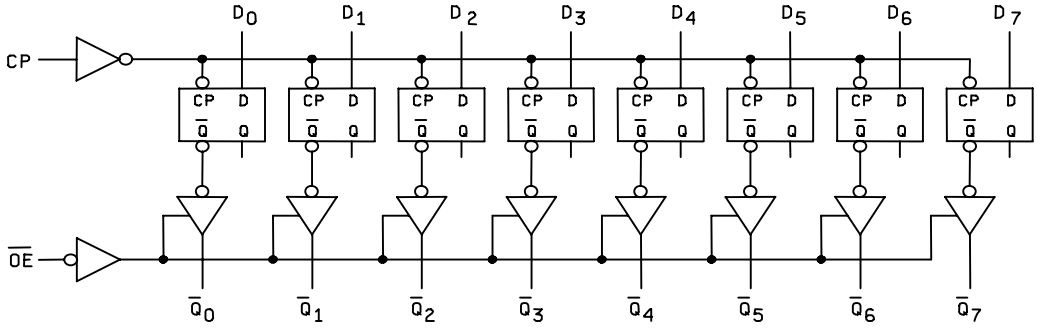


FIGURE 2. Logic diagrams – Continued.

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DEVICE TYPE 07

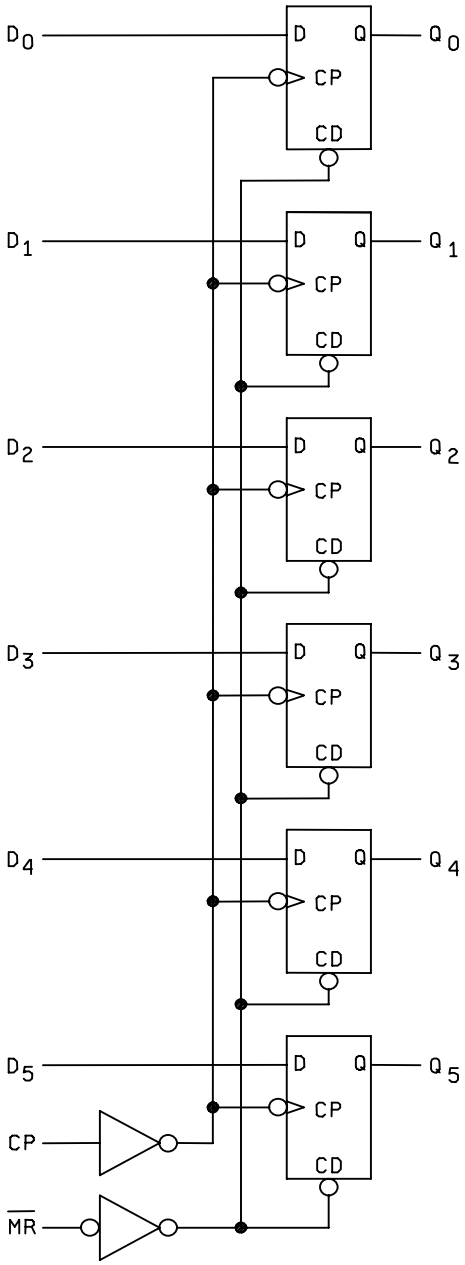
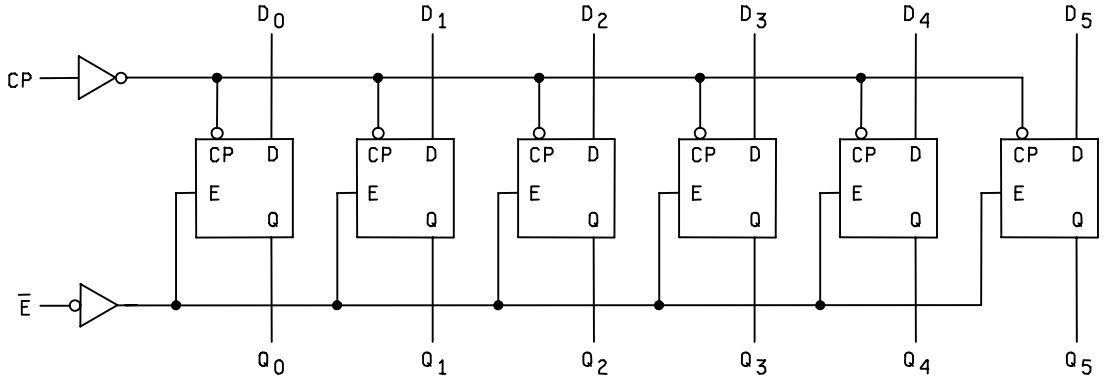


FIGURE 2. Logic diagrams – Continued.

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DEVICE TYPE 08



DEVICE TYPE 09

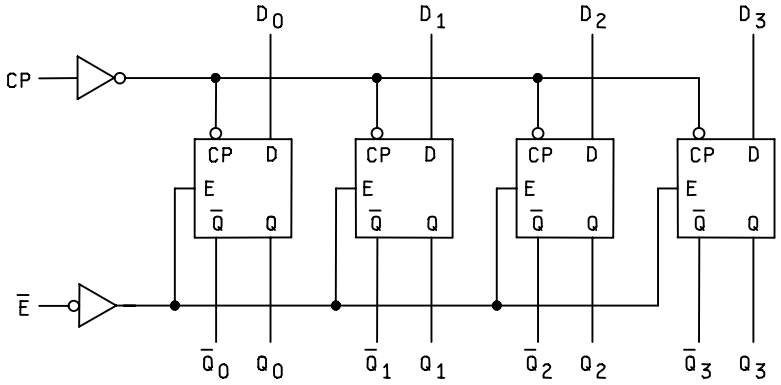
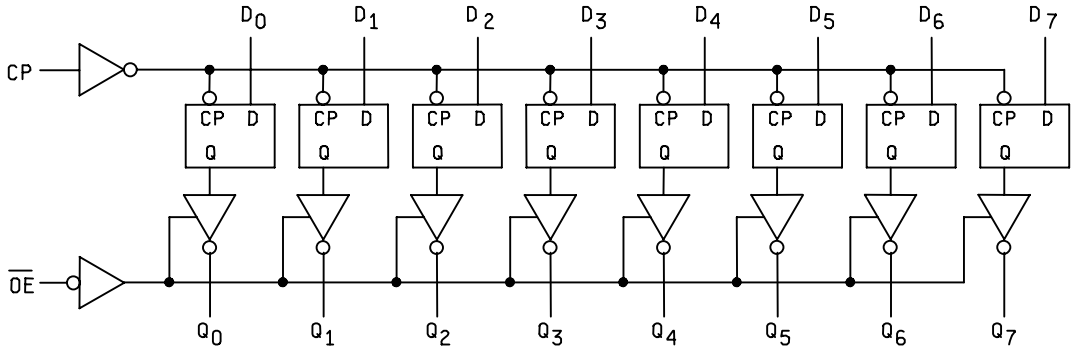


FIGURE 2. Logic diagrams – Continued.

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DEVICE TYPE 10



DEVICE TYPE 11

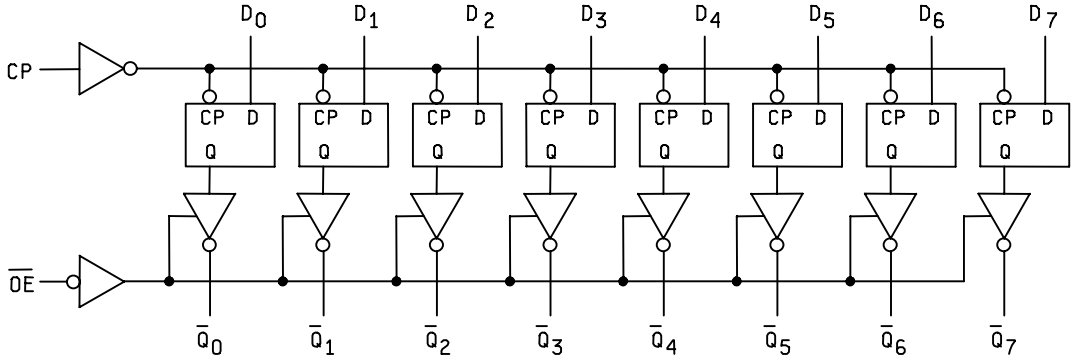


FIGURE 2. Logic diagrams – Continued.

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Device type 01

Input	Outputs	
@ t_n	@ t_n +1	
D	Q	$\overline{Q}$
L	L	H
H	H	L

Device type 02

Input		Outputs	
@ t_n		@ t_{n+1}	
J	$\overline{K}$	Q	$\overline{Q}$
L	H	No change	
L	L	L	H
H	H	H	L
H	L	Toggles	

Device type 03

Input		Outputs
@ t_n		@ t_{n+1}
J	K	Q
L	L	Qn
L	H	L
H	L	H
H	H	$\overline{Q}$ n

Device type 04

Inputs	Outputs	
@ t_n1 $\overline{MR}$ = H	@ t_n +1	
Dn	Qn	$\overline{Q}$ n
L	L	H
H	H	L

t_n = Bit time before clock pulse
t_n +1 = Bit time after clock pulse
H = High voltage level
L = Low voltage level

Device type 05

Inputs		Outputs	
Dn	CP	$\overline{OE}$	Qn
H	⎍	L	H
L	⎍	L	L
X	X	H	Z

Device type 06

Inputs		Outputs	
Dn	CP	$\overline{OE}$	$\overline{Qn}$
H	⎍	L	L
L	⎍	L	H
X	X	H	Z

Device type 07

Inputs			Outputs
$\overline{MR}$	CP	D	Q
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q_0

Device type 08

Inputs			Outputs
$\overline{E}$	CP	Dn	Qn
H	⎍	X	No change
L	⎍	H	H
L	⎍	L	L

H = High voltage level
L = Low voltage level
X = Immaterial

H = High voltage level
L = Low voltage level
X = Immaterial
Z = High impedance
↑ = Transition from low to high level
Q_0 = The level of Q before the indicated steady-state input conditions were established

Device type 09

Inputs			Outputs	
$\overline{E}$	CP	Dn	Qn	$\overline{Q}$ n
H	⎍	X	NC	NC
L	⎍	H	H	L
L	⎍	L	L	H

H = High voltage level
L = Low voltage level
X = Immaterial
NC = No change

FIGURE 3. Truth tables.

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Device type 10

Inputs			Outputs	Function
$\overline{OE}$	CP	D	Q	
H	H	L	Z	Hold
H	H	H	Z	Hold
H	↑	L	Z	Load
H	↑	H	Z	Load
L	↑	L	L	Data available
L	↑	H	H	Data available
L	H	L	NC	No change in data
L	H	H	NC	No change in data

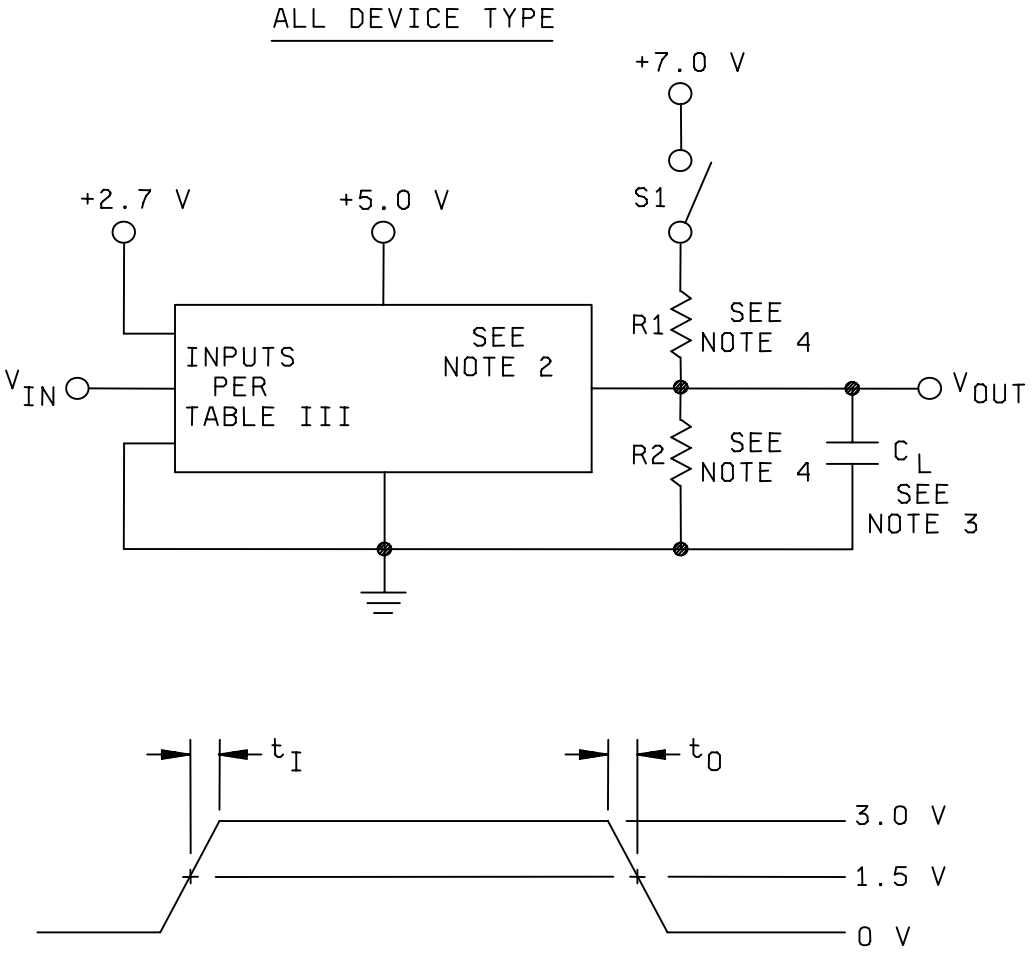
Device type 11

Inputs			Outputs	Function
$\overline{OE}$	CP	D	$\overline{Q}$	
H	H	L	Z	Hold
H	H	H	Z	Hold
H	↑	L	Z	Load
H	↑	H	Z	Load
L	↑	L	H	Data available
L	↑	H	L	Data available
L	H	L	NC	No change in data
L	H	H	NC	No change in data

H = High voltage level
L = Low voltage level
Z = High impedance
↑ = Transition from low to high level
NC = No change

FIGURE 3. Truth tables – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:20Z
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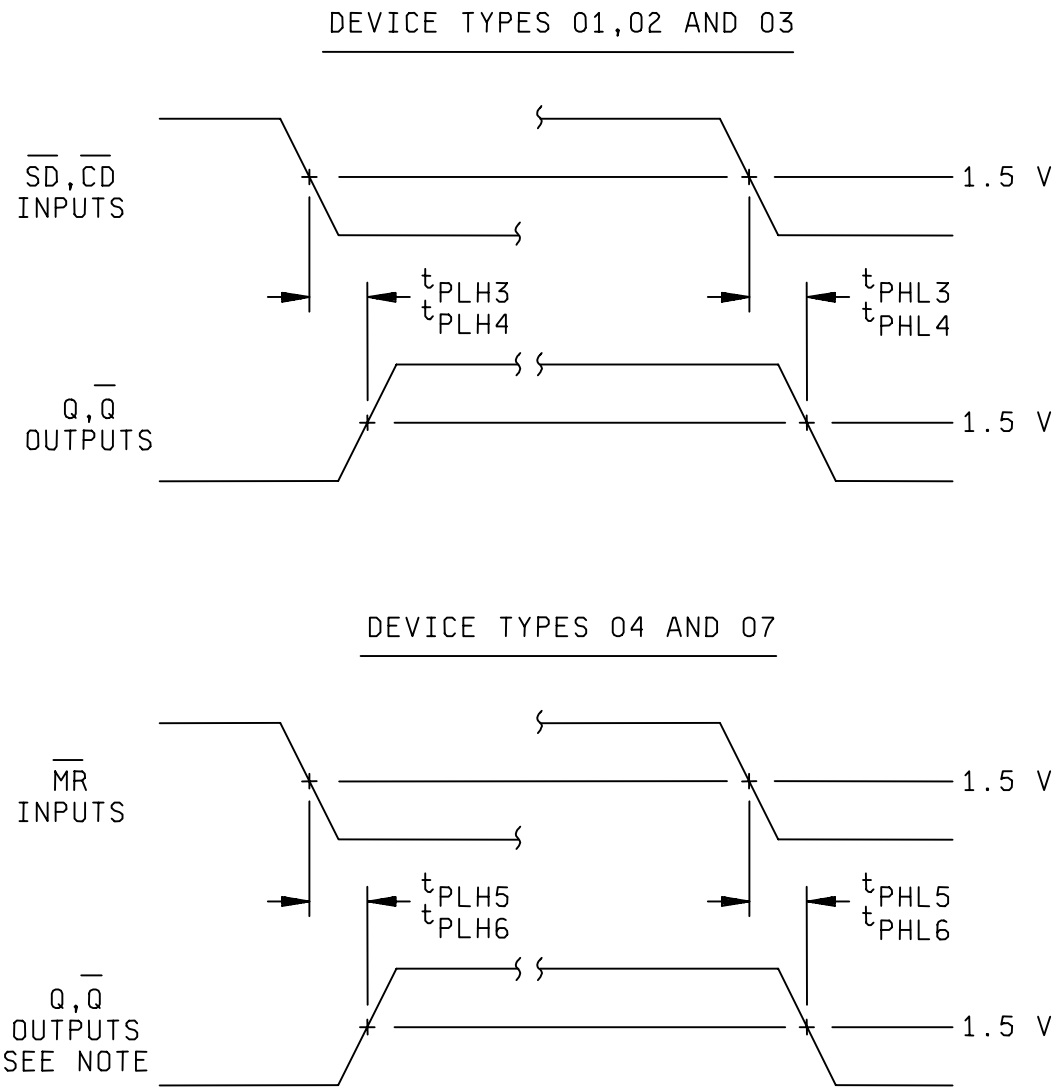


Test type	S1
t_{PLH}	Open
t_{PHL}	Open
t_{PHZ}	Open
t_{PZH}	Open
t_{PLZ}	Closed
t_{PZL}	Closed

NOTES:
1. V_{IN} = input pulse and has the following characteristics: PRR ≤ 1 MHz, $t_1 = t_0 \leq 2.5$ ns.
2. Inputs not under test are at ground.
3 C_L = 50 Pf ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
4 R1 = R2 = 499Ω ±1%.
5 Voltage measurements are to be made with respect to network ground terminal.

FIGURE 4. Switching time waveforms.

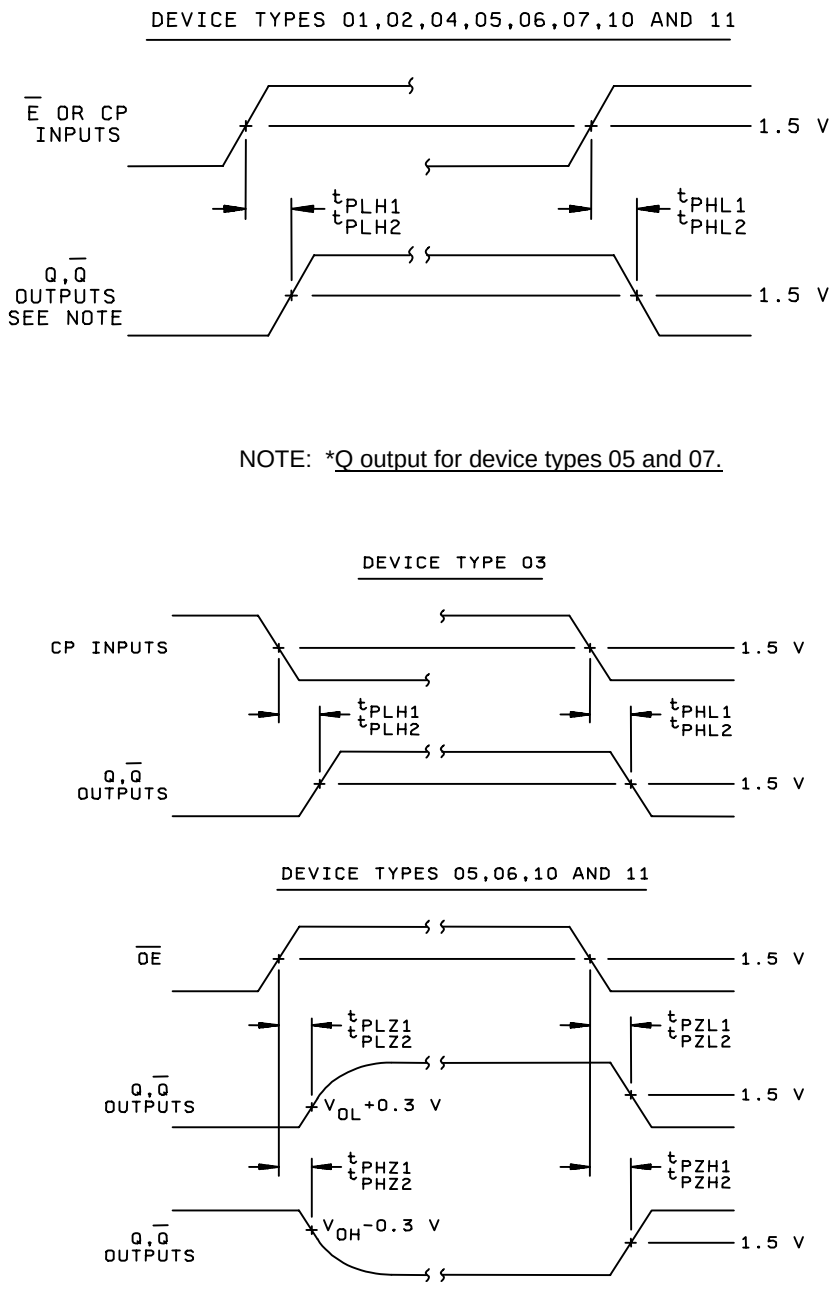
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:20Z
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NOTE: * Q output for device type 07.

FIGURE 4. Switching time waveforms – Continued.

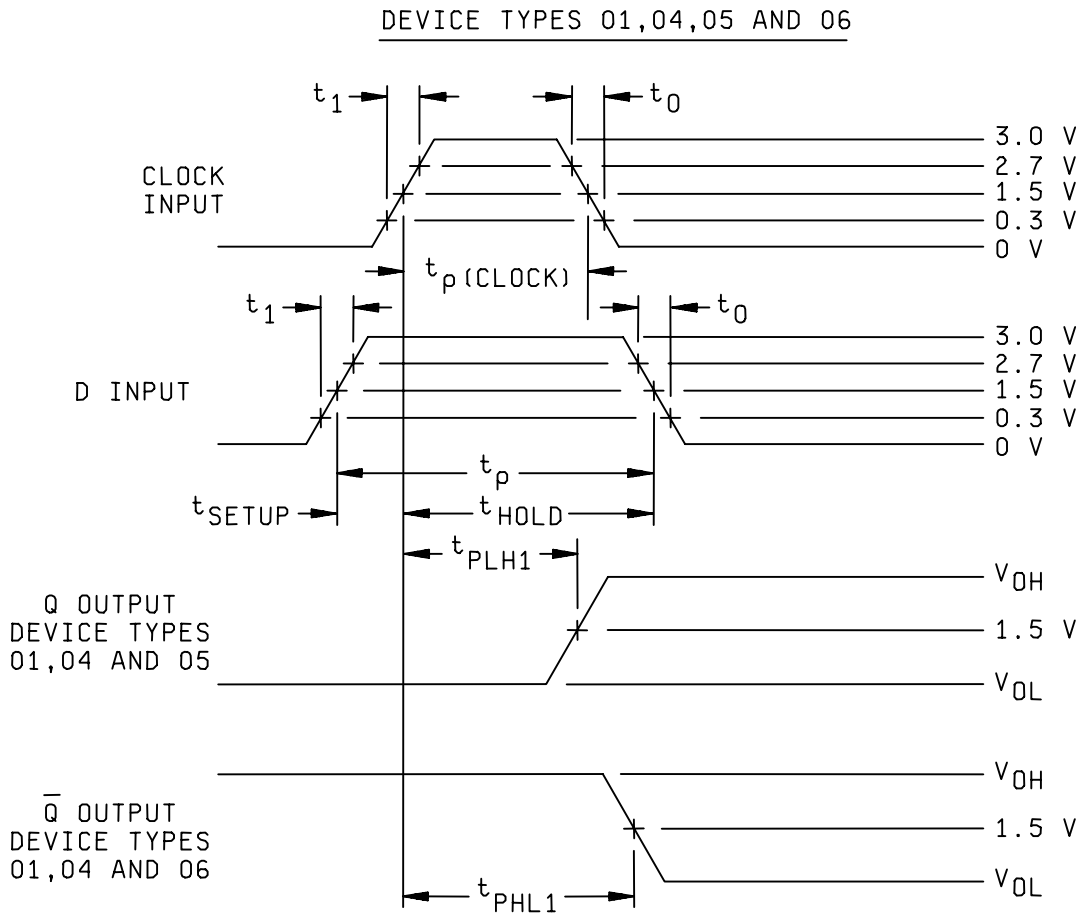
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:20Z
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NOTE: *Q output for device types 05 and 07.

FIGURE 4. Switching time waveforms – Continued.

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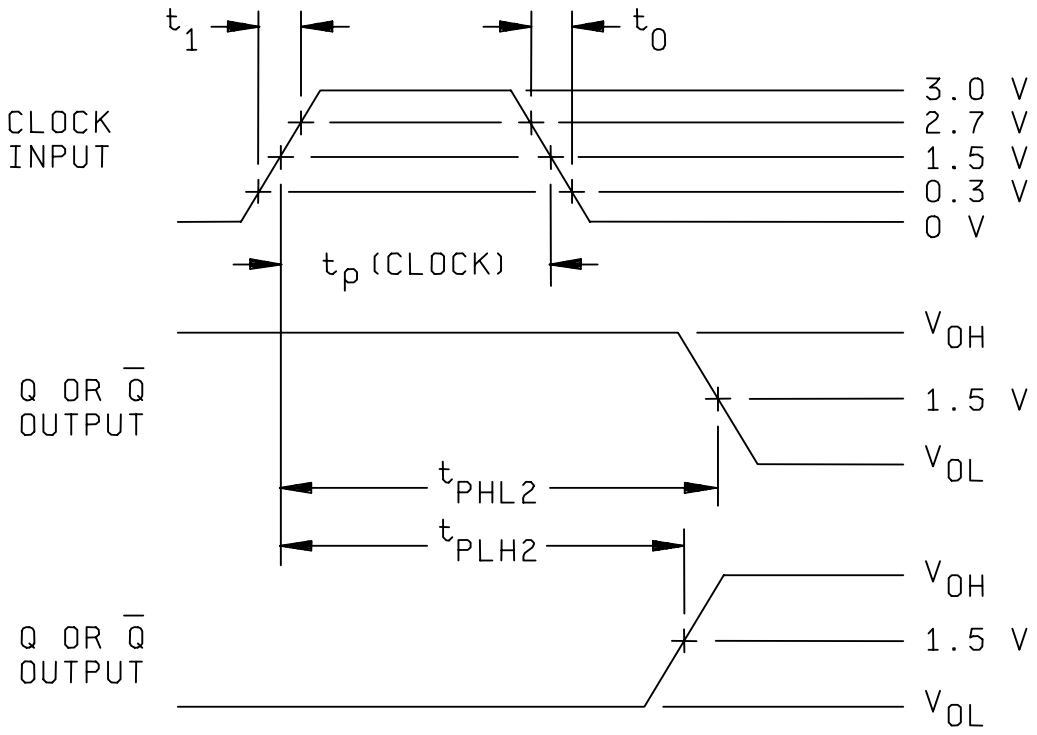


NOTES:
1. $t_1 = t_0 \leq 2.5$ ns.
2. PRR as in table I and III, duty cycle 50 ±15%.
3. When testing f_{MAX}, the output frequency shall be ½ the input frequency.

FIGURE 4. Switching time waveforms – Continued.

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DEVICE TYPE 02

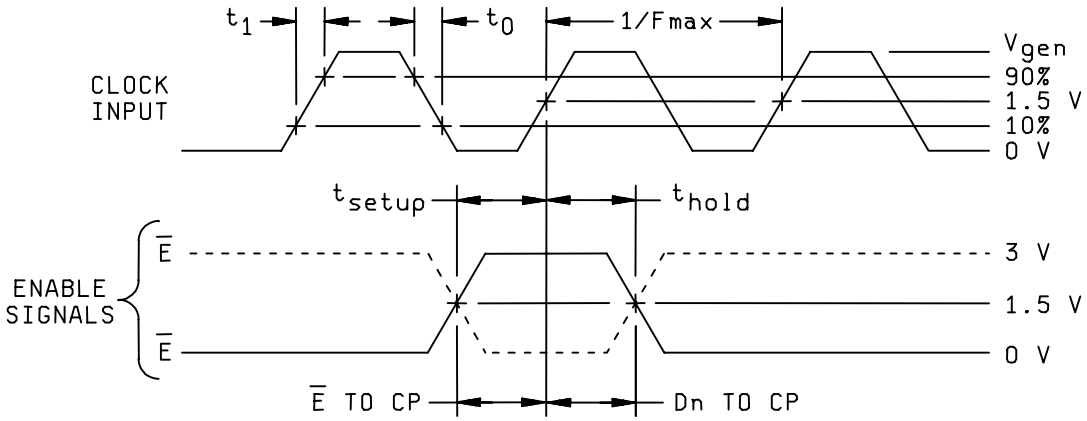


NOTES:
1. $t_1 = t_0 \leq 2.5$ ns
2. PRR as in table I and III, duty cycle 50 ±15%.
3. When testing $f_{MAX,}$ the output frequency shall be ½ the input frequency.

FIGURE 4. Switching time waveforms – Continued.

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DEVICE TYPE 08



NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 = t_0 \leq 2.5$ ns and PRR ≤ 1 MHz.
2. Enable characteristics are:

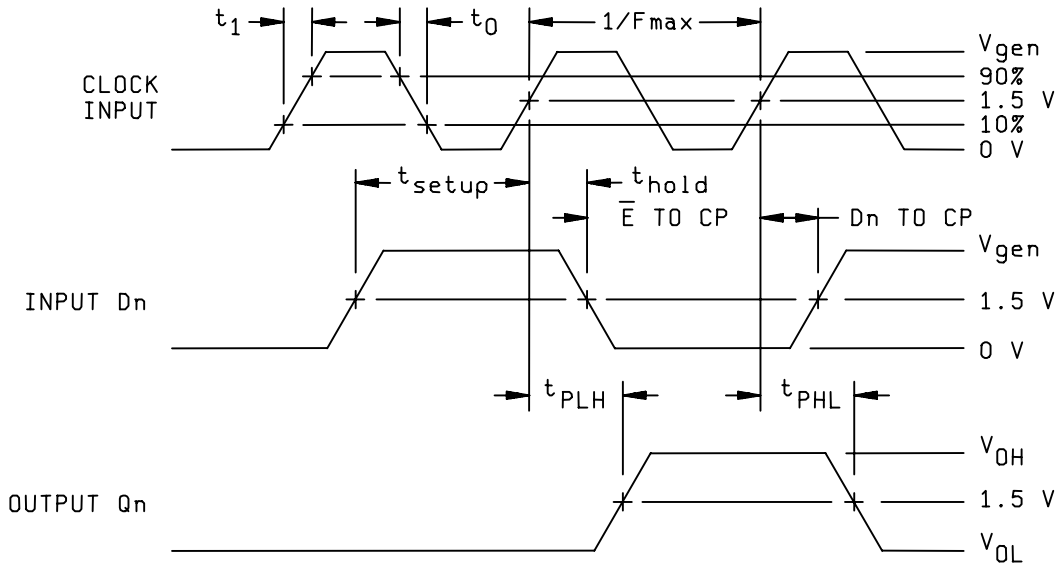
   t_{setup} = $\overline{E}$ to CP = 6 ns ($\overline{E}$ > CP)

   t_{hold} = Dn to CP = 2 ns ($\overline{E}$ > CP)

FIGURE 4. Switching time waveforms – Continued.

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DEVICE TYPE 08

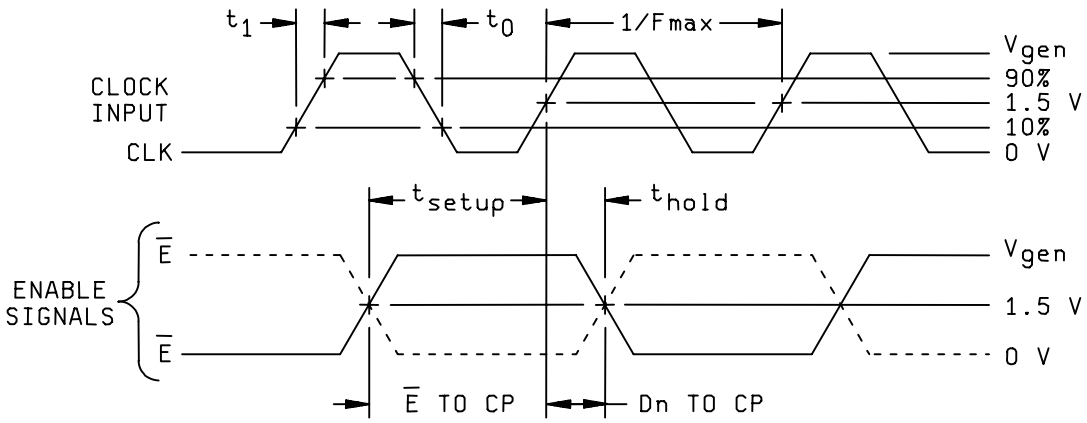


NOTES:
1. Clock input pulse has the following characteristics:
   V_{gen} = 3 ±0.2 V, $t_1 = t_0 \leq 2.5$ ns and PRR ≤ 1 MHz.
2. D input has the following characteristics:
   V_{gen} = 3 ±0.2 V, $\overline{E}$ to CP = t_{setup} = 3 ns minimum (D_n > CLK);
   D_n to CP = t_{hold} = 1 ns minimum (D_N > CLK).
3. For f_{MAX} testing, see table III.

FIGURE 4. Switching time waveforms – Continued.

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DEVICE TYPE 09

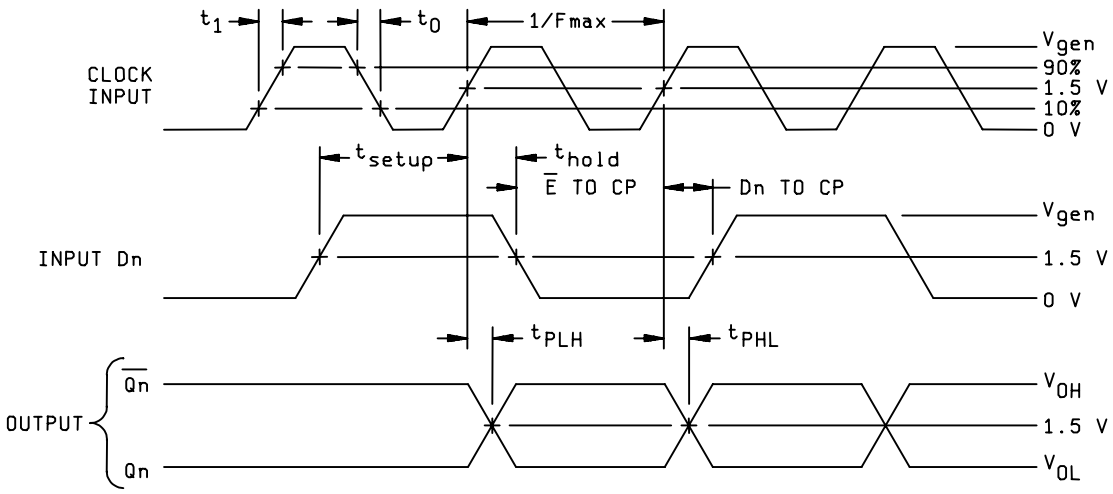


NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 = t_0 \leq 2.5$ ns and PRR ≤ 1 MHz.
2. Enable characteristics are:
   $\overline{E}$ > CP; t_{setup} = $\overline{E}$ to CP = 6 ns;
   t_{hold} = D_n to CP = 2 ns
3. For f_{MAX} testing, see table III.

FIGURE 4. Switching time waveforms – Continued.

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DEVICE TYPE 09



NOTES:
1. Clock input pulse has the following characteristics: V_{gen} = 3 ±0.2 V, $t_1 \leq$ 2.5 ns and PRR ≤ 1 MHz.
2. D input has the following characteristics: V_{gen} = 3 ±0.2 V, t_{setup} = 3 ns minimum, t_{hold} = 1 ns minimum, $\overline{E}$ to CP = t_{setup} (D_n > CLK); D_n to CP = t_{hold} (D_n > CLK).
3. For f_{MAX} testing, see table III.
4. t_{PLH} and t_{PHL} are shown for Q_n only, (CLK > Q_n, $\overline{Q}_n$). The Q_n output will have these reversed and are omitted for clarity.

FIGURE 4. Switching time waveforms – Continued.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{CD}_1$	D_1	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	D_2	$\overline{CD}_2$	V_{CC}				
1 Tc = 25°C	V_{OL}	3007	1	4.5 V	.8 V	2/	2/	20 mA		"							4.5 V	Q_1		.5	V
			2	.8 V			2.0 V	20 mA		"							"	Q_1		"	"
			3	2.0 V			.8 V		20 mA	"							"	$\overline{Q}_1$		"	"
			4							"	20 mA		.8 V			2.0 V	"	$\overline{Q}_2$		"	"
			5							"		20 mA	2.0 V			.8 V	"	Q_2		"	"
			6							"		20 mA	2/	2/	.8 V	4.5 V	"	Q_2		"	"
	V_{OH}	3006	7	2.0 V			.8 V	-1 mA		"							"	Q_1	2.5		"
			8	3/	2.0 V	3/	4.5 V	-1 mA		"							"	Q_1	"		"
			9	.8 V			2.0 V		-1 mA	"							"	$\overline{Q}_1$	"		"
			10							"	-1 mA		2.0 V			.8 V	"	$\overline{Q}_2$	"		"
			11							"		-1 mA	.8 V			2.0 V	"	Q_2	"		"
			12							"		-1 mA	4.5 V	3/	2.0 V	3/	"	Q_2	"		"
	V_{IC}		13	-18 mA						"							"	$\overline{CD}_1$		-1.2	"
			14		-18 mA					"							"	D_1		"	"
			15			-18 mA				"							"	CP_1		"	"
			16				-18 mA			"							"	$\overline{SD}_1$		"	"
			17							"			-18 mA				"	$\overline{SD}_2$		"	"
			18							"				-18 mA			"	CP_2		"	"
			19							"					-18 mA		"	D_2		"	"
			20							"						-18 mA	"	$\overline{CD}_2$		"	"
	I_{IH1}	3010	21	2.7 V	0 V	0 V	4.5 V			"							5.5 V	$\overline{CD}_1$		20	∝A
			22	2.7 V	0 V	4/	"			"							"	$\overline{CD}_1$		"	"
			23	0 V	2.7 V	4.5 V	"			"							"	D_1		"	"
			24	4.5 V	4.5 V	2.7 V	0 V			"							"	CP_1		"	"
			25	0 V	"	2.7 V	4.5 V			"							"	CP_1		"	"
			26	4.5 V	"	4/	2.7 V			"							"	$\overline{SD}_1$		"	"
			27							"			2.7 V	4/	4.5 V	4.5 V	"	$\overline{SD}_2$		"	"
			28							"			0 V	2.7 V	"	4.5 V	"	CP_2		"	"
			29							"			4.5 V	2.7 V	"	0 V	"	CP_2		"	"
			30							"			"	4.5 V	2.7 V	0 V	"	D_2		"	"
			31							"			"	0 V	0 V	2.7 V	"	$\overline{CD}_2$		"	"
			32							"			"	4/	0 V	2.7 V	"	$\overline{CD}_2$		"	"
	I_{IH2}	3010	33	7.0 V	0 V	0 V	4.5 V			"							"	$\overline{CD}_1$		100	"
			34	7.0 V	0 V	4/	"			"							"	$\overline{CD}_1$		"	"
			35	0 V	7.0 V	4.5 V	"			"							"	D_1		"	"
			36	4.5 V	4.5 V	7.0 V	0 V			"							"	CP_1		"	"
			37	0 V	"	7.0 V	4.5 V			"							"	CP_1		"	"
			38	4.5 V	"	4/	7.0 V			"							"	$\overline{SD}_1$		"	"
			39							"			7.0 V	4/	4.5 V	4.5 V	"	$\overline{SD}_2$		"	"
			40							"			0 V	7.0 V	"	4.5 V	"	CP_2		"	"
			41							"			4.5 V	7.0 V	"	0 V	"	CP_2		"	"
			42							"			"	4.5 V	7.0 V	0 V	"	D_2		"	"
			43							"			"	0 V	0 V	7.0 V	"	$\overline{CD}_2$		"	"
			44							"			"	4/	0 V	7.0 V	"	$\overline{CD}_2$		"	"

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	$\overline{CD}_1$	D_1	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	D_2	$\overline{CD}_2$	V_{CC}		Min	Max	
1 Tc = 25°C	I_{IL1}	3009	45	4.5 V	.5 V	4.5 V	0 V			GND							"	D_1	5/	5/	mA
			46	4.5 V	0 V	.5 V	0 V			"							"	CP_1	"	"	"
			47							"			0 V	.5 V	0 V	4.5 V	"	CP_2	"	"	"
			48							"			0 V	4.5 V	.5 V	4.5 V	"	D_2	"	"	"
	I_{IL2}	3009	49	.5 V	4.5 V	4.5 V	0 V			"							"	$\overline{CD}_1$	"	"	"
			50	0 V	0 V	0 V	.5 V			"							"	$\overline{SD}_1$	"	"	"
			51							"			.5 V	0 V	0 V	0 V	"	$\overline{SD}_2$	"	"	"
			52							"			0 V	4.5 V	4.5 V	.5 V	"	$\overline{CD}_2$	"	"	"
	I_{OS}	3011	53				0 V	0 V		"							5.5 V	Q_1	-60	-150	mA
			54	0 V					0 V	"							"	$\overline{Q}_1$	"	"	"
			55							"	0 V					0 V	"	$\overline{Q}_2$	"	"	"
			56							"		0 V	0 V				"	Q_2	"	"	"
	I_{OD}		57	0 V			5.5 V	2.5 V									4.5 V	Q_1	60		"
			58	5.5 V			0 V		2.5 V	"							"	$\overline{Q}_1$	"		"
			59							"	2.5 V		0 V			5.5 V	"	$\overline{Q}_2$	"		"
			60							"		2.5 V	5.5 V			0 V	"	Q_2	"		"
	I_{CC}	3005	61	5.5 V	0 V	0 V	0 V			"			0 V	0 V	0 V	5.5 V	5.5 V	V_{CC}		16	"
	I_{CC}	3005	62	0 V	0 V	0 V	5.5 V			"			5.5 V	0 V	0 V	0 V	5.5 V	V_{CC}		16	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Func-tional tests 6/	3014	63	A	B	B	B	H	L	GND	L	H	B	B	B	A	7/	All Outputs			
			64	"	B	A	B	"	"	"	"	"	B	A	B	"	"				
			65	"	A	A	A	"	"	"	"	"	A	A	A	"	"				
			66	"	A	B	"	"	"	"	"	"	"	B	A	"	"				
			67	"	B	B	"	"	"	"	"	"	"	B	B	"	"				
			68	"	"	A	"	L	H	"	H	L	"	A	"	"	"				
			69	"	"	B	"	"	"	"	"	"	"	B	"	"	"				
			70	"	A	B	"	"	"	"	"	"	"	B	A	"	"				
			71	"	"	A	"	H	L	"	L	H	"	A	"	"	"				
			72	B	"	A	"	L	H	"	H	L	"	A	"	B	"				
			73	"	"	B	"	"	"	"	"	"	"	B	"	"	"				
			74	"	"	A	"	"	"	"	"	"	"	A	"	"	"				
			75	A	"	"	"	"	"	"	"	"	"	"	"	A	"				
			76	A	"	"	B	H	L	"	L	H	B	"	"	A	"				
			77	B	"	"	B	H	H	"	H	H	B	"	"	B	"				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																				
9 T_C = +25°C	f_{MAX} 8/	3003	78	2.7 V	IN	IN	2.7 V	OUT		GND							5.0 V	Q_1	100		MHz
			79	2.7 V	IN	IN	2.7 V		OUT	"							"	$\overline{Q}_1$	"		"
			80							"	OUT		2.7 V	IN	IN	2.7 V	"	$\overline{Q}_2$	"		"
			81							"		OUT	2.7 V	IN	IN	2.7 V	"	Q_1	"		"

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TABLE III. Group A inspection for device type 01 - Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	$\overline{CD}_1$	D_1	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	D_2	$\overline{CD}_2$	V_{CC}		Min	Max	
9	t_{PLH1}	3003 Fig. 4	82	2.7 V	IN	IN	2.7 V	OUT		GND							5.0 V	CP_1 to Q_1	3.8	6.8	ns
T_C = +25°C	t_{PHL1}		83	"	"	"	"	OUT		"							"	CP_1 to Q_1	4.4	8.0	"
	t_{PLH2}		84	"	"	"	"		OUT	"							"	CP_1 to $\overline{Q}_1$	3.8	6.8	"
	t_{PHL2}		85	"	"	"	"		OUT	"							"	CP_1 to $\overline{Q}_1$	4.4	8.0	"
	t_{PLH1}	3003 Fig. 4	86							GND		OUT	2.7 V	IN	IN	2.7 V	5.0 V	CP_2 to Q_2	3.8	6.8	ns
	t_{PHL1}		87							"		OUT	"	"	"	"	"	CP_2 to Q_2	4.4	8.0	"
	t_{PLH2}		88							"	OUT		"	"	"	"	"	CP_2 to $\overline{Q}_2$	3.8	6.8	"
	t_{PHL2}		89							"	OUT		"	"	"	"	"	CP_2 to $\overline{Q}_2$	4.4	8.0	"
	t_{PHL3}		90	IN	2.7 V	2.7 V	IN	OUT		"							"	$\overline{CD}_1$ to Q_1	3.5	9.0	"
	t_{PHL4}		91	"	"	0 V	"	OUT		"							"	$\overline{CD}_1$ to Q_1	3.5	9.0	"
	t_{PHL3}		92	"	"	2.7 V	"		OUT	"							"	$\overline{CD}_1$ to $\overline{Q}_1$	3.2	6.1	"
	t_{PHL4}		93	"	"	0 V	"		OUT	"							"	$\overline{CD}_1$ to $\overline{Q}_1$	"	6.1	"
	t_{PHL3}		94	"	"	2.7 V	"	OUT		"							"	$\overline{SD}_1$ to Q_1	"	6.1	"
	t_{PHL4}		95	"	"	0 V	"	OUT		"							"	$\overline{SD}_1$ to Q_1	"	6.1	"
	t_{PHL3}		96	"	2.7 V	2.7 V	"		OUT	"							"	$\overline{SD}_1$ to $\overline{Q}_1$	3.5	9.0	"
	t_{PHL4}		97	"	2.7 V	0 V	"		OUT	"							"	$\overline{SD}_1$ to $\overline{Q}_1$	"	9.0	"
	t_{PHL3}		98							"		OUT	IN	2.7 V	2.7 V	IN	"	$\overline{CD}_2$ to Q_2	"	9.0	"
	t_{PHL4}		99							"		OUT	"	0 V	"	"	"	$\overline{CD}_2$ to Q_2	"	9.0	"
	t_{PHL3}		100							"	OUT		"	2.7 V	"	"	"	$\overline{CD}_2$ to $\overline{Q}_2$	3.2	6.1	"
	t_{PHL4}		101							"	OUT		"	0 V	"	"	"	$\overline{CD}_2$ to $\overline{Q}_2$	"	6.1	"
	t_{PHL3}		102							"		OUT	"	2.7 V	"	"	"	$\overline{SD}_2$ to Q_2	"	6.1	"
	t_{PHL4}		103							"		OUT	"	0 V	"	"	"	$\overline{SD}_2$ to Q_2	"	6.1	"

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`TABLE III. Group A inspection for device type 01 - Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	$\overline{CD}_1$	D_1	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	D_2	$\overline{CD}_2$	V_{CC}		Min	Max	
9	t_{PHL3}	3003 Fig. 4	104							GND	OUT		IN	2.7 V	2.7 V	IN	5.0 V	$\overline{SD}_2$ to $\overline{Q}_2$	3.5	9.0	"
T_C = +25°C	t_{PHL4}		105							"	OUT		"	0 V	"	"	"	$\overline{SD}_2$ to $\overline{Q}_2$	3.5	9.0	"
10	Same tests, and terminal conditions as for subgroup 9, except T_C = +125°C and use limits from table I.																				
11	Same tests, and terminal conditions as for subgroup 9, except T_C = -55°C and use limits from table I.																				

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1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 4.5 V, 0 V, 4.5 V to $\overline{SD}$ X, then apply 4.5 V, 0 V, 4.5 V to CPX, then make measurement. Alternate clock: Apply all voltages, then apply 4.5 V, 0 V, 4.5 V to $\overline{SD}$ X, then apply 0 V, 4.5 V, 0 V to CPX, then make measurement.

3/ Apply all voltages, then apply 4.5 V, 0 V, 4.5 V to $\overline{CD}$ X, then apply 4.5 V, 0 V, 4.5 V to CPX, then make measurement. Alternate clock: Apply all voltages, then apply 4.5 V, 0 V, 4.5 V to $\overline{CD}$ X, then apply 0 V, 4.5 V, 0 V, to CPX, then make measurement.

4/ Apply all voltages, then apply 0, 4.5 V, 0, to CPX then make measurement.

5/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits			
	A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	-.03/-.60
I_{IL2}	-.75/-1.80	-.09/-1.80	-.09/-1.80	-.09/-1.80

6/ A = 2.5 V
B = 0.5 V
H ≥ 1.5 V
L ≤ 1.5 V

7/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

8/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency. f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein.

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{CD}_1$	J_1	$\overline{K}_1$	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	$\overline{K}_2$	J_2	$\overline{CD}_2$	V_{CC}				
1 Tc = 25°C	V_{OL}	3007	1	.8 V				2.0 V	20 mA		GND								4.5 V	Q_1		.5	V
			2	4.5 V	0 V	.8 V	2/	2/	"		"								"	Q_1		"	"
			3	4.5 V	.8 V	0 V	2/	2/	"		"								"	Q_1		"	"
			4	2.0 V				.8 V		20 mA	"								"	$\overline{Q}_1$		"	"
			5	3/	2.0 V	4.5 V	3/	4.5 V		"	"								"	$\overline{Q}_1$		"	"
			6	3/	4.5 V	2.0 V	3/	4.5 V		"	"								"	$\overline{Q}_1$		"	"
			7								"	20 mA		.8 V				2.0 V	"	$\overline{Q}_2$		"	"
			8								"	"		4.5 V	3/	4.5 V	2.0 V	3/	"	$\overline{Q}_2$		"	"
			9								"	"			3/	2.0 V	4.5 V	3/	"	$\overline{Q}_2$		"	"
			10								"		20 mA	2.0 V				.8 V	"	Q_2		"	"
			11								"		"	2/	2/	.8 V	0 V	4.5 V	"	Q_2		"	"
			12								"		"	2/	2/	0 V	.8 V	4.5 V	"	Q_2		"	"
	V_{OH}		13	2.0 V				4/	-1 mA		"								"	Q_1	2.5		"
			14	4/				2.0 V		-1 mA	"								"	$\overline{Q}_1$	"		"
			15								"	-1 mA		2.0 V				4/	"	$\overline{Q}_2$	"		"
			16								"		-1 mA	4/				2.0 V	"	Q_2	"		"
	V_{IC}		17	-18 mA							"								"	$\overline{CD}_1$		-1.2	"
			18		-18 mA						"								"	J_1		"	"
			19			-18 mA					"								"	$\overline{K}_1$		"	"
			20				-18 mA				"								"	CP_1		"	"
			21					-18 mA			"								"	$\overline{SD}_1$		"	"
			22								"			-18 mA					"	$\overline{SD}_2$		"	"
			23								"				-18 mA				"	CP_2		"	"
			24								"					-18 mA			"	$\overline{K}_2$		"	"
			25								"						-18 mA		"	J_2		"	"
			26								"							-18 mA	"	$\overline{CD}_2$		"	"
	I_{IH1}	3010	27	2.7 V	0 V	0 V	4/				"								5.5 V	$\overline{CD}_1$		20	∝A
			28	0 V	2.7 V						"								"	J_1		"	"
			29	"		2.7 V					"								"	$\overline{K}_1$		"	"
			30	"			2.7 V				"						"		"	CP_1		"	"
			31				2.7 V	0 V			"						"		"	CP_1		"	"
			32		4.5 V		5/	2.7 V			"						"		"	$\overline{SD}_1$		"	"
			33								"			2.7 V	5/		4.5 V		"	$\overline{SD}_2$		"	"
			34								"			0 V	2.7 V				"	CP_2		"	"
			35								"				2.7 V			0 V	"	CP_2		"	"
			36								"					2.7 V		"	"	$\overline{K}_2$		"	"
			37								"						2.7 V	"	"	J_2		"	"
			38								"				4/	0 V	0 V	2.7 V	"	$\overline{CD}_2$		"	"

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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{CD}_1$	J_1	$\overline{K}_1$	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	$\overline{K}_2$	J_2	$\overline{CD}_2$	V_{CC}		Min	Max	
1	I_{IH2}	3010	39	7.0 V	0 V	0 V	4/				GND								5.5 V	$\overline{CD}_1$		100	"
Tc = 25°C			40	0 V	7.0 V						"								"	J_1		"	"
			41	"		7.0 V					"								"	$\overline{K}_1$		"	"
			42	"			7.0 V				"								"	CP_1		"	"
			43				7.0 V	0 V			"								"	CP_1		"	"
			44		4.5 V		5/	7.0 V			"								"	$\overline{SD}_1$		"	"
			45								"			7.0 V	5/		4.5		"	$\overline{SD}_2$		"	"
			46								"			0 V	7.0 V				"	CP_2		"	"
			47								"				7.0 V			0 V	"	CP_2		"	"
			48								"					7.0 V		"	"	$\overline{K}_2$		"	"
			49								"						7.0 V	"	"	J_2		"	"
			50								"				4/	0 V	0 V	7.0 V	"	$\overline{CD}_2$		"	"
	I_{IL1}	3009	51	4/	.5 V		0 V				"								"	J_1	6/	6/	mA
			52	4.5 V		.5 V	0 V	0 V			"								"	$\overline{K}_1$	"	"	"
			53	4.5 V		0 V	.5 V	0 V			"								"	CP_1	"	"	"
			54								"			0 V	.5 V	0 V		4.5 V	"	CP_2	"	"	"
			55								"			0 V	0 V	.5 V		4.5 V	"	$\overline{K}_2$	"	"	"
			56								"				0 V		.5 V	4/	"	J_2	"	"	"
	I_{IL2}	3009	57	.5 V	4.5 V	4.5 V	4.5 V	0 V			"								"	$\overline{CD}_1$	"	"	"
			58	0 V	0 V	0 V		.5 V			"								"	$\overline{SD}_1$	"	"	"
			59								"			.5 V		0 V	0 V	0 V	"	$\overline{SD}_2$	"	"	"
			60								"			0 V	4.5 V	4.5 V	4.5 V	.5 V	"	$\overline{CD}_2$	"	"	"
	I_{OS}	3011	61					0 V	0 V		"								"	Q_1	-60	-150	"
			62	0 V						0 V	"								"	$\overline{Q}_1$	"	"	"
			63								"	0 V						0 V	"	$\overline{Q}_2$	"	"	"
			64								"		0 V	0 V					"	Q_2	"	"	"
	I_{OD}		65	0 V				5.5 V	2.5 V		"								4.5 V	Q_1	60		"
			66	5.5 V				0 V		2.5 V	"								"	$\overline{Q}_1$	"		"
			67								"	2.5 V		0 V				5.5 V	"	$\overline{Q}_2$	"		"
			68								"		2.5 V	5.5 V				0 V	"	Q_2	"		"
	I_{CC}	3005	69	5.5 V		0 V	0 V	0 V			"			0 V	0 V	0 V		5.5 V	5.5 V	V_{CC}		17	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device types 02.

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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{CD}_1$	J_1	$\overline{K}_1$	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	$\overline{K}_2$	J_2	$\overline{CD}_2$	V_{CC}		Min	Max	
7 Tc = +25°C	Functional tests 8/	3014	70	A	A	B	A	B	H	L	GND	L	H	B	A	B	A	A	7/	All outputs			
			71	A	"	"	"	A	H	L	"	L	H	A	"	"	"	A	"				
			72	B	"	"	"	A	L	H	"	H	L	"	"	"	"	B	"	"			
			73	A	"	"	"	A	L	H	"	H	L	"	"	"	"	A	"	"			
			74	"	"	"	"	B	H	L	"	L	H	B	"	"	"	"	"	"			
			75	"	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"	"			
			76	"	B	A	"	B	"	"	"	"	"	B	"	A	B	"	"	"			
			77	"	"	A	"	A	"	"	"	"	"	A	"	A	"	"	"	"			
			78	"	"	B	"	"	"	"	"	"	"	"	"	B	"	"	"	"			
			79	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"			
			80	"	"	"	A	"	L	H	"	H	L	"	A	"	"	"	"	"			
			81	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"			
			82	"	"	"	A	"	"	"	"	"	"	"	A	"	"	"	"	"			
			83	"	"	A	"	"	"	"	"	"	"	"	"	A	"	"	"	"			
			84	"	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"			
			85	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"			
			86	"	"	"	A	"	H	L	"	L	H	"	A	"	"	"	"	"			
			87	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"			
			88	"	"	"	A	"	"	"	"	"	"	"	A	"	"	"	"	"			
			89	"	B	"	"	B	"	"	"	"	"	B	"	"	B	"	"	"			
			90	"	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"	"			
			91	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"			
			92	"	"	"	A	"	"	"	"	"	"	"	A	"	"	"	"	"			
			93	B	"	"	"	"	L	H	"	H	L	"	"	"	"	B	"	"			
			94	A	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"			
			95	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"			
			96	"	"	"	A	"	"	"	"	"	"	"	A	"	"	"	"	"			
			97	"	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"			
			98	"	"	B	"	"	"	"	"	"	"	"	"	B	"	"	"	"			
			99	"	"	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"			
			100	"	"	"	A	"	H	L	"	L	H	"	A	"	"	"	"	"			
			101	"	"	"	B	"	H	L	"	L	H	"	B	"	"	"	"	"			
			102	"	"	"	A	"	L	H	"	H	L	"	A	"	"	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and T_C = -55°C.																						
9 Tc = +25°C	f_{MAX}	3003 Fig. 4	103	2.7 V	2.7 V	0	IN	2.7 V	OUT		"								5.0 V	Q_1	90		MHz
	9/		104	2.7 V	2.7 V	0	IN	2.7 V		OUT	"								"	$\overline{Q}_1$	"		"
			105								"	OUT		2.7 V	IN	0	2.7 V	2.7 V	"	$\overline{Q}_2$	"		"
			106								"		OUT	2.7 V	IN	0	2.7 V	2.7 V	"	Q_2	"		"
	t_{PLH1}		107	2.7 V	2.7 V	0 V	IN	2.7 V	OUT		"								"	CP_1 to Q_1	3.8	7.0	ns
	t_{PLH1}		108								"		OUT	2.7 V	IN	0 V	2.7 V	2.7 V	"	CP_2 to Q_2	3.8	7.0	"
	t_{PHL1}		109	2.7 V	2.7 V	0	IN	2.7 V	OUT		"								"	CP_1 to Q_1	4.4	8.0	"
	t_{PHL1}		110								"		OUT	2.7 V	IN	0 V	2.7 V	2.7 V	"	CP_2 to Q_2	4.4	8.0	"
	t_{PLH2}		111	2.7 V	2.7 V	0	IN	2.7 V		OUT	"								"	CP_1 to $\overline{Q}_1$	3.8	7.0	"
	t_{PLH2}		112								"	OUT		2.7 V	IN	0 V	2.7 V	2.7 V	"	CP_2 to $\overline{Q}_2$	3.8	7.0	"
	t_{PHL2}		113	2.7 V	2.7 V	0	IN	2.7 V		OUT	"								"	CP_1 to $\overline{Q}_1$	4.4	8.0	"
	t_{PHL2}		114								"	OUT		2.7 V	IN	0 V	2.7 V	2.7 V	"	CP_2 to $\overline{Q}_2$	4.4	8.0	"

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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{CD}_1$	J_1	K_1	CP_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	GND	$\overline{Q}_2$	Q_2	$\overline{SD}_2$	CP_2	K_2	J_2	$\overline{CD}_2$	V_{CC}				
9	t_{PLH3}	3003	115	IN	2.7 V	0 V	2.7 V	2.7 V		OUT	"								5.0 V	$\overline{CD}_1$ to $\overline{Q}_1$	3.2	7.0	"
Tc = +25°C		Fig. 4	116	2.7 V	2.7 V	0 V	2.7 V	IN	OUT		"								"	$\overline{SD}_1$ to Q_1	"	"	"
			117								"	OUT		2.7 V	2.7 V	0 V	2.7 V	IN	"	$\overline{CD}_2$ to $\overline{Q}_2$	"	"	"
			118								"		OUT	IN	2.7 V	0 V	2.7 V	2.7 V	"	$\overline{SD}_2$ to Q_2	"	"	"
	t_{PHL3}		119	IN	2.7 V	0 V	2.7 V	2.7 V	OUT		"								"	$\overline{CD}_1$ to Q_1	3.5	9.0	"
			120	2.7 V				IN		OUT	"								"	$\overline{SD}_1$ to $\overline{Q}_1$	"	"	"
			121								"		OUT	2.7 V	2.7 V	0 V	2.7 V	IN	"	$\overline{CD}_2$ to Q_2	"	"	"
			122								"	OUT		IN				2.7 V	"	$\overline{SD}_2$ to $\overline{Q}_2$	"	"	"
	t_{PLH4}		123	IN	2.7 V	0 V	0 V	2.7 V		OUT	"								"	$\overline{CD}_1$ to $\overline{Q}_1$	3.2	7.0	"
			124	2.7 V				IN	OUT		"								"	$\overline{SD}_1$ to Q_1	"	"	"
			125								"	OUT		2.7 V	0 V	0 V	2.7 V	IN	"	$\overline{CD}_2$ to $\overline{Q}_2$	"	"	"
			126								"		OUT	IN				2.7 V	"	$\overline{SD}_2$ to Q_2	"	"	"
	t_{PHL4}		127	IN	2.7 V	0 V	0 V	2.7 V	OUT		"								"	$\overline{CD}_1$ to Q_1	3.5	9.0	"
			128	2.7 V				IN		OUT	"								"	$\overline{SD}_1$ to $\overline{Q}_1$	"	"	"
			129								"		OUT	2.7 V	0 V	0 V	2.7 V	IN	"	$\overline{CD}_2$ to Q_2	"	"	"
			130								"	OUT		IN				2.7 V	"	$\overline{SD}_2$ to $\overline{Q}_2$	"	"	"
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C and use limits from table I.																						

See footnotes at end of device type 02.

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1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 3 V, 0 V, 3 V to $\overline{SD}$ X, then apply 3 V, 0 V, 3 V to CPX, then make measurement. For circuit C devices, apply all voltages then apply 4.5 V, 0 V, 4.5 V to $\overline{SD}$ X. then apply 4.5 V, 0 V, 4.5 V to CPX, then make the measurement.

3/ Apply all voltages, then apply 3 V, 0 V, 3 V to $\overline{CD}$ X, then apply 3 V, 0 V, 3 V to CPX, then make measurement. For circuit C devices, apply all voltages then apply 4.5 V, 0 V, 4.5 V to $\overline{CP}$ X, then make the measurement.

4/ Apply all voltages, then apply 0 V, 4.5 V to CPX, CDX, or SDX, then make measurement.

5/ Apply all voltages, then apply 0 V, 4.5 V, 0 V to CP_1 test 32 and 44 and CP_2 test 33 and 45 then make measurement.

6/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits			
	A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	-.03/-.60
I_{IL2}	-.75/-1.80	-.09/-1.80	-.09/-1.80	-.09/-1.80

7/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

8/ A = 2.5 V
B = 0.5 V
H ≥ 1.5 V
L ≤ 1.5 V

9/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{CP}_1$	K_1	J_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	$\overline{Q}_2$	GND	Q_2	$\overline{SD}_2$	J_2	$\overline{K}_2$	$\overline{CP}_2$	$\overline{CD}_2$	$\overline{CD}_1$	V_{CC}	terminal	Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	2/	.8 V	2.0 V	2.0 V	-1 mA			GND							2.0 V	4.5 V	Q_1	2.5		V
			2								"	-1mA	2.0 V	2.0 V	.8 V	2/	2.0 V		"	Q_2	"		"
			3	2/	2.0 V	.8 V	2.0 V		-1 mA		"							2.0 V	"	$\overline{Q}_1$	"		"
			4							-1 mA	"		2.0 V	.8 V	2.0 V	2/	2.0 V		"	$\overline{Q}_2$	"		"
			5	4.5 V	4.5 V	4.5 V	.8 V	-1 mA			"							2.0 V	"	Q_1	"		"
			6								"	-1 mA	.8 V	4.5 V	4.5 V	4.5 V	2.0 V		"	Q_2	"		"
			7	4.5 V	4.5 V	4.5 V	2.0 V		-1 mA		"							.8 V	"	$\overline{Q}_1$	"		"
			8							-1 mA	"		2.0 V	4.5 V	4.5 V	4.5 V	.8 V		"	$\overline{Q}_2$	"		"
	V_{OL}	3007	9	2/	.8 V	2.0 V	2.0 V		20 mA		"							2.0 V	"	$\overline{Q}_1$		.50	"
			10							20 mA	"		2.0 V	2.0 V	.8 V	2/	2.0 V		"	$\overline{Q}_2$		"	"
			11	2/	2.0 V	.8 V	2.0 V	20 mA			"							2.0 V	"	Q_1		"	"
			12								"	20 mA	2.0 V	.8 V	2.0 V	2/	2.0 V		"	Q_2		"	"
			13	4.5 V	4.5 V	4.5 V	.8 V		20 mA		"							2.0 V	"	$\overline{Q}_1$		"	"
			14							20 mA	"		.8 V	4.5 V	4.5 V	4.5 V	2.0 V		"	$\overline{Q}_2$		"	"
			15	4.5 V	4.5 V	4.5 V	2.0 V	20 ma			"							.8 V	"	Q_1		"	"
			16								"	20 mA	2.0 V	4.5 V	4.5 V	4.5 V	.8 V		"	Q_2		"	"
	V_{IC}		17	-18 mA							"								"	$\overline{CP}_1$		-1.2	"
			18		-18 mA						"								"	K_1		"	"
			19			-18 mA					"								"	J_1		"	"
			20				-18 mA				"								"	$\overline{SD}_1$		"	"
			21								"		-18 mA						"	$\overline{SD}_2$		"	"
			22								"			-18 mA					"	J_2		"	"
			23								"				-18 mA				"	K_2		"	"
			24								"					-18 mA			"	$\overline{CP}_2$		"	"
			25	2.7 V	0 V	0 V	4.5 V				"						-18 mA		"	$\overline{CD}_2$		"	"
			26	2.7 V	0 V	4/	"				"							-18 mA	"	$\overline{CD}_1$		"	"
	I_{IL1}	3009	27	4.5 V	.5 V	4.5 V	2/				"							4.5 V	5.5 V	K_1	3/	3/	mA
			28								"		2/	4.5 V	.5 V	4.5 V	4.5 V		"	K_2	"	"	"
			29	4.5 V	4.5 V	.5 V	4.5 V				"							2/	"	J_1	"	"	"
			30								"		4.5 V	.5 V	4.5 V	4.5 V	2/		"	J_2	"	"	"
	I_{IL2}	3009	31	4.5 V	4.5 V	4.5 V	4.5 V				"							.5 V	"	$\overline{CD}_1$	"	"	"
			32								"		4.5 V	4.5 V	4.5 V	4.5 V	.5 V		"	$\overline{CD}_2$	"	"	"
			33	4.5 V	4.5 V	4.5 V	.5 V				"							4.5 V	"	$\overline{SD}_1$	"	"	"
			34								"		.5 V	4.5 V	4.5 V	4.5 V	4.5 V		"	$\overline{SD}_2$	"	"	"
	I_{IL3}	3009	35	.5 V	4.5 V	4.5 V	4.5 V				"							2/	"	$\overline{CP}_1$	"	"	"
		"	36								"		4.5 V	4.5 V	4.5 V	.5 V	2/		"	$\overline{CP}_2$	"	"	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{CP}_1$	K_1	J_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	$\overline{Q}_2$	GND	Q_2	$\overline{SD}_2$	J_2	$\overline{K}_2$	$\overline{CP}_2$	$\overline{CD}_2$	$\overline{CD}_1$	V_{CC}		Min	Max	
1 Tc = 25°C	I_{IH1}	3010	37	GND	2.7 V	4.5 V	GND				GND							4.5 V	5.5 V	K_1		20	∞A
			38								"		GND	4.5 V	2.7 V	GND	4.5 V		"	K_2		"	"
			39	GND	4.5 V	2.7 V	4.5 V				"					"		GND	"	J_1		"	"
			40								"		4.5 V	2.7 V	4.5 V	"	GND		"	J_2		"	"
			41	GND	4.5 V	GND	2/				"					"		2.7 V	"	$\overline{CD}_1$		"	"
			42								"		2/	GND	4.5 V	"	2.7 V		"	$\overline{CD}_2$		"	"
			43	GND	GND	4.5 V	2.7 V				"					"		2/	"	$\overline{SD}_1$		"	"
			44								"		2.7 V	4.5 V	GND	"	2/		"	$\overline{SD}_2$		"	"
			45	2.7 V	GND	GND	GND				"					"		GND	"	$\overline{CP}_1$		"	"
			46								"		GND	GND	GND	2.7 V	GND		"	$\overline{CP}_2$		"	"
	I_{IH2}	3010	47	GND	7.0 V	4.5 V	GND				"							4.5 V	"	K_1		100	∞A
			48								"		GND	4.5 V	7.0 V	GND	4.5 V		"	K_2		"	"
			49	GND	4.5 V	7.0 V	4.5 V				"							GND	"	J_1		"	"
			50								"		4.5 V	7.0 V	4.5 V	GND	GND		"	J_2		"	"
			51	GND	4.5 V	GND	2/				"							7.0 V	"	$\overline{CD}_1$		"	"
			52								"		2/	GND	4.5 V	GND	7.0 V		"	$\overline{CD}_2$		"	"
			53	GND	GND	4.5 V	7.0 V				"							2/	"	$\overline{SD}_1$		"	"
			54								"		7.0 V	4.5 V	GND	GND	2/		"	$\overline{SD}_2$		"	"
			55	7.0 V	GND	GND	GND				"							GND	"	$\overline{CP}_1$		"	"
			56								"		GND	GND	GND	7.0 V	GND		"	$\overline{CP}_2$		"	"
	I_{OS}	3011	57	4.5 V	4.5 V	4.5 V	GND	0 V			"							4.5 V	"	Q_1	-60	-150	mA
			58								"	0 V	GND	4.5 V	4.5 V	4.5 V	4.5 V		"	Q_2	"	"	"
			59	4.5 V	4.5 V	4.5 V	4.5 V		0 V		"							GND	"	$\overline{Q}_1$	"	"	"
			60							0 V	"		4.5 V	4.5 V	4.5 V	4.5 V	GND		"	$\overline{Q}_2$	"	"	"
	I_{OD}		61	5.5 V	5.5 V	5.5 V	5.5 V	2.5 V			"		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0 V	4.5 V	Q_1	60		"
			62	"	"	"	0 V		2.5 V		"		5.5 V	"	"	"	"	5.5 V	"	$\overline{Q}_1$	"		"
			63	"	"	"	5.5 V			2.5 V	"		0 V	"	"	"	"	"	"	$\overline{Q}_2$	"		"
			64	"	"	"	5.5 V				"	2.5 V	5.5 V	"	"	"	0 V	"	"	Q_2	"		"
	I_{CC}	3005	65	4.5 V	4.5 V	4.5 V	GND				"		GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	5.5 V	V_{CC}		19	"
	I_{CC}	3005	66	4.5 V	4.5 V	4.5 V	4.5 V				"		4.5 V	4.5 V	4.5 V	4.5 V	GND	GND	5.5 V	V_{CC}		19	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{CP}_1$	K_1	J_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	$\overline{Q}_2$	GND	Q_2	$\overline{SD}_2$	J_2	K_2	$\overline{CP}_2$	$\overline{CD}_2$	$\overline{CD}_1$	V_{CC}		Min	Max	
7 Tc = 25°C	Functional tests 5/	3006	67	B	B	A	A	L	H	H	GND	L	A	A	B	B	B	B	4/	All outputs			
			68	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"			
			69	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"			
			70	B	A	B	B	H	L	L	"	H	B	B	A	B	A	A	"	"			
			71	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"			
			72	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"			
			73	"	B	"	A	L	H	H	"	L	A		B	"	B	B	"	"			
			74	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"	"			
			75	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"			
			76	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"			
			77	"	"	"	B	H	L	L	"	H	B	"	"	"	"	"	"	"			
			78	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"	"	"			
			79	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"			
			80	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"			
			81	"	"	A	"	L	H	H	"	L	"	A	"	"	B	B	"	"			
			82	"	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"	"			
			83	A	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"			
			84	B	"	"	"	H	L	L	"	H	"	"	"	B	"	"	"	"			
			85	"	A	B	B	"	"	"	"	"	B	B	A	"	"	"	"	"			
			86	"	"	"	A	"	"	"	"	"	A	"	"	"	"	"	"	"			
			87	A	"	"	"	"	"	"	"	"	A	"	"	A	"	"	"	"			
			88	B	"	"	"	L	H	H	"	L	A	"	"	B	"	"	"	"			
			89	B	"	"	A	L	H	H	"	"	"	"	"	"	B	B	"	"			
			90	B	"	"	"	"	"	"	"	"	"	"	"	"	A	A	"	"			
			91	A	"	A	"	"	"	"	"	"	"	A	"	A	"	"	"	"			
			92	B	"	A	"	H	L	L	"	H	"	A	"	B	"	"	"	"			
			93	A	"	"	"	H	L	L	"	H	"	"	"	A	"	"	"	"			
			94	B	"	"	"	L	H	H	"	L	"	"	"	B	"	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C	f_{MAX} 6/	3003 Fig. 4	95	IN	2.7 V	2.7 V	2.7 V	OUT			GND							2.7 V	5.0 V	Q_1	90		MHz
			96	IN	2.7 V	2.7 V	2.7 V		OUT		"							2.7 V	"	$\overline{Q}_1$	"		"
			97							OUT	"		2.7 V	2.7 V	2.7 V	IN	2.7 V		"	$\overline{Q}_2$	"		"
			98								"	OUT		2.7 V	2.7 V	IN	2.7 V		"	Q_1	"		"
	t_{PLH1}		99	IN	2.7 V	2.7 V	2.7 V	OUT			GND							IN	5.0 V	$\overline{CP}_1$ to Q_1	3.3	7.7	ns
			100								"	OUT	2.7 V	2.7 V	2.7 V	IN	IN		"	$\overline{CP}_2$ to Q_2	"	"	"
	t_{PLH2}		101	IN	2.7 V	2.7 V	IN		OUT		"							2.7 V	"	$\overline{CP}_1$ to $\overline{Q}_1$	"	"	"
			102							OUT	"		IN	2.7 V	2.7 V	IN	2.7 V		"	$\overline{CP}_2$ to $\overline{Q}_2$	"	"	"
	t_{PHL1}		103	IN	2.7 V	2.7 V	IN	OUT			"							2.7 V	"	$\overline{CP}_1$ to Q_1	"	"	"
			104								"	OUT	IN	2.7 V	2.7 V	IN	2.7 V		"	$\overline{CP}_2$ to Q_2	"	"	"
	t_{PHL2}		105	IN	2.7 V	2.7 V	2.7 V		OUT		"							IN	"	$\overline{CP}_1$ to $\overline{Q}_1$	"	"	"
			106							OUT	"		2.7 V	2.7 V	2.7 V	IN	IN		"	$\overline{CP}_2$ to $\overline{Q}_2$	"	"	"
	t_{PLH3}		107	2.7 V	2.7 V	2.7 V	IN	OUT			"							IN	"	$\overline{SD}_1$ to Q_1	2.0	7.0	"
			108								"	OUT	IN	2.7 V	2.7 V	2.7 V	IN		"	$\overline{SD}_2$ to Q_2	"	"	"
			109	2.7 V	2.7 V	2.7 V	IN		OUT		"							IN	"	$\overline{CD}_1$ to $\overline{Q}_1$	"	"	"
			110							OUT	"		IN	2.7 V	2.7 V	2.7 V	IN		"	$\overline{CD}_2$ to $\overline{Q}_2$	"	"	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{CP}_1$	K_1	J_1	$\overline{SD}_1$	Q_1	$\overline{Q}_1$	$\overline{Q}_2$	GND	Q_2	$\overline{SD}_2$	J_2	K_2	$\overline{CP}_2$	$\overline{CD}_2$	$\overline{CD}_1$	V_{CC}				
9	t_{PHL3}	3003	111	2.7 V	2.7 V	2.7 V	IN	OUT			"							IN	"	$\overline{CD}_1$ to Q_1	2.5	7.7	"
Tc = +25°C		Fig. 4	112								"	OUT	IN	2.7 V	2.7 V	2.7 V	IN		"	$\overline{CD}_2$ to Q_2	"	"	"
			113	2.7 V	2.7 V	2.7 V	IN		OUT		"							IN	"	$\overline{SD}_1$ to $\overline{Q}_1$	"	"	"
			114							OUT	"		IN	2.7 V	2.7 V	2.7 V	IN		"	$\overline{SD}_2$ to $\overline{Q}_2$	"	"	"
	t_{PLH4}		115	0 V	2.7 V	2.7 V	IN	OUT			"							IN	"	$\overline{SD}_1$ to Q_1	2.0 V	7.0	"
			116								"	OUT	IN	2.7 V	2.7 V	0 V	IN		"	$\overline{SD}_2$ to Q_2	"	"	"
			117	0 V	2.7 V	2.7 V	IN		OUT		"							IN	"	$\overline{CD}_1$ to $\overline{Q}_1$	"	"	"
			118							OUT	"		IN	2.7 V	2.7 V	0 V	IN		"	$\overline{CD}_2$ to $\overline{Q}_2$	"	"	"
	t_{PHL4}		119	0 V	2.7 V	2.7 V	IN	OUT			"							IN	"	$\overline{CD}_1$ to Q_1	2.5	7.7	"
			120								"	OUT	IN	2.7 V	2.7 V	0 V	IN		"	$\overline{CD}_2$ to Q_2	"	"	"
			121	0 V	2.7 V	2.7 V	IN		OUT		"							IN	"	$\overline{SD}_1$ to $\overline{Q}_1$	"	"	"
			122							OUT	"		IN	2.7 V	2.7 V	0 V	IN		"	$\overline{SD}_2$ to $\overline{Q}_2$	"	"	"
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests and terminal conditions as subgroup 9, except T_C = -55°C and use limits from table I.																						

See footnotes at end of device type 03.

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1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 3 V, 0 V, 3 V to CPX, $\overline{\text{CDX}}$, or $\overline{\text{SDX}}$ (as required) then make measurement.

3/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits			
	A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	0.0 mA/-0.6 mA
I_{IL2}	-.75/-3.0	-.09/-3.0	-.12/-3.0	0.0 mA/-3.0 mA
I_{IL3}	-1.25/-3.0	-.12/-2.40	-.12/-3.0	0.0 mA/-2.4 mA

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

5/ A = 2.5 V
B = 0.5 V
H ≥ 1.5 V, L ≤ 1.5 V.

6/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 1/ and X / Test no.	1 / 2 / $\overline{MR}$	2 / 3 / Q_0	3 / 4 / $\overline{Q}_0$	4 / 5 / D_0	5 / 7 / D_1	6 / 8 / $\overline{Q}_1$	7 / 9 / Q_1	8 / 10 / GND	9 / 12 / CP	10 / 13 / Q_2	11 / 14 / $\overline{Q}_2$	12 / 15 / D_2	13 / 17 / D_3	14 / 18 / $\overline{Q}_3$	15 / 19 / Q_3	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 Tc = 25°C	V_{OL}	3007	1	.8 V	20 mA						GND								4.5 V	Q_0		.5	V
			2	"						20 mA	"								"	Q_1		"	"
			3	"							"		20 mA						"	Q_2		"	"
			4	"							"							20 mA	"	Q_3		"	"
			5	2/		20 mA	2.0 V				"	2/							"	$\overline{Q}_0$		"	"
			6	"				2.0 V	20 mA		"	"							"	$\overline{Q}_1$		"	"
			7	"							"	"		20 mA	2.0 V				"	$\overline{Q}_2$		"	"
			8	"							"	"				2.0 V	20 mA		"	$\overline{Q}_3$		"	"
	V_{OH}	3006	9	"	-1 mA		2.0 V				"	"							"	Q_0	2.5		"
			10	"		-1 mA	.8 V				"	"							"	$\overline{Q}_0$	"		"
			11	"				.8 V	-1 mA		"	"							"	$\overline{Q}_1$	"		"
			12	"				2.0 V		-1 mA	"	"							"	Q_1	"		"
			13	"							"	"	-1 mA		2.0 V				"	Q_2	"		"
			14	"							"	"		-1 mA	.8 V				"	$\overline{Q}_2$	"		"
			15	"							"	"				.8 V	-1 mA		"	$\overline{Q}_3$	"		"
			16	"							"	"				2.0 V		-1 mA	"	Q_3	"		"
	V_{IC}		17	-18 mA							"								"	$\overline{MR}$		-1.2	"
			18				-18 mA				"								"	D_0		"	"
			19					-18 mA			"								"	D_1		"	"
			20								"	18 mA							"	CP		"	"
			21								"				-18 mA				"	D_2		"	"
			22								"					-18 mA			"	D_3		"	"
	I_{IH1}	3010	23	2.7 V							"								5.5 V	$\overline{MR}$		20	∝A
			24				2.7 V				"								"	D_0		"	"
			25					2.7 V			"								"	D_1		"	"
			26								"	2.7 V							"	CP		"	"
			27								"				2.7 V				"	D_2		"	"
			28								"					2.7 V			"	D_3		"	"
	I_{IH2}	3010	29	7.0 V							"								"	$\overline{MR}$		100	"
			30				7.0 V				"								"	D_0		"	"
			31					7.0 V			"								"	D_1		"	"
			32								"	7.0 V							"	CP		"	"
			33								"				7.0 V				"	D_2		"	"
			34								"					7.0 V			"	D_3		"	"
	I_{IL1}	3009	35	.5 V							"								"	$\overline{MR}$	3/	3/	mA
			36				.5 V				"								"	D_0	"	"	"
			37					.5 V			"								"	D_1	"	"	"
			38								"	.5 V							"	CP	"	"	"
			39								"				.5 V				"	D_2	"	"	"
			40								"					.5 V			"	D_3	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{MR}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}				
1 Tc = 25°C	I_{OS}	3011	41	2/	0 V		4.5 V				GND	2/							5.5 V	Q_0	-60	-150	mA
			42	"				4.5 V		0 V	"	"							"	Q_1	"	"	"
			43	"							"	"	0 V		4.5 V				"	Q_2	"	"	"
			44	"							"	"				4.5 V		0 V	"	Q_3	"	"	"
			45	0 V		0 V					"	4.5 V							"	$\overline{Q}_0$	"	"	"
			46	"					0 V		"	"							"	$\overline{Q}_1$	"	"	"
			47	"							"	"		0 V					"	$\overline{Q}_2$	"	"	"
			48	"							"	"					0 V		"	$\overline{Q}_3$	"	"	"
	I_{OD}	3006	49	0 V	2.5 V						"								4.5 V	Q_0	60		"
			50	"						2.5 V	"								"	Q_1	"		"
			51	"							"		2.5 V						"	Q_2	"		"
			52	"							"							2.5 V	"	Q_3	"		"
			53	5.5 V		2.5 V	5.5 V	5.5 V			"	4/			5.5 V	5.5 V			"	$\overline{Q}_0$	"		"
			54	"			"	"	2.5 V		"	"			"	"			"	$\overline{Q}_1$	"		"
			55	"			"	"			"	"		2.5 V	"	"			"	$\overline{Q}_2$	"		"
			56	"			"	"			"	"			"	"	2.5 V		"	$\overline{Q}_3$	"		"
	I_{CC}	3005	57	4.5 V			4.5 V	4.5 V				"			4.5 V	4.5 V			5.5 V	V_{CC}		34	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{MR}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}				
7 T_C = 25°C	Func-Tional Tests 6/	3014	58	B	L	H	A	A	H	L	"	B	L	H	A	A	H	L	5/	All			
			59	B	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	outputs			
			60	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			61	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"	"				
			62	"	"	"	A	A	"	"	"	"	"	"	A	A	"	"	"	"				
			63	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"				
			64	"	H	L	"	"	L	H	"	A	H	L	"	"	L	H	"				
			65	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			66	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
			67	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"				
			68	"	"	"	A	A	"	"	"	"	"	"	A	A	"	"	"				
			69	"	"	"	A	A	"	"	"	B	"	"	A	A	"	"	"				
			70	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"				
			71	"	"	"	A	A	"	"	"	"	"	"	A	A	"	"	"				
			72	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"				
			73	"	L	H	"	"	H	L	"	A	L	H	"	"	H	L	"				
			74	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			75	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
			76	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			77	"	"	"	A	A	"	"	"	"	"	"	A	A	"	"	"				
			78	"	"	"	B	B	"	"	"	"	"	"	B	B	"	"	"				
			79	"	"	"	A	A	"	"	"	"	"	"	A	A	"	"	"				
			80	"	H	L	A	A	L	H	"	A	H	L	A	A	L	H	"				
			81	"	H	L	B	B	L	H	"	"	H	L	B	B	L	H	"				
			82	B	L	H	"	"	H	L	"	"	L	H	"	"	H	L	"				
			83	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			84	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																						
9/ T_C = +25°C	f_{MAX}	3003 Fig. 4	85	2.7 V	OUT		IN				GND	IN							5.0 V	Q_0	100		MHz
	7/		86	"		OUT	IN				"	"								$\overline{Q}_0$	"		"
			87	"				IN	OUT		"	"								$\overline{Q}_1$	"		"
			88	"				IN		OUT	"	"								Q_1	"		"
			89	"							"	"	OUT		IN					Q_2	"		"
			90	"							"	"		OUT	IN					$\overline{Q}_2$	"		"
			91	"							"	"				IN	OUT			$\overline{Q}_3$	"		"
			92	"							"	"				IN		OUT		Q_3	"		"
	t_{PLH1}		93	"	OUT		IN				"	IN								CP to Q_0	4.0	6.5	ns
			94	"				IN		OUT	"	"								CP to Q_1	"	"	"
			95	"							"	"	OUT		IN					CP to Q_2	"	"	"
			96	"							"	"				IN		OUT		CP to Q_2	"	"	"
	t_{PLH2}		97	"		OUT	IN				"	"								CP to $\overline{Q}_0$	"	"	"
			98	"				IN	OUT		"	"								CP to $\overline{Q}_1$	"	"	"
			99	"							"	"		OUT	IN					CP to $\overline{Q}_2$	"	"	"
			100	"							"	"				IN	OUT			CP to $\overline{Q}_3$	"	"	"
	t_{PHL1}		101	"	OUT		IN				"	"								CP to Q_0	"	8.5	"
			102	"				IN		OUT	"	"								CP to Q_1	"	"	"
			103	"							"	"	OUT		IN					CP to Q_2	"	"	"
			104	"							"	"				IN		OUT		CP to Q_2	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{MR}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}			
9/ T_C = +25°C	t_{PHL2}	3003 Fig. 4	105	2.7 V		OUT	IN				GND	IN							5.0 V	CP to $\overline{Q}_0$	4.0	8.5	ns
			106	"				IN	OUT		"	"							"	CP to $\overline{Q}_1$	"	"	"
			107	"							"	"		OUT	IN				"	CP to $\overline{Q}_2$	"	"	"
			108	"							"	"				IN	OUT		"	CP to $\overline{Q}_3$	"	"	"
	t_{PHL5}	3003 Fig. 4	109	IN	OUT		2.7 V				"	2.7 V							"	$\overline{MR}$ to Q_0	4.5	11.5	"
			110	"				2.7 V		OUT	"	"							"	$\overline{MR}$ to Q_1	"	"	"
			111	"							"	"	OUT		2.7 V				"	$\overline{MR}$ to Q_2	"	"	"
			112	"							"	"				2.7 V		OUT	"	$\overline{MR}$ to Q_3	"	"	"
	t_{PLH5}		113	"		OUT	2.7 V				"	"							"	$\overline{MR}$ to $\overline{Q}_0$	4.0	8.0	"
			114	"				2.7 V	OUT		"	"							"	$\overline{MR}$ to $\overline{Q}_1$	"	"	"
			115	"							"	"		OUT	2.7 V				"	$\overline{MR}$ to $\overline{Q}_2$	"	"	"
			116	"							"	"				2.7 V	OUT		"	$\overline{MR}$ to $\overline{Q}_3$	"	"	"
	t_{PHL6}		117	"	OUT		2.7 V				"	0.0 V							"	$\overline{MR}$ to Q_0	4.5	11.5	"
			118	"				2.7 V		OUT	"	"							"	$\overline{MR}$ to Q_1	"	"	"
			119	"							"	"	OUT		2.7 V				"	$\overline{MR}$ to Q_2	"	"	"
			120	"							"	"				2.7 V		OUT	"	$\overline{MR}$ to Q_3	"	"	"
	t_{PLH6}		121	"		OUT	2.7 V				"	"							"	$\overline{MR}$ to $\overline{Q}_0$	4.0	8.0	"
			122	"				2.7 V	OUT		"	"							"	$\overline{MR}$ to $\overline{Q}_1$	"	"	"
			123	"							"	"		OUT	2.7 V				"	$\overline{MR}$ to $\overline{Q}_2$	"	"	"
			124	"							"	"				2.7 V	OUT		"	$\overline{MR}$ to $\overline{Q}_3$	"	"	"
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests and terminal conditions as subgroup 9, except T_C = -55°C and use limits from table I.																						

See footnotes at end of device type 04.

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1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 3 V, 0 V, 3 V to $\overline{MR}$, then apply 3 V, 0 V, 3 V to CP, then make measurement.

3/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits			
	A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	0/-.30

4/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP then make measurement.

5/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

6/ A = 2.5 V
B = 0.5 V
H ≥ 1.5 V
L ≤ 1.5 V

7/ f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein.

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TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1 $\overline{OE}$	2 Q_0	3 D_0	4 D_1	5 Q_1	6 Q_2	7 D_2	8 D_3	9 Q_3	10 GND
1 Tc = 25°C	V_{OH}	3006	1	-.8 V	-1 mA	2.0 V							"
			2	"			2.0 V	-1 mA					"
			3	"					-1 mA	2.0 V			"
			4	"							2.0 V	-1 mA	"
			5	"									"
			6	"									"
			7	"									"
			8	"									"
	V_{OL}	3007	9	"	20 mA	.8 V							"
			10	"			.8 V	20 mA					"
			11	"					20 mA	.8 V			"
			12	"							.8 V	20 mA	"
			13	"									"
			14	"									"
			15	"									"
			16	"									"
	V_{IC}		17	-18 mA									"
			18			-18 mA							"
			19				-18 mA						"
			20							-18 mA			"
			21								-18 mA		"
			22										"
			23										"
			24										"
			25										"
			26										"
	I_{IH1}	3010	27	2.7 V									"
			28			2.7 V							"
			29				2.7 V						"
			30							2.7 V			"
			31								2.7 V		"
			32										"
			33										"
			34										"
			35										"
			36										"
	I_{IH2}	3010	37	7.0 V									"
			38			7.0 V							"
			39				7.0 V						"
			40							7.0 V			"
			41								7.0 V		"
			42										"
			43										"
			44										"
			45										"
			46										"
	I_{IL1}	3009	47	.5 V									"
			48			.5 V							"
			49				.5 V						"
			50							.5 V			"
			51								.5 V		"
			52										"
			54										"
			55										"
			56										"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	11	12	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	CP	Q_4	D_4	D_5	Q_5	Q_6	D_6	D_7	Q_7	V_{CC}	terminal	Min	Max	
1 Tc = +25°C	V_{OH}	3006	1	1/									4.5 V	Q_0	2.5		V
			2	"									"	Q_1	"		"
			3	"									"	Q_2	"		"
			4	"									"	Q_3	"		"
			5	"	-1 mA	2.0 V							"	Q_4	"		"
			6	"			2.0 V	-1 mA					"	Q_5	"		"
			7	"					-1 mA	2.0 V			"	Q_6	"		"
			8	"							2.0 V	-1 mA	"	Q_7	"		"
	V_{OL}	3007	9	"									"	Q_0		.5	"
			10	"									"	Q_1		"	"
			11	"									"	Q_2		"	"
			12	"									"	Q_3		"	"
			13	"	20 mA	.8 V							"	Q_4		"	"
			14	"			.8 V	20 mA					"	Q_5		"	"
			15	"					20 mA	.8 V			"	Q_6		"	"
			16	"							.8 V	20 mA	"	Q_7		"	"
	V_{IC}		17										"	$\overline{OE}$		-1.2	"
			18										"	D_0		"	"
			19										"	D_1		"	"
			20										"	D_2		"	"
			21										"	D_3		"	"
			22			-18 mA							"	D_4		"	"
			23				-18 mA						"	D_5		"	"
			24							-18 mA			"	D_6		"	"
			25								-18 mA		"	D7		"	"
			26	-18 mA									"	CP		"	"
	I_{IH1}	3010	27										5.5 V	$\overline{OE}$		20	∞A
			28										"	D_0		"	"
			29										"	D_1		"	"
			30										"	D_2		"	"
			31										"	D_3		"	"
			32			2.7 V							"	D_4		"	"
			33				2.7 V						"	D_5		"	"
			34							2.7 V			"	D_6		"	"
			35								2.7 V		"	D7		"	"
			36	2.7 V									"	CP		"	"
	I_{IH2}	3010	37										"	$\overline{OE}$		100	"
			38										"	D_0		"	"
			39										"	D_1		"	"
			40										"	D_2		"	"
			41										"	D_3		"	"
			42			7.0 V							"	D_4		"	"
			43				7.0 V						"	D_5		"	"
			44							7.0 V			"	D_6		"	"
			45								7.0 V		"	D7		"	"
			46	7.0 V									"	CP		"	"
	I_{IL1}	3009	47										"	$\overline{OE}$	2/	2/	mA
			48										"	D_0	"	"	"
			49										"	D_1	"	"	"
			50										"	D_2	"	"	"
			51										"	D_3	"	"	"
			52			.5 V							"	D_4	"	"	"
			53				.5 V						"	D_5	"	"	"
			54							.5 V			"	D_6	"	"	"
			55								.5 V		"	D7	"	"	"
			56	.5 V									"	CP	"	"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1 $\overline{OE}$	2 Q_0	3 D_0	4 D_1	5 Q_1	6 Q_2	7 D_2	8 D_3	9 Q_3	10 GND
1 T_C = 25°C	I_{OS}	3011	57	0 V	0 V	4.5 V							"
			58	"			4.5 V	0 V					"
			59	"					0 V	4.5 V			"
			60	"							4.5 V	0 V	"
			61	"									"
			62	"									"
			63	"									"
			64	"									"
	I_{OD}		65	"	2.5 V	0 V							"
			66	"			0 V	2.5 V					"
			67	"					2.5 V	0 V			"
			68	"							0 V	2.5 V	"
			69	"									"
			70	"									"
			71	"									"
			72	"									"
	I_{OZH}	3011	73	2.0 V	2.7 V	0 V							"
			74	"			0 V	2.7 V					"
			75	"					2.7 V	0 V			"
			76	"							0 V	2.7 V	"
			77	"									"
			78	"									"
			79	"									"
			80	"									"
	I_{OZL}	3011	81	"	.5 V	4.5 V							"
			82	"			4.5 V	.5 V					"
			83	"					.5 V	4.5 V			"
			84	"							4.5 V	.5 V	"
			85	"									"
			86	"									"
			87	"									"
			88	"									"
	I_{CCZ}	3011	89	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V		"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.												
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.												
7 T_C = +25°C	Func tional tests 3/	3014	90	B	X	B	B	X	X	B	B	X	"
			91	"	L	B	B	L	L	B	B	L	"
			92	"	"	A	A	"	"	A	A	"	"
			93	"	"	"	"	"	"	"	"	"	"
			94	"	H	"	"	H	H	"	"	H	"
			95	"	"	B	B	"	"	B	B	"	"
			96	"	"	"	"	"	"	"	"	"	"
			97	"	L	"	"	L	L	"	"	L	"
			98	"	"	A	A	"	"	A	A	"	"
			99	"	"	B	B	"	"	B	B	"	"
			100	"	L	A	A	"	"	A	A	"	"
			101	"	H	A	A	H	H	A	A	H	"
			102	"	"	B	B	"	"	B	B	"	"
			103	"	"	A	A	"	"	A	A	"	"
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.												

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Case 2, R, S, and X Test no.	11 CP	12 Q_4	13 D_4	14 D_5	15 Q_5	16 Q_6	17 D_6	18 D_7	19 Q_7	20 V_{CC}	Measured terminal	Limits Min	Limits Max	
1 Tc = +25°C	I_{OS}	3011	57	1/									5.5 V	Q_0	-60	-150	mA
			58	"									"	Q_1	"	"	"
			59	"									"	Q_2	"	"	"
			60	"									"	Q_3	"	"	"
			61	"	0 V	4.5 V							"	Q_4	"	"	"
			62	"			4.5 V	0 V					"	Q_5	"	"	"
			63	"					0 V	4.5 V			"	Q_6	"	"	"
			64	"							4.5 V	0 V	"	Q_7	"	"	"
	I_{OD}		65	"									4.5 V	Q_0	35		"
			66	"									"	Q_1	"		"
			67	"									"	Q_2	"		"
			68	"									"	Q_3	"		"
			69	"	2.5 V	0 V							"	Q_4	"		"
			70	"			0 V	2.5					"	Q_5	"		"
			71	"					2.5 V	0 V			"	Q_6	"		"
			72	"							0V	2.5 V	"	Q_7	"		"
	I_{OZH}	3011	73	"									5.5 V	Q_0		+ 50	∝A
			74	"									"	Q_1		"	"
			75	"									"	Q_2		"	"
			76	"									"	Q_3		"	"
			77	"	2.7 V	0 V							"	Q_4		"	"
			78	"			0 V	2.7 V					"	Q_5		"	"
			79	"					2.7 V	0 V			"	Q_6		"	"
			80	"							0 V	2.7 V	"	Q_7		"	"
	I_{OZL}	3011	81	"									"	Q_0		-50	"
			82	"									"	Q_1		"	"
			83	"									"	Q_2		"	"
			84	"									"	Q_3		"	"
			85	"	.5 V	4.5 V							"	Q_4		"	"
			86	"			4.5 V	.5 V					"	Q_5		"	"
			87	"					.5 V	4.5 V			"	Q_6		"	"
			88	"							4.5 V	.5 V	"	Q_7		"	"
	I_{CCZ}	3011	89	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V		"	V_{CC}		86	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																
7 T_C = +25°C	Func-Tional Tests 3/	3014	90	B	X	B	B	X	X	B	B	X	4/	All outputs			
			91	A	L	B	B	L	L	B	B	L	"				
			92	A	"	A	A	"	"	A	A	"	"				
			93	B	"	"	"	"	"	"	"	"	"	"			
			94	A	H	"	"	H	H	"	"	H	"	"			
			95	A	"	B	B	"	"	B	B	"	"	"			
			96	B	"	"	"	"	"	"	"	"	"	"			
			97	A	L	"	"	L	L	"	"	L	"	"			
			98	"	"	A	A	"	"	A	A	"	"	"			
			99	"	"	B	B	"	"	B	B	"	"	"			
			100	B	"	A	A	"	"	A	A	"	"	"			
			101	A	H	A	A	H	H	A	A	H	"	"			
			102	"	"	B	B	"	"	B	B	"	"	"			
			103	"	"	A	A	"	"	A	A	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	Q_0	D_0	D_1	Q_1	Q_2	D_2	D_3	Q_3	GND
9 Tc = +25°C	f_{MAX} 5/	3003 Fig. 4	104	0 V	OUT	IN							"
			105	"			IN	OUT					"
			106	"					OUT	IN			"
			107	"							IN	OUT	"
			108	"									"
			109	"									"
			110	"									"
			111	"									"
	t_{PLH1}		112	"	OUT	IN							"
			113	"			IN	OUT					"
			114	"					OUT	IN			"
			115	"							IN	OUT	"
			116	"									"
			117	"									"
			118	"									"
			119	"									"
	t_{PHL1}		120	"	OUT	IN							"
			121	"			IN	OUT					"
			122	"					OUT	IN			"
			123	"							IN	OUT	"
			124	"									"
			125	"									"
			126	"									"
			127	"									"
	t_{PLZ1}		128	IN	OUT	0 V							"
			129	"			0 V	OUT					"
			130	"					OUT	0 V			"
			131	"							0 V	OUT	"
			132	"									"
			133	"									"
			134	"									"
			135	"									"
	t_{PHZ1}		136	"	OUT	2.7 V							"
			137	"			2.7 V	OUT					"
			138	"					OUT	2.7 V			"
			139	"							2.7 V	OUT	"
			140	"									"
			141	"									"
			142	"									"
			143	"									"
	t_{PZL1}		144	"	OUT	0 V							"
			145	"			0 V	OUT					"
			146	"					OUT	0 V			"
			147	"							0 V	OUT	"
			148	"									"
			149	"									"
			150	"									"
			151	"									"
	t_{PZH1}		152	"	OUT	2.7 V							"
			153	"			2.7 V	OUT					"
			154	"					OUT	2.7 V			"
			155	"							2.7 V	OUT	"
			156	"									"
			157	"									"
			158	"									"
			159	"									"
10	Same tests, and terminal conditions, as subgroup 9, except T_C = +125°C and use limits from table I.												
11	Same tests, and terminal conditions, as subgroup 9, except T_C = -55°C and use limits from table I.												

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	11	12	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	CP	Q_4	D_4	D_5	Q_5	Q_6	D_6	D_7	Q_7	V_{CC}	terminal	Min	Max	
9 Tc = +25°C	f_{MAX} 5/	3003 Fig. 4	104	IN									5.0 V	Q_0	80		MHz
			105	"									"	Q_1	"		"
			106	"									"	Q_2	"		"
			107	"									"	Q_3	"		"
			108	"	OUT	IN							"	Q_4	"		"
			109	"			IN	OUT					"	Q_5	"		"
			110	"					OUT	IN			"	Q_6	"		"
			111	"							IN	OUT	"	Q_7	"		"
	t_{PLH1}		112	IN									"	CP to Q_0	3.0	8.5	ns
			113	"									"	CP to Q_1	"	"	"
			114	"									"	CP to Q_2	"	"	"
			115	"									"	CP to Q_3	"	"	"
			116	"	OUT	IN							"	CP to Q_4	"	"	"
			117	"			IN	OUT					"	CP to Q_5	"	"	"
			118	"					OUT	IN			"	CP to Q_6	"	"	"
			119	"							IN	OUT	"	CP to Q_7	"	"	"
	t_{PHL1}		120	IN									"	CP to Q_0	"	"	"
			121	"									"	CP to Q_1	"	"	"
			122	"									"	CP to Q_2	"	"	"
			123	"									"	CP to Q_3	"	"	"
			124	"	OUT	IN							"	CP to Q_4	"	"	"
			125	"			IN	OUT					"	CP to Q_5	"	"	"
			126	"					OUT	IN			"	CP to Q_6	"	"	"
			127	"							IN	OUT	"	CP to Q_7	"	"	"
	t_{PLZ1}		128	1/									"	$\overline{OE}$ to Q_0	1.5	6.5	"
			129	"									"	$\overline{OE}$ to Q_1	"	"	"
			130	"									"	$\overline{OE}$ to Q_2	"	"	"
			131	"									"	$\overline{OE}$ to Q_3	"	"	"
			132	"	OUT	0 V							"	$\overline{OE}$ to Q_4	"	"	"
			133	"			0 V	OUT					"	$\overline{OE}$ to Q_5	"	"	"
			134	"					OUT	0 V			"	$\overline{OE}$ to Q_6	"	"	"
			135	"							0 V	OUT	"	$\overline{OE}$ to Q_7	"	"	"
	t_{PHZ1}		136	"									"	$\overline{OE}$ to Q_0	"	7.5	"
			137	"									"	$\overline{OE}$ to Q_1	"	"	"
			138	"									"	$\overline{OE}$ to Q_2	"	"	"
			139	"									"	$\overline{OE}$ to Q_3	"	"	"
			140	"	OUT	2.7 V							"	$\overline{OE}$ to Q_4	"	"	"
			141	"			2.7 V	OUT					"	$\overline{OE}$ to Q_5	"	"	"
			142	"					OUT	2.7 V			"	$\overline{OE}$ to Q_6	"	"	"
			143	"							2.7 V	OUT	"	$\overline{OE}$ to Q_7	"	"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	11	12	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	CP	Q_4	D_4	D_5	Q_5	Q_6	D_6	D_7	Q_7	V_{CC}	terminal	Min	Max	
9	t_{PZL1}	3003	144	1/									5.5 V	$\overline{OE}$ to Q_0	2.0	7.5	ns
Tc = +25°C		Fig. 4	145	"									"	$\overline{OE}$ to Q_1	"	"	"
			146	"									"	$\overline{OE}$ to Q_2	"	"	"
			147	"									"	$\overline{OE}$ to Q_3	"	"	"
			148	"	OUT	0 V							"	$\overline{OE}$ to Q_4	"	"	"
			149	"			0 V	OUT					"	$\overline{OE}$ to Q_5	"	"	"
			150	"					OUT	0 V			"	$\overline{OE}$ to Q_6	"	"	"
			151	"							0 V	OUT	"	$\overline{OE}$ to Q_7	"	"	"
	t_{PZH1}		152	"									"	$\overline{OE}$ to Q_0	"	11.5	"
			153	"									"	$\overline{OE}$ to Q_1	"	"	"
			154	"									"	$\overline{OE}$ to Q_2	"	"	"
			155	"									"	$\overline{OE}$ to Q_3	"	"	"
			156	"	OUT	2.7 V							"	$\overline{OE}$ to Q_4	"	"	"
			157	"			2.7 V	OUT					"	$\overline{OE}$ to Q_5	"	"	"
			158	"					OUT	2.7 V			"	$\overline{OE}$ to Q_6	"	"	"
			159	"							2.7 V	OUT	"	$\overline{OE}$ to Q_7	"	"	"
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = +125°C and use limits from table I.																
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C and use limits from table I.																

See footnotes at end of device type 05.

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1/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP, then make measurement.

2/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits			
	A	B	C	D
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60	0/-.30

3/ A = 2.5 V
B = 0.5 V
H ≥ 1.5 V
L ≤ 1.5 V

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V

5/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency. f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein.

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TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1	2	3	4	5	6	7	8	9	10
				$\overline{OE}$	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	GND
1 Tc = +25°C	V_{OH}	3006	1	.8 V	-1 mA	.8 V							"
			2				.8 V	-1 mA					"
			3						-1 mA	.8 V			"
			4								.8 V	-1 mA	"
			5										"
			6										"
			7										"
			8										"
	V_{OL}	3007	9	.8 V	20 mA	2.0 V							"
			10				2.0 V	20 mA					"
			11						20 mA	2.0 V			"
			12								2.0 V	20 mA	"
			13										"
			14										"
			15										"
			16										"
	V_{IC}		17	-18 mA									"
			18			-18 mA							"
			19				-18 mA						"
			20							-18 mA			"
			21								-18 mA		"
			22										"
			22										"
			23										"
			24										
			25										
			26										
	I_{IH1}	3010	27	2.7 V									"
			28			2.7 V							"
			29				2.7 V						"
			30							2.7 V			"
			31								2.7 V		"
			32										"
			33										"
			34										"
			35										"
			36										"
	I_{IH2}	3010	37	7.0 V									"
			38			7.0 V							"
			39				7.0 V						"
			40							7.0 V			"
			41								7.0 V		"
			42										"
			43										"
			44										"
			45										"
			46										"
	I_{IL1}	3009	47	.5 V									"
			48			.5 V							"
			49				.5 V						"
			50							.5 V			"
			51								.5 V		"
			52										"
			53										"
			54										"
			55										"
			56										"

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X Test no.	11 CP	12 $\overline{Q}_4$	13 D_4	14 D_5	15 $\overline{Q}_5$	16 $\overline{Q}_6$	17 D_6	18 D_7	19 $\overline{Q}_7$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	
1 Tc = +25°C	V_{OH}	3006	1	1/									4.5 V	$\overline{Q}_0$	2.5		V
			2	"									"	$\overline{Q}_1$	"		"
			3	"									"	$\overline{Q}_2$	"		"
			4	"									"	$\overline{Q}_3$	"		"
			5	"	-1 mA	.8 V							"	$\overline{Q}_4$	"		"
			6	"			.8 V	-1 mA					"	$\overline{Q}_5$	"		"
			7	"					-1mA	.8 V			"	$\overline{Q}_6$	"		"
			8	"							.8 V	-1 mA	"	$\overline{Q}_7$	"		"
	V_{OL}	3007	9	"									"	$\overline{Q}_0$		-.5	"
			10	"									"	$\overline{Q}_1$		"	"
			11	"									"	$\overline{Q}_2$		"	"
			12	"									"	$\overline{Q}_3$		"	"
			13	"	20 mA	2.0 V							"	$\overline{Q}_4$		"	"
			14	"			2.0 V	20 mA					"	$\overline{Q}_5$		"	"
			15	"					20 mA	2.0 V			"	$\overline{Q}_6$		"	"
			16	"							2.0 V	20 mA	"	$\overline{Q}_7$		"	"
	V_{IC}		17										"	$\overline{OE}$		-1.2	"
			18										"	D_0		"	"
			19										"	D_1		"	"
			20										"	D_2		"	"
			21										"	D_3		"	"
			22			-18 mA							"	D_4		"	"
			23				-18 mA						"	D_5		"	"
			24							-18 mA			"	D_6		"	"
			25								-18 mA		"	D_7		"	"
			26	-18 mA									"	CP		"	"
	I_{IH1}	3010	27										5.5 V	$\overline{OE}$		20	∝A
			28										"	D_0		"	"
			29										"	D_1		"	"
			30										"	D_2		"	"
			31										"	D_3		"	"
			32			2.7 V							"	D_4		"	"
			33				2.7 V						"	D_5		"	"
			34							2.7 V			"	D_6		"	"
			35								2.7 V		"	D_7		"	"
			36	2.7 V									"	CP		"	"
	I_{IH2}	3010	37											$\overline{OE}$		100	"
			38											D_0		"	"
			39											D_1		"	"
			40											D_2		"	"
			41											D_3		"	"
			42			7.0 V								D_4		"	"
			43				7.0 V							D_5		"	"
			44							7.0 V				D_6		"	"
			45								7.0 V			D_7		"	"
			46	7.0 V										CP		"	"

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	11	12	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	CP	$\overline{Q}_4$	D_4	D_5	$\overline{Q}_5$	$\overline{Q}_6$	D_6	D_7	$\overline{Q}_7$	V_{CC}	terminal	Min	Max	
1	I_{IL1}	3009	47										5.5 V	$\overline{OE}$	2/	2/	mA
Tc = +25°C			48										"	D_0	"	"	"
			49										"	D_1	"	"	"
			50										"	D_2	"	"	"
			51										"	D_3	"	"	"
			52			.5 V							"	D_4	"	"	"
			53				.5 V						"	D_5	"	"	"
			54							.5 V			"	D_6	"	"	"
			55								.5 V		"	D_7	"	"	"
			56	.5 V									"	CP	"	"	"

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	GND
1 Tc = +25°C	I_{OS}	3011	57	0 V	0 V	0 V							"
			58	"			0 V	0 V					"
			59	"					0 V	0 V			"
			60	"							0 V	0 V	"
			61	"									"
			62	"									"
			63	"									"
			64	"									"
	I_{OD}		65	0 V	2.5 V	5.5 V							"
			66	"			5.5 V	2.5 V					"
			67	"					2.5 V	5.5 V			"
			68	"							5.5 V	2.5 V	"
			69	"									"
			70	"									"
			71	"									"
			72	"									"
	I_{OZH}	3011	73	2.0 V	2.7 V	4.5 V							"
			74	"			4.5 V	2.7 V					"
			75	"					2.7 V	4.5 V			"
			76	"							4.5 V	2.7 V	"
			77	"									"
			78	"									"
			79	"									"
			80	"									"
	I_{OZL}	3011	81	"	.5 V	0 V							"
			82	"			0 V	.5 V					"
			83	"					.5 V	0 V			"
			84	"							0 V	.5 V	"
			85	"									"
			86	"									"
			87	"									"
			88	"									"
	I_{CCZ}	3011	89	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V		"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.												
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.												

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X Test no.	11 CP	12 $\overline{Q}_4$	13 D_4	14 D_5	15 $\overline{Q}_5$	16 $\overline{Q}_6$	17 D_6	18 D_7	19 $\overline{Q}_7$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	
1	I_{OS}	3011	57	1/									5.5 V	$\overline{Q}_0$	-60	-150	mA
Tc = +25°C			58	"									"	$\overline{Q}_1$	"	"	"
			59	"									"	$\overline{Q}_2$	"	"	"
			60	"									"	$\overline{Q}_3$	"	"	"
			61	"	0 V	0 V							"	$\overline{Q}_4$	"	"	"
			62	"			0 V	0 V					"	$\overline{Q}_5$	"	"	"
			63	"					0 V	0 V			"	$\overline{Q}_6$	"	"	"
			64	"							0 V	0 V	"	$\overline{Q}_7$	"	"	"
	I_{OD}		65	"									4.5 V	$\overline{Q}_0$	35		"
			66	"									"	$\overline{Q}_1$	"		"
			67	"									"	$\overline{Q}_2$	"		"
			68	"									"	$\overline{Q}_3$	"		"
			69	"	2.5 V	5.5 V							"	$\overline{Q}_4$	"		"
			70	"			5.5 V	2.5 V					"	$\overline{Q}_5$	"		"
			71	"					2.5 V	5.5 V			"	$\overline{Q}_6$	"		"
			72	"							5.5 V	2.5 V	"	$\overline{Q}_7$	"		"
	I_{OZH}	3011	73	"									5.5 V	$\overline{Q}_0$		+50	∝A
			74	"									"	$\overline{Q}_1$		"	"
			75	"									"	$\overline{Q}_2$		"	"
			76	"									"	$\overline{Q}_3$		"	"
			77	"	2.7 V	4.5 V							"	$\overline{Q}_4$		"	"
			78	"			4.5 V	2.7 V					"	$\overline{Q}_5$		"	"
			79	"					2.7 V	4.5 V			"	$\overline{Q}_6$		"	"
			80	"							4.5 V	2.7 V	"	$\overline{Q}_7$		"	"
	I_{OZL}	3011	81	"									"	$\overline{Q}_0$		-50	"
			82	"									"	$\overline{Q}_1$		"	"
			83	"									"	$\overline{Q}_2$		"	"
			84	"									"	$\overline{Q}_3$		"	"
			85	"	.5 V	0 V							"	$\overline{Q}_4$		"	"
			86	"			0 V	2.5 V					"	$\overline{Q}_5$		"	"
			87	"					.5 V	0 V			"	$\overline{Q}_6$		"	"
			88	"							0 V	.5 V	"	$\overline{Q}_7$		"	"
	I_{CCZ}	3011	89	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V		"	V_{CC}		86	mA
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	GND
7 Tc = +25°C	Functional 3/	3014	90.1	B	X	A	A	X	X	B	A	X	GND
			90.2	"	L	A	A	L	H	B	A	L	"
			90.3	"	L	A	B	L	H	B	A	L	"
			90.4	"	L	A	B	H	H	B	A	L	"
			90.5	"	L	A	B	H	H	A	A	L	"
			90.6	"	L	A	B	H	L	A	A	L	"
			90.7	"	L	A	B	H	L	A	B	L	"
			90.8	"	L	A	B	H	L	A	B	H	"
			90.9	"	L	A	B	H	L	A	B	H	"
			90.10	"	L	A	B	H	L	A	B	H	"
			90.11	"	L	A	B	H	L	A	B	H	"
			90.12	"	L	A	B	H	L	A	B	H	"
			90.13	"	L	A	B	H	L	A	B	H	"
			90.14	"	L	A	B	H	L	A	B	H	"
			90.15	"	L	A	B	H	L	A	B	H	"
			90.16	"	L	A	B	H	L	A	B	H	"
			90.17	"	L	B	A	H	L	A	B	H	"
			90.18	"	H	B	A	L	L	A	B	H	"
			90.19	"	H	B	A	L	L	A	B	H	"
			90.20	"	H	B	A	L	L	A	B	H	"
			90.21	"	H	B	A	L	L	B	B	H	"
			90.22	"	H	B	A	L	H	B	B	H	"
			90.23	"	H	B	A	L	H	B	A	H	"
			90.24	"	H	B	A	L	H	B	A	L	"
			90.25	"	H	B	A	L	H	B	A	L	
			90.26	"	H	B	A	L	H	B	A	L	
			90.27	"	H	B	A	L	H	B	A	L	
			90.28	"	H	B	A	L	H	B	A	L	"
			90.29	"	H	B	A	L	H	B	A	L	"
			90.30	"	H	B	A	L	H	B	A	L	"
			90.31	"	H	B	A	L	H	B	A	L	"
			90.32	"	H	B	A	L	H	B	A	L	"
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.												

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X Test no.	11 CP	12 $\overline{Q}_4$	13 D_4	14 D_5	15 $\overline{Q}_5$	16 $\overline{Q}_6$	17 D_6	18 D_7	19 $\overline{Q}_7$	20 V_{CC}	Measured terminal
7 Tc = +25°C	Functional 3/	3014	90.1	B	X	B	A	X	X	B	A	X	4/	All outputs
			90.2	A	H	B	A	L	H	B	A	L	"	"
			90.3	B	H	B	A	L	H	B	A	L	"	"
			90.4	A	H	B	A	L	H	B	A	L	"	"
			90.5	B	H	B	A	L	H	B	A	L	"	"
			90.6	A	H	B	A	L	H	B	A	L	"	"
			90.7	B	H	B	A	L	H	B	A	L	"	"
			90.8	A	H	B	A	L	H	B	A	L	"	"
			90.9	B	H	A	A	L	H	B	A	L	"	"
			90.10	A	L	A	A	L	H	B	A	L	"	"
			90.11	B	L	A	B	L	H	B	A	L	"	"
			90.12	A	L	A	B	H	H	B	A	L	"	"
			90.13	B	L	A	B	H	H	A	A	L	"	"
			90.14	A	L	A	B	H	L	A	A	L	"	"
			90.15	B	L	A	B	H	L	A	B	L	"	"
			90.16	A	L	A	B	H	L	A	B	H	"	"
			90.17	B	L	A	B	H	L	A	B	H	"	"
			90.18	A	L	A	B	H	L	A	B	H	"	"
			90.19	B	L	A	B	H	L	A	B	H	"	"
			90.20	A	L	A	B	H	L	A	B	H	"	"
			90.21	B	L	A	B	H	L	A	B	H	"	"
			90.22	A	L	A	B	H	L	A	B	H	"	"
			90.23	B	L	A	B	H	L	A	B	H	"	"
			90.24	A	L	A	B	H	L	A	B	H	"	"
			90.25	B	L	B	B	H	L	A	B	H	"	"
			90.26	A	H	B	B	H	L	A	B	H	"	"
			90.27	B	H	B	A	H	L	A	B	H	"	"
			90.28	A	H	B	A	L	L	A	B	H	"	"
			90.29	B	H	B	A	L	L	B	B	H	"	"
			90.30	A	H	B	A	L	H	B	B	H	"	"
			90.31	B	H	B	A	L	H	B	A	H	"	"
			90.32	A	H	B	A	L	H	B	A	L	"	"
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = +125°C and -55°C.													

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 – Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1	2	3	4	5	6	7	8	9	10
				$\overline{OE}$	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	GND
9	f_{MAX}	3003	103	0 V	OUT	IN							GND
Tc = +25°C	5/	Fig. 4	104	"			IN	OUT					"
			105	"					OUT	IN			"
			106	"							IN	OUT	"
			107	"									"
			108	"									"
			109	"									"
			110	"									"
	t_{PLH2}		111	"	OUT	IN							"
			112	"			IN	OUT					"
			113	"					OUT	IN			"
			114	"							IN	OUT	"
			115	"									"
			116	"									"
			117	"									"
			118	"									"
	t_{PHL2}		119	"	OUT	IN							"
			120	"			IN	OUT					"
			121	"					OUT	IN			"
			122	"							IN	OUT	"
			123	"									"
			124	"									"
			125	"									"
			126	"									"
	t_{PLZ2}		127	IN	OUT	2.7 V							"
			128	"			2.7 V	OUT					"
			129	"					OUT	2.7 V			"
			130	"							2.7 V	OUT	"
			131	"									"
			132	"									"
			133	"									"
			134	"									"
	t_{PHZ2}		135	"	OUT	0 V							"
			136	"			0 V	OUT					"
			137	"					OUT	0 V			"
			138	"							0 V	OUT	"
			139	"									"
			140	"									"
			141	"									"
			142	"									"
	t_{PZL2}		143	"	OUT	2.7 V							"
			144	"			2.7 V	OUT					"
			145	"					OUT	2.7 V			"
			146	"							2.7 V	OUT	"
			147	"									"
			148	"									"
			149	"									"
			150	"									"
	t_{PZH2}		151	"	OUT	0 V							"
			152	"			0 V	OUT					"
			153	"					OUT	0 V			"
			154	"							0 V	OUT	"
			155	"									"
			156	"									"
			157	"									"
			158	"									"

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Casess 2, R, S, and X	11	12	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	CP	$\overline{Q}_4$	D_4	D_5	$\overline{Q}_5$	$\overline{Q}_6$	D_6	D_7	$\overline{Q}_7$	V_{CC}	terminal	Min	Max	Unit
9	f_{MAX}	3005	103	IN									5.0 V	$\overline{Q}_0$	80		MHz
Tc = +25°C	5/	Fig. 4	104	"									"	$\overline{Q}_1$	"		"
			105	"									"	$\overline{Q}_2$	"		"
			106	"									"	$\overline{Q}_3$	"		"
			107	"	OUT	IN							"	$\overline{Q}_4$	"		"
			108	"			IN	OUT					"	$\overline{Q}_5$	"		"
			109	"					OUT	IN			"	$\overline{Q}_6$	"		"
			110	"							IN	OUT	"	$\overline{Q}_7$	"		"
	t_{PLH2}		111	"									"	CP to $\overline{Q}_0$	4.0	8.5	ns
			112	"									"	CP to $\overline{Q}_1$	"	"	"
			113	"									"	CP to $\overline{Q}_2$	"	"	"
			114	"									"	CP to $\overline{Q}_3$	"	"	"
			115	"	OUT	IN							"	CP to $\overline{Q}_4$	"	"	"
			116	"			IN	OUT					"	CP to $\overline{Q}_5$	"	"	"
			117	"					OUT	IN			"	CP to $\overline{Q}_6$	"	"	"
			118	"							IN	OUT	"	CP to $\overline{Q}_7$	"	"	"
	t_{PHL2}	3011	119	"									"	CP to $\overline{Q}_0$	"	"	"
			120	"									"	CP to $\overline{Q}_1$	"	"	"
			121	"									"	CP to $\overline{Q}_2$	"	"	"
			122	"									"	CP to $\overline{Q}_3$	"	"	"
			123	"	OUT	IN							"	CP to $\overline{Q}_4$	"	"	"
			124	"			IN	OUT					"	CP to $\overline{Q}_5$	"	"	"
			125	"					OUT	IN			"	CP to $\overline{Q}_6$	"	"	"
			126	"							IN	OUT	"	CP to $\overline{Q}_7$	"	"	"
	t_{PLZ2}	3011	127	1/									"	$\overline{OE}$ to $\overline{Q}_0$	1.5	5.5	"
			128	"									"	$\overline{OE}$ to $\overline{Q}_1$	"	"	"
			129	"									"	$\overline{OE}$ to $\overline{Q}_2$	"	"	"
			130	"									"	$\overline{OE}$ to $\overline{Q}_3$	"	"	"
			131	"	OUT	2.7 V							"	$\overline{OE}$ to $\overline{Q}_4$	"	"	"
			132	"			2.7 V	OUT					"	$\overline{OE}$ to $\overline{Q}_5$	"	"	"
			133	"					OUT	2.7 V			"	$\overline{OE}$ to $\overline{Q}_6$	"	"	"
			134	"							2.7 V	OUT	"	$\overline{OE}$ to $\overline{Q}_7$	"	"	"

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	11	12	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	CP	$\overline{Q}_4$	D_4	D_5	$\overline{Q}_5$	$\overline{Q}_6$	D_6	D_7	$\overline{Q}_7$	V_{CC}	terminal	Min	Max	Unit
9	t_{PHZ2}	3005	135	1/									5.0 V	$\overline{OE}$ to $\overline{Q}_0$	1.5	7.0	ns
Tc = +25°C		Fig. 4	136	"									"	$\overline{OE}$ to $\overline{Q}_1$	"	"	"
			137	"									"	$\overline{OE}$ to $\overline{Q}_2$	"	"	"
			138	"									"	$\overline{OE}$ to $\overline{Q}_3$	"	"	"
			139	"	OUT	0 V							"	$\overline{OE}$ to $\overline{Q}_4$	"	"	"
			140	"			0 V	OUT					"	$\overline{OE}$ to $\overline{Q}_5$	"	"	"
			141	"					OUT	0 V			"	$\overline{OE}$ to $\overline{Q}_6$	"	"	"
			142	"							0 V	OUT	"	$\overline{OE}$ to $\overline{Q}_7$	"	"	"
	t_{PZL2}		143	"									"	$\overline{OE}$ to $\overline{Q}_0$	2.0	7.5	"
			144	"									"	$\overline{OE}$ to $\overline{Q}_1$	"	"	"
			145	"									"	$\overline{OE}$ to $\overline{Q}_2$	"	"	"
			146	"									"	$\overline{OE}$ to $\overline{Q}_3$	"	"	"
			147	"	OUT	2.7 V							"	$\overline{OE}$ to $\overline{Q}_4$	"	"	"
			148	"			2.7 V	OUT					"	$\overline{OE}$ to $\overline{Q}_5$	"	"	"
			149	"					OUT	2.7 V			"	$\overline{OE}$ to $\overline{Q}_6$	"	"	"
			150	"							2.7 V	OUT	"	$\overline{OE}$ to $\overline{Q}_7$	"	"	"
	t_{PZH2}		151	"									"	$\overline{OE}$ to $\overline{Q}_0$	"	11.5	"
			152	"									"	$\overline{OE}$ to $\overline{Q}_1$	"	"	"
			153	"									"	$\overline{OE}$ to $\overline{Q}_2$	"	"	"
			154	"									"	$\overline{OE}$ to $\overline{Q}_3$	"	"	"
			155	"	OUT	0 V							"	$\overline{OE}$ to $\overline{Q}_4$	"	"	"
			156	"			0 V	OUT					"	$\overline{OE}$ to $\overline{Q}_5$	"	"	"
			157	"					OUT	0 V			"	$\overline{OE}$ to $\overline{Q}_6$	"	"	"
			158	"							0 V	OUT	"	$\overline{OE}$ to $\overline{Q}_7$	"	"	"

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X	1	2	3	4	5	6	7	8	9	10
			Test no.	$\overline{OE}$	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	GND
10	Same tests, and terminal conditions, and limits as subgroup 9, except T_C = +125°C and use limits from table I.												
11	Same tests, and terminal conditions, and limits as subgroup 9, except T_C = -55°C and use limits from table I.												

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	CP	$\overline{Q}_4$	D_4	D_5	$\overline{Q}_5$	$\overline{Q}_6$	D_6	D_7	$\overline{Q}_7$	V_{CC}		Min	Max	
10	Same tests, and terminal conditions, and limits as subgroup 9, except T_C = +125°C and use limits from table I.																
11	Same tests, and terminal conditions, and limits as subgroup 9, except T_C = -55°C and use limits from table I.																

1/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP, then make measurement.

2/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits		
	A	B	C
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60

3/ Output voltage shall be either:

A = 2.5 V
B = 0.5 V
H ≥ 1.5 V
L ≤ 1.5 V
X = Don't care

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

5/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency. f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 1/ and X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{MR}$	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	0.0 V							GND	0.0 V							4.5 V	Setup			V
			2	5.5 V	-1 mA	2.0 V	2.0 V		2.0 V		"	2/		2.0 V		2.0 V	2.0 V		"	Q_0	2.5		"
			3	"		"	"	-1 mA	"		"	0.0 V		"		"	"		"	Q_1	"		"
			4	"		"	"		"	-1 mA	"	"		"		"	"		"	Q_2	"		"
			5	"		"	"		"		"	"	-1 mA	"		"	"		"	Q_3	"		"
			6	"		"	"		"		"	"		"	-1 mA	"	"		"	Q_4	"		"
			7	"		"	"		"		"	"		"		"	"	-1 mA	"	Q_5	"		"
			8	"		5.5 V	5.5 V		5.5 V		"	2/		5.5 V		5.5 V	5.5 V		"	Setup			
			9	2.0 V	-1 mA	"	"		"		"	"		"		"	"		"	Q_0	2.5		V
			10	"		"	"	-1 mA	"		"	"		"		"	"		"	Q_1	"		"
			11	"		"	"		"	-1 mA	"	"		"		"	"		"	Q_2	"		"
			12	"		"	"		"		"	"	-1 mA	"		"	"		"	Q_3	"		"
			13	"		"	"		"		"	"		"	-1 mA	"	"		"	Q_4	"		"
			14	"		"	"		"		"	"		"		"	"	-1 mA	"	Q_5	"		"
	V_{OL}	3007	15	5.5 V		"	"		"		"	"		"		"	"		"	Setup			
			16	"	20 mA	0.8 V	0.8 V		0.8 V		"	"		0.8 V		0.8 V	0.8 V		"	Q_0		0.5	V
			17	"		"	"	20 mA	"		"	0.0 V		"		"	"		"	Q_1		"	"
			18	"		"	"		"	20 mA	"	"		"		"	"		"	Q_2		"	"
			19	"		"	"		"		"	"	20 mA	"		"	"		"	Q_3		"	"
			20	"		"	"		"		"	"		"	20 mA	"	"		"	Q_4		"	"
			21	"		"	"		"		"	"		"		"	"	20 mA	"	Q_5		"	"
			22	"		5.5 V	5.5 V		5.5 V		"	2/		5.5 V		5.5 V	5.5 V		"	Setup			
			23	0.8 V	20 mA	"	"		"		"			"		"	"		"	Q_0		0.5	V
			24	"		"	"	20 mA	"		"			"		"	"		"	Q_1		"	"
			25	"		"	"		"	20 mA	"			"		"	"		"	Q_2		"	"
			26	"		"	"		"		"		20 mA	"		"	"		"	Q_3		"	"
			27	"		"	"		"		"			"	20 mA	"	"		"	Q_4		"	"
			28	"		"	"		"		"			"		"	"	20 mA	"	Q_5		"	"
	V_{IC}		29	-18 mA							"								"	$\overline{MR}$		-1.2	"
			30			-18 mA					"								"	D_0		"	"
			31				-18 mA				"								"	D_1		"	"
			32						-18 mA		"								"	D_2		"	"
			33								"	-18 mA							"	CP		"	"
			34								"			-18 mA					"	D_3		"	"
			35								"					-18 mA			"	D_4		"	"
			36								"						-18 mA		"	D_5		"	"
	I_{IL1}	3009	37								"						0.5 V		5.5 V	D_5	3/	3/	mA
			38								"					0.5 V			"	D_4	"	"	"
			39								"			0.5 V					"	D_3	"	"	"
			40						0.5 V		"								"	D_2	"	"	"
			41				0.5 V				"								"	D_1	"	"	"
			42			0.5 V					"								"	D_0	"	"	"
	I_{IL4}	3009	43	0.5 V							"								"	$\overline{MR}$	"	"	"
			44								"	0.5 V							"	CP	"	"	"
	I_{IH1}	3010	45	2.7 V							"								"	$\overline{MR}$		20	∝A
			46			2.7 V					"								"	D_0		"	"
			47				2.7 V				"								"	D_1		"	"
			48						2.7 V		"								"	D_2		"	"
			49								"	2.7 V							"	CP		"	"
			50								"			2.7 V					"	D_3		"	"
			51								"					2.7 V			"	D_4		"	"
			52								"						2.7 V		"	D_5		"	"

See footnotes at end of device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{MR}$	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}		Min	Max	
1 Tc = +25°C	I_{IH2}	3010	53								"						7.0 V		5.5 V	D_5		100	∝A
			54								"					7.0 V			"	D_4		"	"
			55								"			7.0 V					"	D_3		"	"
			56								"	7.0 V							"	CP		"	"
			57						7.0 V		"								"	D_2		"	"
			58				7.0 V				"								"	D_1		"	"
			59			7.0 V					"								"	D_0		"	"
			60	7.0 V							"								"	MR		"	"
	I_{OS}	3011	61	4.5 V	GND	4.5 V					"	4/							"	Q_0	-60	-150	mA
			62	"			4.5 V	GND			"	"							"	Q_1	"	"	"
			63	"					4.5 V	GND	"	"							"	Q_2	"	"	"
			64	"							"	"	GND	4.5 V					"	Q_3	"	"	"
			65	"							"	"			GND	4.5 V			"	Q_4	"	"	"
			66	"							"	"					4.5 V	GND	"	Q_5	"	"	"
	I_{OD}		67	0.0 V	2.5 V						"	"							4.5 V	Q_0	60		"
			68	"				2.5 V			"	"							"	Q_1	"		"
			69	"						2.5 V	"	"							"	Q_2	"		"
			70	"							"	"	2.5 V						"	Q_3	"		"
			71	"							"	"			2.5 V				"	Q_4	"		"
			72	"							"	"						2.5 V	"	Q_5	"		"
	I_{CC}	3005	73	5.5 V		5.5 V	5.5 V		5.5 V		"	"		5.5 V		5.5 V	5.5 V		5.5 V	V_{CC}		45	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 5/ Tc = +25°C	Truth Table tests	3014	74	B	L	A	A	L	A	L	GND	A	L	A	L	A	A	L	6/	All outputs			
			75	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"			
			76	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"			
			77	A	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"			
			78	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"			
			79	"	H	"	"	H	"	H	"	A	H	"	H	"	"	H	"	"			
			80	"	"	B	B	"	B	"	"	A	"	B	"	B	B	"	"	"			
			81	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"			
			82	"	L	"	"	L	"	L	"	A	L	"	L	"	"	L	"	"			
			83	"	L	A	A	L	A	L	"	B	L	A	"	A	A	L	"	"			
			84	"	H	"	"	H	"	H	"	A	H	"	H	"	"	H	"	"			
			85	"	H	"	"	H	"	H	"	B	H	"	H	"	"	H	"	"			
			86	B	L	"	"	L	"	L	"	B	L	"	L	"	"	L	"	"			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 Tc = +25°C	f_{MAX} 7/		87	2.7 V	OUT	IN					GND	IN							5.0 V	Q_1	90		MHz
			88	"			IN	OUT			"								"	Q_2	"		"
			89	"					IN	OUT	"								"	Q_3	"		"
			90	"							"		OUT	IN					"	Q_4	"		"
			91	"							"				OUT	IN			"	Q_5	"		"
			92	"													IN	OUT	"	Q_6			
	t_{PLH1}	3003	93	"	OUT	IN					"	IN							"	CP to Q_0	1.5	9.0	ns
			94	"			IN	OUT			"	"							"	CP to Q_1	"	"	"
			95	"					IN	OUT	"	"							"	CP to Q_2	"		
			96	"							"	"	OUT	IN					"	CP to Q_3	"	"	"
			97	"							"	"			OUT	IN			"	CP to Q_4	"	"	"
			98	"							"	"					IN	OUT	"	CP to Q_5	"	"	"

See footnotes at end of device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{MR}$	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}				
9 Tc = +25°C	t_{PHL1}	3003	99	2.7 V							GND	IN					IN	OUT	5.0 V	CP to Q_5	1.5 V	11.0	ns
			100	"							"	"			OUT	IN			"	CP to Q_4	"	"	"
			101	"							"	"	OUT	IN					"	CP to Q_3	"	"	"
			102	"					IN	OUT	"	"							"	CP to Q_2	"	"	"
			103	"			IN	OUT			"	"							"	CP to Q_1	"	"	"
			104	"	OUT	IN					"	"							"	CP to Q_0	"	"	"
	t_{PHL5}		105	IN							"	2.7 V					2.7 V	OUT	"	$\overline{MR}$ to Q_5	"	15.0	"
			106	"							"	"			OUT	2.7 V			"	$\overline{MR}$ to Q_4	"	"	"
			107	"							"	"	OUT	2.7 V					"	$\overline{MR}$ to Q_3	"	"	"
			108	"					2.7 V	OUT	"	"							"	$\overline{MR}$ to Q_2	"	"	"
			109	"			2.7 V	OUT			"	"							"	$\overline{MR}$ to Q_1	"	"	"
			110	"	OUT	2.7 V					"	"							"	$\overline{MR}$ to Q_0	"	"	"
	t_{PHL6}		111	IN							"	0.0 V					2.7 V	OUT	5.0 V	$\overline{MR}$ to Q_5	1.0	15.0	ns
			112	"							"	"			OUT	2.7 V			"	$\overline{MR}$ to Q_4	"	"	"
			113	"							"	"	OUT	2.7 V					"	$\overline{MR}$ to Q_3	"	"	"
			114	"					2.7 V	OUT	"	"							"	$\overline{MR}$ to Q_2	"	"	"
			115	"			2.7 V	OUT			"	"							"	$\overline{MR}$ to Q_1	"	"	"
			116	"	OUT	2.7 V					"	"							"	$\overline{MR}$ to Q_0	"	"	"
10	f_{MAX} 7/			Same tests and terminal conditions as for subgroup 9, except T_C = +125°C, and limits as shown.																	70		MHz
	t_{PLH1} t_{PHL1} t_{PHL5} t_{PHL6}																				1.0 " " "	11.0 13.0 17.0 17.0	ns " " "
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of device type 07.

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1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 0 V, 3 V, 0 V to CP, then make measurement.

3/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits		
	A	B	C
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60
I_{IL4}	-.25/-.60	-.06/-1.2	-.06/-1.2

4/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP, then make measurement:

5/ A = 2.5 V
B = 0.5 V
H ≥ 1.5 V
L ≤ 1.5 V

6/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V

7/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter.
For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein.

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TABLE III. Group A inspection for device type 08.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{E}$	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}				
1 Tc = +25°C	V_{OL}	3007	1	.8 V	20 mA	.8 V	.8 V		.8 V		GND	2/		.8 V		.8 V	.8 V		4.5 V	Q_0		.5	V
			2	"		"	"	20 mA	"		"	"		"		"	"		"	Q_1		"	"
			3	"		"	"		"	20 mA	"	"		"		"	"		"	Q_2		"	"
			4	"		"	"		"		"	"	20 mA	"		"	"		"	Q_3		"	"
			5	"		"	"		"		"	"		"	20 mA	"	"		"	Q_4		"	"
			6	"		"	"		"		"	"		"		"	"	20 mA	"	Q_5		"	"
	V_{OH}	3006	7	.8 V	-1 mA	2.0 V	2.0 V		2.0 V		"	"		2.0 V		2.0 V	2.0 V		"	Q_0	2.5 V		"
			8	"		"	"	-1 mA	"		"	"		"		"	"		"	Q_1			"
			9	"		"	"		"	-1 mA	"	"		"		"	"		"	Q_2	"		"
			10	"		"	"		"		"	"	-1 mA	"		"	"		"	Q_3	"		"
			11	"		"	"		"		"	"		"	-1 mA	"	"		"	Q_4	"		"
			12	"		"	"		"		"	"		"		"	"	-1 mA	"	Q_5	"		"
	V_{IC}		13	-18 mA							"								"	$\overline{E}$		-1.2	"
			14			-18 mA					"								"	D_0		"	"
			15				-18 mA				"								"	D_1		"	"
			16						-18 mA		"								"	D_2		"	"
			17								"	-18 mA							"	CP		"	"
			18								"			-18 mA					"	D_3		"	"
			19								"					-18 mA			"	D_4		"	"
			20								"						-18 mA		"	D_5		"	"
	I_{IH1}		21	2.7 V							"								5.5 V	$\overline{E}$		20	∝A
			22			2.7 V					"								"	D_0		"	"
			23				2.7 V				"								"	D_1		"	"
			24						2.7 V		"								"	D_2		"	"
			25								"	2.7 V							"	CP		"	"
			26								"			2.7 V					"	D_3		"	"
			27								"					2.7 V			"	D_4		"	"
			28								"						2.7 V		"	D_5		"	"
	I_{IH2}	3010	29	7.0 V							"								"	$\overline{E}$		100	"
			30			7.0 V					"								"	D_0		"	"
			31				7.0 V				"								"	D_1		"	"
			32						7.0 V		"								"	D_2		"	"
			33								"	7.0 V							"	CP		"	"
			34								"			7.0 V					"	D_3		"	"
			35								"					7.0 V			"	D_4		"	"
			36								"						7.0 V		"	D_5		"	"
	I_{IL1}	3009	37	.5 V							"								"	$\overline{E}$	3/	3/	mA
			38			.5 V					"								"	D_0	"	"	"
			39				.5 V				"								"	D_1	"	"	"
			40						.5 V		"								"	D_2	"	"	"
			41								"	.5 V							"	CP	"	"	"
			42								"			.5 V					"	D_3	"	"	"
			43								"					.5 V			"	D_4	"	"	"
			44								"						.5 V		"	D_5	"	"	"
	I_{OS}	3011	45	GND	GND	4.5 V	4.5 V		4.5 V		"	2/		4.5 V		4.5 V	4.5 V		"	Q_0	-60	-150	"
			46	"		"	"	GND	"		"	"		"		"	"		"	Q_1	"	"	"
			47	"		"	"		"	GND	"	"		"		"	"		"	Q_2	"	"	"
			48	"		"	"		"		"	"	GND	"		"	"		"	Q_3	"	"	"
			49	"		"	"		"		"	"		"	GND	"	"		"	Q_4	"	"	"
			50	"		"	"		"		"	"		"		"	"	GND	"	Q_5	"	"	"

See footnotes at end of device type 08.

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F / Cases 2 and X 1/ / Test no.	1 / 2 / $\overline{E}$	2 / 3 / Q_0	3 / 4 / D_0	4 / 5 / D_1	5 / 7 / Q_1	6 / 8 / D_2	7 / 9 / Q_2	8 / 10 / GND	9 / 12 / CP	10 / 13 / Q_3	11 / 14 / D_3	12 / 15 / Q_4	13 / 17 / D_4	14 / 18 / D_5	15 / 19 / Q_5	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 Tc = +25°C	I_{OD}		51	GND	2.5 V	GND	GND		GND		GND	2/		GND		GND	GND		4.5 V	Q_0	60		mA
			52	"		"	"	2.5 V	"		"	"		"		"	"		"	Q_1	"		"
			53	"		"	"		"	2.5 V	"	"		"		"	"		"	Q_2	"		"
			54	"		"	"		"		"	"	2.5 V	"		"	"		"	Q_3	"		"
			55	"		"	"		"		"	"		"	2.5 V	"	"		"	Q_4	"		"
			56	"		"	"		"		"	"		"		"	"	2.5 V	"	Q_5	"		"
	I_{CC}	3005	57	"		"	"		"		"	"		"		"	"		5.5 V	V_{CC}		45	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 Tc = +25°C	Truth table tests 4/	3014	58	B	L	B	B	L	B	L	GND	A	L	B	L	B	B	L	5/	All outputs			
			59	"	"	B	"	"	B	"	"	B	"	"	"	B	"	"	"				
			60	"	"	A	"	"	A	"	"	B	"	"	"	A	"	"	"				
			61	"	H	"	"	"	"	H	"	A	"	"	H	"	"	"	"				
			62	"	"	"	A	"	"	"	"	A	"	A	"	"	A	"	"				
			63	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			64	"	"	"	"	H	"	"	"	A	H	"	"	"	"	H	"				
			65	A	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
			66	A	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			67	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			68	"	"	B	"	"	B	"	"	"	"	"	"	B	"	"	"				
			69	"	L	"	"	"	"	L	"	A	"	"	L	"	"	"	"				
			70	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			71	"	"	"	"	"	"	"	"	B	"	B	"	"	B	"	"				
			72	"	"	"	B	L	"	"	"	A	L	"	"	"	"	L	"				
			73	A	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"				
			74	"	"	A	A	"	A	"	"	"	"	A	"	A	A	"	"				
			75	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			76	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
			77	B	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
			78	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			79	"	H	"	"	H	"	H	"	A	H	"	H	"	"	H	"				
			80	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			81	"	"	B	B	"	B	"	"	"	"	B	"	B	B	"	"				
			82	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			83	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
			84	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			85	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			86	"	L	"	"	L	"	L	"	A	L	"	L	"	"	L	"				
			87	"	"	A	A	"	A	"	"	A	"	A	"	A	A	"	"				
			88	"	"	A	A	"	A	"	"	B	"	A	"	A	A	"	"				
8	Same tests, terminal conditions, for as subgroup 7, except T_C = +125°C and -55°C.																						
9 Tc = +25°C	f_{MAX} 6/	3003	89	GND	OUT	IN					GND	IN							5.0 V	Q_0	80		MHz
			90	"			IN	OUT			"	"							"	Q_1	"		"
			91	"					IN	OUT	"	"							"	Q_2	"		"
			92	"							"	"	OUT	IN					"	Q_3	"		"
			93	"							"	"			OUT	IN			"	Q_4	"		"
			94	"							"	"					IN	OUT	"	Q_5	"		"

See footnotes at end of device type 08.

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{E}$	Q_0	D_0	D_1	Q_1	D_2	Q_2	GND	CP	Q_3	D_3	Q_4	D_4	D_5	Q_5	V_{CC}		Min	Max	
9 Tc = +25°C	t_{PHL1}	3003	95	GND	OUT	IN					GND	IN							5.0 V	Q_0	3.0	8.5	ns
			96	"			IN	OUT			"	"							"	Q_1	"	"	"
			97	"					IN	OUT	"	"							"	Q_2	"	"	"
			98	"							"	"	OUT	IN					"	Q_3	"	"	"
			99	"							"	"			OUT	IN			"	Q_4	"	"	"
			100	"							"	"					IN	OUT	"	Q_5	"	"	"
	t_{PLH1}	3003	101	"	OUT	IN					"	"							5.0 V	Q_0	2.5	7.5	"
			102	"			IN	OUT			"	"							"	Q_1	"	"	"
			103	"					IN	OUT	"	"							"	Q_2	"	"	"
			104	"							"	"	OUT	IN					"	Q_3	"	"	"
			105	"							"	"			OUT	IN			"	Q_4	"	"	"
			106	"							"	"					IN	OUT	"	Q_5	"	"	"
10	f_{MAX} 6/	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C.																			60		MHz
	t_{PHL1}																				2.5	10.5	ns
	t_{PLH1}																				2.0	9.5	
11	Same tests and terminal conditions as for subgroup 10, except T_C = -55°C.																						

1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 0 V, 3 V, 0 V to clock pulse, then make measurement.

3/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits		
	A	B	C
I_{IL1}	-.25/-.60	-.03/-.60	

4/ Inputs A = 2.5 V
B = 0.5 V
Outputs H ≥ 1.5 V
L ≤ 1.5 V

5/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V

6/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 09.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{E}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}		Min	Max	
1 T_C = +25°C	V_{OH}	3007	1	.8 V	20 mA		.8 V	.8 V			GND	2/			.8 V	.8 V			4.5 V	Q_0		.5	V
			2	"			"	"		20 mA	"	"			"	"			"	Q_1		"	"
			3	"			"	"			"	"	20 mA		"	"			"	Q_2		"	"
			4	"			"	"			"	"			"	"		20 mA	"	Q_3		"	"
			5	"		20 mA	2.0 V	2.0 V			"	"			2.0 V	2.0 V			"	$\overline{Q}_0$		"	"
			6	"			"	"	20 mA		"	"			"	"			"	$\overline{Q}_1$		"	"
			7	"			"	"			"	"		20 mA	"	"			"	$\overline{Q}_2$		"	"
			8	"			"	"			"	"			"	"	20 mA		"	$\overline{Q}_3$		"	"
	V_{OH}	3006	9	.8 V	-1 mA		2.0 V	2.0 V			"	3/			2.0 V	2.0 V			"	Q_0	2.5		"
			10	"			"	"		-1 mA	"	"			"	"			"	Q_1	"		"
			11	"			"	"			"	"	-1 mA		"	"			"	Q_2	"		"
			12	"			"	"			"	"			"	"		-1 mA	"	Q_3	"		"
			13	"		-1 mA	.8 V	.8 V			"	"			.8 V	.8 V			"	$\overline{Q}_0$	"		"
			14	"			"	"	-1 mA		"	"			"	"			"	$\overline{Q}_1$	"		"
			15	"			"	"			"	"		-1 mA	"	"			"	$\overline{Q}_2$	"		"
			16	"			"	"			"	"			"	"	-1 mA		"	$\overline{Q}_3$	"		"
	V_{IC}		17	-18 mA							"								"	$\overline{E}$		-1.2	"
			18				-18 mA				"								"	D_0		"	"
			19					-18 mA			"								"	D_1		"	"
			20								"	-18 mA							"	CP		"	"
			21								"				-18 mA				"	D_2		"	"
			22								"					-18 mA			"	D_1		"	"
	I_{IH1}	3010	23	2.7 V							"								5.5 V	$\overline{E}$		20	∝A
			24				2.7 V				"								"	D_0		"	"
			25					2.7 V			"								"	D_1		"	"
			26								"	2.7 V							"	CP		"	"
			27								"				2.7 V				"	D_2		"	"
			28								"					2.7 V			"	D_1		"	"
	I_{IH2}	3010	29	7.0 V							"								"	$\overline{E}$		100	"
			30				7.0 V				"								"	D_0		"	"
			31					7.0 V			"								"	D_1		"	"
			32								"	7.0 V							"	CP		"	"
			33								"				7.0 V				"	D_2		"	"
			34								"					7.0 V			"	D_3		"	"
	I_{IL1}	3009	35	.5 V							"								"	$\overline{E}$	4/	4/	mA
			36				.5 V				"								"	D_0	"	"	"
			37					.5 V			"								"	D_1	"	"	"
			38								"	.5 V							"	CP	"	"	"
			39								"				.5 V				"	D_2	"	"	"
			40								"					.5 V			"	D_3	"	"	"
	I_{OS}	3011	41	GND	GND		4.5 V	4.5 V			"	5/			4.5 V	4.5 V			"	Q_0	-60	-150	"
			42	"			"	"		GND	"	"			"	"			"	Q_1	"	"	"
			43	"			"	"			"	"	GND		"	"			"	Q_2	"	"	"
			44	"			"	"			"	"			"	"		GND	"	Q_3	"	"	"
			45	"		GND	GND	GND			"	"			GND	GND			"	$\overline{Q}_0$	"	"	"
			46	"			"	"	GND		"	"			"	"			"	$\overline{Q}_1$	"	"	"
			47	"			"	"			"	"		GND	"	"			"	$\overline{Q}_2$	"	"	"
			48	"			"	"			"	"			"	"	GND		"	$\overline{Q}_3$	"	"	"

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{E}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}				
1 Tc = +25°C	I_{OD}		49	GND	2.5 V		GND	GND			GND	2/			GND	GND			4.5 V	Q_0	60		mA
			50	"			"	"		2.5 V	"	"			"	"			"	Q_1	"		"
			51	"			"	"			"	"	2.5 V		"	"			"	Q_2	"		"
			52	"			"	"			"	"			"	"		2.5 V	"	Q_3	"		"
			53	"		2.5 V	4.5 V	4.5 V			"	"			4.5 V	4.5 V			"	$\overline{Q}_0$	"		"
			54	"			"	"	2.5 V		"	"			"	"			"	$\overline{Q}_1$	"		"
			55	"			"	"			"	"		2.5 V	"	"				$\overline{Q}_2$	"		"
			56	"			"	"			"	"			"	"				$\overline{Q}_3$	"		"
	I_{CC}	3005	57	"			GND	GND			"	5/			GND	GND			5.5 V	V_{CC}		40	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 Tc = +25°C	Truth table tests 6/	3014	58	B	L	H	B	B	H	L	GND	A	L	H	B	B	H	L	7/	All outputs			
			59	"	"	"	B	"	"	"	"	B	"	"	"	"	"	"	"				
			60	"	"	"	A	"	"	"	"	B	"	"	"	"	"	"	"				
			61	"	H	L	"	"	"	"	"	A	"	"	"	"	"	"	"				
			62	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			63	"	"	"	B	A	"	"	"	B	"	"	"	"	"	"	"				
			64	"	L	H	"	A	L	H	"	A	"	"	"	"	"	"	"				
			65	"	"	"	"	B	L	H	"	B	"	"	A	"	"	"	"				
			66	"	"	"	"	"	H	L	"	A	H	L	"	"	"	"	"				
			67	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			68	"	"	"	"	"	"	"	"	B	"	"	B	A	"	"	"				
			69	"	"	"	"	"	"	"	"	A	L	H	"	"	L	H	"				
			70	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			71	"	"	"	"	"	"	"	"	B	"	"	"	B	"	"	"				
			72	"	"	"	"	"	"	"	"	A	"	"	"	B	H	L	"				
			73	"	"	"	A	A	"	"	"	A	"	"	A	A	"	"	"				
			74	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			75	"	H	L	"	"	L	H	"	A	H	L	"	"	L	H	"				
			76	"	"	"	B	B	"	"	"	A	"	"	B	B	"	"	"				
			77	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			78	"	L	H	"	"	H	L	"	A	L	H	"	"	H	L	"				
			79	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			80	A	"	"	A	A	"	"	"	B	"	"	A	A	"	"	"				
			81	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
			82	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			83	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			84	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			85	"	H	L	"	"	L	H	"	A	H	L	"	"	L	H	"				
			86	A	"	"	B	B	"	"	"	A	"	"	B	B	"	"	"				
			87	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			88	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
			89	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			90	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			91	B	L	H	"	"	H	L	"	A	L	H	"	"	H	L	"				
			92	A	"	"	A	A	"	"	"	A	"	"	A	A	"	"	"				
			93	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			94	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
			95	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"				
			96	B	H	L	"	"	L	H	"	A	H	L	"	"	L	H	"				
8	Same test and terminal conditions as for subgroup 7, except T_C = +125°C = -55°C.																						

See footnotes at end of device type 09.

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2 and X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{E}$	Q_0	$\overline{Q}_0$	D_0	D_1	$\overline{Q}_1$	Q_1	GND	CP	Q_2	$\overline{Q}_2$	D_2	D_3	$\overline{Q}_3$	Q_3	V_{CC}				
9 Tc = +25°C	f_{MAX} 8/	3003	97	GND	OUT		IN				GND	IN							5.0 V	Q_0	90		MHz
			98	"				IN		OUT	"	"							"	Q_1	"		"
			99	"							"	"	OUT		IN				"	Q_2	"		"
			100	"							"	"				IN		OUT	"	Q_3	"		"
			101	"		OUT	IN				"	"							"	$\overline{Q}_0$	"		"
			102	"				IN	OUT		"	"							"	$\overline{Q}_1$	"		"
			103	"							"	"		OUT	IN				"	$\overline{Q}_2$	"		"
			104	"							"	"				IN	OUT		"	$\overline{Q}_3$	"		"
	t_{PHL1}	3003	105	"	OUT		IN				GND	IN							"	Q_0	3.0	9.0	ns
			106	"				IN		OUT	"	"							"	Q_1	"	"	"
			107	"							"	"	OUT		IN				"	Q_2	"	"	"
			108	"							"	"				IN		OUT	"	Q_3	"	"	"
			109	"		OUT	IN				"	"							"	$\overline{Q}_0$	"	"	"
			110	"				IN	OUT		"	"							"	$\overline{Q}_1$	"	"	"
			111	"							"	"		OUT	IN				"	$\overline{Q}_2$	"	"	"
			112	"							"	"				IN	OUT		"	$\overline{Q}_3$	"	"	"
	t_{PLH1}	3003	113	"	OUT		IN				"	IN							"	D_0	2.5	6.5	"
			114	"				IN		OUT	"	"							"	Q_1	"	"	"
			115	"							"	"	OUT		IN				"	Q_2	"	"	"
			116	"							"	"				IN		OUT	"	Q_3	"	"	"
			117	"		OUT	IN				"	"							"	$\overline{Q}_0$	"	"	"
			118	"				IN	OUT		"	"							"	$\overline{Q}_1$	"	"	"
			119	"							"	"		OUT	IN				"	$\overline{Q}_2$	"	"	"
			120	"							"	"				IN	OUT		"	$\overline{Q}_3$	"	"	"
10	f_{MAX} 6/	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																			70		MHz
	t_{PHL1}																				2.5	10.5	ns
	t_{PLH1}																				2.0	8.5	ns
11	Same tests, terminal conditions, and limits, as subgroup 10, except T_C = -55°C.																						

See footnotes at end of device type 09.

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1/ Cases 2 and X pins not referenced are N/C.

2/ Apply all voltages, then apply 0 V, 3 V, 0 V to clock pulse, then make measurement.

3/ Apply all voltages, then apply 3 V, 0 V, 3 V to clock pulse, then make measurement.

4/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuits		
	A	B	C
I_{IL1}	-.25/-.60	-.03/-.60	

5/ Apply all voltages, then apply 0 V, 3 V to clock pulse, then make measurement.

6/ Inputs A = 2.5 V
B = 0.5 V
Outputs H ≥ 1.5 V
L ≤ 1.5 V

5/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V

6/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 10 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1 $\overline{OE}$	2 D_0	3 D_1	4 D_2	5 D_3	6 D_4	7 D_5	8 D_6	9 D_7	10 GND	11 CP	12 Q_7
1 Tc = +25°C	V_{OH}	3006	1	0.8 V	2.0 V								"	1/	
			2	"		2.0 V							"	"	
			3	"			2.0 V						"	"	
			4	"				2.0 V					"	"	
			5	"					2.0 V				"	"	
			6	"						2.0 V			"	"	
			7	"							2.0 V		"	"	
			8	"								2.0 V	"	"	-1.0 mA
	V_{OL}	3007	9	"	0.8 V								"	"	
			10	"		0.8 V							"	"	
			11	"			0.8 V						"	"	
			12	"				0.8 V					"	"	
			13	"					0.8 V				"	"	
			14	"						0.8 V			"	"	
			15	"							0.8 V		"	"	
			16	"								0.8 V	"	"	20 mA
	V_{IC}		17	-18 mA									"		
			18		-18 mA								"		
			19			-18 mA							"		
			20				-18 mA						"		
			21					-18 mA					"		
			22						-18 mA				"		
			23							-18 mA			"		
			24								-18 mA		"		
			25									-18 mA	"		
			26										"	-18 mA	
	I_{IH1}	3010	27	2.7 V									"		
			28		2.7 V								"		
			29			2.7 V							"		
			30				2.7 V						"		
			31					2.7 V					"		
			32						2.7 V				"		
			33							2.7 V			"		
			34								2.7 V		"		
			35									2.7 V	"		
			36										"	2.7 V	
	I_{IH2}	3010	37	7.0 V									"		
			38		7.0 V								"		
			39			7.0 V							"		
			40				7.0 V						"		
			41					7.0 V					"		
			42						7.0 V				"		
			43							7.0 V			"		
			44								7.0 V		"		
			45									7.0 V	"		
			46										"	7.0 V	
	I_{IL}	3009	47	0.5 V									"		
			48		0.5 V								"		
			49			0.5 V							"		
			50				0.5 V						"		
			51					0.5 V					"		
			52						0.5 V				"		
			53							0.5 V			"		
			54								0.5 V		"		
			55									0.5 V	"		
			56										"	0.5 V	

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}		Min	Max	
1 Tc = +25°C	V_{OH}	3006	1							-1.0 mA	4.5 V	Q_0	2.5		V
			2						-1.0 mA		"	Q_1	"		"
			3					-1.0 mA			"	Q_2	"		"
			4				-1.0 mA				"	Q_3	"		"
			5			-1.0 mA					"	Q_4	"		"
			6		-1.0 mA						"	Q_5	"		"
			7	-1.0 mA							"	Q_6	"		"
			8								"	Q_7	"		"
	V_{OL}	3007	9							20 mA	"	Q_0		0.5	"
			10						20 mA		"	Q_1		"	"
			11					20 mA			"	Q_2		"	"
			12				20 mA				"	Q_3		"	"
			13			20 mA					"	Q_4		"	"
			14		20 mA						"	Q_5		"	"
			15	20 mA							"	Q_6		"	"
			16								"	Q_7		"	"
	V_{IC}		17								"	$\overline{OE}$		-1.2	"
			18								"	D_0		"	"
			19								"	D_1		"	"
			20								"	D_2		"	"
			21								"	D_3		"	"
			22								"	D_4		"	"
			23								"	D_5		"	"
			24								"	D_6		"	"
			25								"	D_7		"	"
			26								"	CP		"	"
	I_{IH1}	3010	27								5.5 V	$\overline{OE}$		20	∝A
			28								"	D_0		"	"
			29								"	D_1		"	"
			30								"	D_2		"	"
			31								"	D_3		"	"
			32								"	D_4		"	"
			33								"	D_5		"	"
			34								"	D_6		"	"
			35								"	D_7		"	"
			36								"	CP		"	"
	I_{IH2}	3010	37								"	$\overline{OE}$		100	"
			38								"	D_0		"	"
			39								"	D_1		"	"
			40								"	D_2		"	"
			41								"	D_3		"	"
			42								"	D_4		"	"
			43								"	D_5		"	"
			44								"	D_6		"	"
			45								"	D_7		"	"
			46								"	CP		"	"
	I_{IL}	3009	47								"	$\overline{OE}$	2/	2/	mA
			48								"	D_0	"	"	"
			49								"	D_1	"	"	"
			50								"	D_2	"	"	"
			51								"	D_3	"	"	"
			52								"	D_4	"	"	"
			53								"	D_5	"	"	"
			54								"	D_6	"	"	"
			55								"	D_7	"	"	"
			56								"	CP	"	"	"

See footnotes at end of device type 10.

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TABLE III. Group A inspection for device type 10 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1 $\overline{OE}$	2 D_0	3 D_1	4 D_2	5 D_3	6 D_4	7 D_5	8 D_6	9 D_7	10 GND	11 CP	12 Q_7
1 Tc = +25°C	I_{OS}	3011	57	0.0 V	5.5 V								GND	1/	
			58	"		5.5 V							"	"	
			59	"			5.5 V						"	"	
			60	"				5.5 V					"	"	
			61	"					5.5 V				"	"	
			62	"						5.5 V			"	"	
			63	"							5.5 V		"	"	
			64	"								5.5 V	"	"	0.0 V
	I_{OZH}	3011	65	2.0 V	0.0 V								"	"	
			66	"		0.0 V							"	"	
			67	"			0.0 V						"	"	
			68	"				0.0 V					"	"	
			69	"					0.0 V				"	"	
			70	"						0.0 V			"	"	
			71	"							0.0 V		"	"	
			72	"								0.0 V	"	"	2.7 V
	I_{OZL}	3011	73	"	5.5 V								"	"	
			74	"		5.5 V							"	"	
			75	"			5.5 V						"	"	
			76	"				5.5 V					"	"	
			77	"					5.5 V				"	"	
			78	"						5.5 V			"	"	
			79	"							5.5 V		"	"	
			80	"								5.5 V	"	"	0.5 V
	I_{CCZ}	3011	81	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	
	I_{CCL}	3011	82	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"	"	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.														
7 Tc = +25°C	Func-tional 3/	3014	83	B	A	B	A	B	A	B	A	B	GND	1/	L
			84	"	B	A	B	A	B	A	B	A	"	A	L
			85	"	B	A	B	A	B	A	B	A	"	B	L
			86	"	A	B	A	B	A	B	A	B	"	B	L
			87	"	B	A	B	A	B	A	B	A	"	B	L
			88	"	B	A	B	A	B	A	B	A	"	1/	H
			89	"	A	B	A	B	A	B	A	B	"	A	H
			90	"	A	B	A	B	A	B	A	B	"	B	H
			91	"	B	A	B	A	B	A	B	A	"	B	H
			92	"	A	B	A	B	A	B	A	B	"	B	H
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.														

See footnotes at end of device type 10.

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X Test no.	13 Q_6	14 Q_5	15 Q_4	16 Q_3	17 Q_2	18 Q_1	19 Q_0	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 Tc = +25°C	I_{OS}	3011	57							0.0 V	5.5 V	Q_0	-60	-150	mA
			58						0.0 V		"	Q_1	"	"	"
			59					0.0 V			"	Q_2	"	"	"
			60				0.0 V				"	Q_3	"	"	"
			61			0.0 V					"	Q_4	"	"	"
			62		0.0 V						"	Q_5	"	"	"
			63	0.0 V							"	Q_6	"	"	"
			64								"	Q_7	"	"	"
	I_{OZH}	3011	65							2.7 V	"	Q_0		50	∝A
			66						2.7 V		"	Q_1		"	"
			67					2.7 V			"	Q_2		"	"
			68				2.7 V				"	Q_3		"	"
			69			2.7 V					"	Q_4		"	"
			70		2.7 V						"	Q_5		"	"
			71	2.7 V							"	Q_6		"	"
			72								"	Q_7		"	"
	I_{OZL}	3011	73							0.5 V	"	Q_0		-50	"
			74						0.5 V		"	Q_1		"	"
			75					0.5 V			"	Q_2		"	"
			76				0.5 V				"	Q_3		"	"
			77			0.5 V					"	Q_4		"	"
			78		0.5 V						"	Q_5		"	"
			79	0.5 V							"	Q_6		"	"
			80								"	Q_7		"	"
	I_{CCZ}	3011	81								"	V_{CC}		90	mA
	I_{CCL}	3011	82								"	V_{CC}		86	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.														
7 Tc = +25°C	Func-tional 3/	3014	83	H	L	H	L	H	L	H	4/	All outputs			
			84	H	L	H	L	H	L	H	"	"			
			85	H	L	H	L	H	L	H	"	"			
			86	H	L	H	L	H	L	H	"	"			
			87	H	L	H	L	H	L	H	"	"			
			88	L	H	L	H	L	H	L	"	"			
			89	L	H	L	H	L	H	L	"	"			
			90	L	H	L	H	L	H	L	"	"			
			91	L	H	L	H	L	H	L	"	"			
			92	L	H	L	H	L	H	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.														

See footnotes at end of device type 10.

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TABLE III. Group A inspection for device type 10 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1 $\overline{OE}$	2 D_0	3 D_1	4 D_2	5 D_3	6 D_4	7 D_5	8 D_6	9 D_7	10 GND	11 CP	12 Q_7
9	f_{MAX}	3003	93	0.0 V	IN								GND	IN	
Tc = +25°C	5/	Fig. 4	94	"		IN							"	"	
			95	"			IN						"	"	
			96	"				IN					"	"	
			97	"					IN				"	"	
			98	"						IN			"	"	
			99	"							IN		"	"	
			100	"								IN	"	"	OUT
	t_{PLH1}	3003	101	"	IN								"	"	
			102	"		IN							"	"	
			103	"			IN						"	"	
			104	"				IN					"	"	
			105	"					IN				"	"	
			106	"						IN			"	"	
			107	"							IN		"	"	
			108	"								IN	"	"	OUT
	t_{PHL1}	3003	109	"	IN								"	"	
			110	"		IN							"	"	
			111	"			IN						"	"	
			112	"				IN					"	"	
			113	"					IN				"	"	
			114	"						IN			"	"	
			115	"							IN		"	"	
			116	"								IN	"	"	OUT
	t_{PHZ1}	3003	117	IN	2.7 V								"	"	
			118	"		2.7 V							"	"	
			119	"			2.7 V						"	"	
			120	"				2.7 V					"	"	
			121	"					2.7 V				"	"	
			122	"						2.7 V			"	"	
			123	"							2.7 V		"	"	
			124	"								2.7 V	"	"	OUT
	t_{PLZ1}	3003	125	"	0.0 V								"	"	
			126	"		0.0 V							"	"	
			127	"			0.0 V						"	"	
			128	"				0.0 V					"	"	
			129	"					0.0 V				"	"	
			130	"						0.0 V			"	"	
			131	"							0.0 V		"	"	
			132	"								0.0 V	"	"	OUT
	t_{PZH1}	3003	133	"	2.7 V								"	"	
			134	"		2.7 V							"	"	
			135	"			2.7 V						"	"	
			136	"				2.7 V					"	"	
			137	"					2.7 V				"	"	
			138	"						2.7 V			"	"	
			139	"							2.7 V		"	"	
			140	"								2.7 V	"	"	OUT

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X Test no.	13 Q_6	14 Q_5	15 Q_4	16 Q_3	17 Q_2	18 Q_1	19 Q_0	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9	f_{MAX}	3003	93							OUT	5.0 V	Q_0	100		MHz
Tc = +25°C	5/	Fig. 4	94						OUT		"	Q_1	"		"
			95					OUT			"	Q_2	"		"
			96				OUT				"	Q_3	"		"
			97			OUT					"	Q_4	"		"
			98		OUT						"	Q_5	"		"
			99	OUT							"	Q_6	"		"
			100								"	Q_7	"		"
	t_{PLH1}	3003	101							OUT	"	CP to Q_0	2.5	8.5	ns
		Fig. 4	102						OUT		"	CP to Q_1	"	"	"
			103					OUT			"	CP to Q_2	"	"	"
			104				OUT				"	CP to Q_3	"	"	"
			105			OUT					"	CP to Q_4	"	"	"
			106		OUT						"	CP to Q_5	"	"	"
			107	OUT							"	CP to Q_6	"	"	"
			108								"	CP to Q_7	"	"	"
	t_{PHL1}	3003	109							OUT	"	CP to Q_0	"	"	"
		Fig. 4	110						OUT		"	CP to Q_1	"	"	"
			111					OUT			"	CP to Q_2	"	"	"
			112				OUT				"	CP to Q_3	"	"	"
			113			OUT					"	CP to Q_4	"	"	"
			114		OUT						"	CP to Q_5	"	"	"
			115	OUT							"	CP to Q_6	"	"	"
			116								"	CP to Q_7	"	"	"
	t_{PHZ1}	3003	117							OUT	"	$\overline{OE}$ to Q_0	1.0	6.0	"
		Fig. 4	118						OUT		"	$\overline{OE}$ to Q_1	"	"	"
			119					OUT			"	$\overline{OE}$ to Q_2	"	"	"
			120				OUT				"	$\overline{OE}$ to Q_3	"	"	"
			121			OUT					"	$\overline{OE}$ to Q_4	"	"	"
			122		OUT						"	$\overline{OE}$ to Q_5	"	"	"
			123	OUT							"	$\overline{OE}$ to Q_6	"	"	"
			124								"	$\overline{OE}$ to Q_7	"	"	"
	t_{PLZ1}	3003	125							OUT	"	$\overline{OE}$ to Q_0	"	5.5	"
		Fig. 4	126						OUT		"	$\overline{OE}$ to Q_1	"	"	"
			127					OUT			"	$\overline{OE}$ to Q_2	"	"	"
			128				OUT				"	$\overline{OE}$ to Q_3	"	"	"
			129			OUT					"	$\overline{OE}$ to Q_4	"	"	"
			130		OUT						"	$\overline{OE}$ to Q_5	"	"	"
			131	OUT							"	$\overline{OE}$ to Q_6	"	"	"
			132								"	$\overline{OE}$ to Q_7	"	"	"

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}	terminal	Min	Max	Unit
9	t_{PZH1}	3003	133							OUT	"	$\overline{OE}$ to Q_0	2.5	9.0	"
Tc = +25°C		Fig. 4	134						OUT		"	$\overline{OE}$ to Q_1	"	"	"
			135					OUT			"	$\overline{OE}$ to Q_2	"	"	"
			136				OUT				"	$\overline{OE}$ to Q_3	"	"	"
			137			OUT					"	$\overline{OE}$ to Q_4	"	"	"
			138		OUT						"	$\overline{OE}$ to Q_5	"	"	"
			139	OUT							"	$\overline{OE}$ to Q_6	"	"	"
			140								"	$\overline{OE}$ to Q_7	"	"	"

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TABLE III. Group A inspection for device type 10 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1 $\overline{OE}$	2 D_0	3 D_1	4 D_2	5 D_3	6 D_4	7 D_5	8 D_6	9 D_7	10 GND	11 CP	12 Q_7
9 Tc = +25°C	t_{PZL1}	3003	141	IN	0.0 V								GND	IN	
			142	"		0.0 V							"	"	
			143	"			0.0 V						"	"	
			144	"				0.0 V					"	"	
			145	"					0.0 V				"	"	
			146	"						0.0 V			"	"	
			147	"							0.0 V		"	"	
			148	"								0.0 V	"	"	OUT
10	Same tests as subgroup 9 except T_C = +125°C, use limits from table I.														
11	Same tests as subgroup 10 except T_C = -55°C, use limits from table I.														

See footnotes at end of device type 10.

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}	terminal	Min	Max	Unit
9	t_{PZL1}	3003	141							OUT	5.0 V	$\overline{OE}$ to Q_0	3.0	9.0	ns
Tc = +25°C			142						OUT		"	$\overline{OE}$ to Q_1	"	"	"
			143					OUT			"	$\overline{OE}$ to Q_2	"	"	"
			144				OUT				"	$\overline{OE}$ to Q_3	"	"	"
			145			OUT					"	$\overline{OE}$ to Q_4	"	"	"
			146		OUT						"	$\overline{OE}$ to Q_5	"	"	"
			147	OUT							"	$\overline{OE}$ to Q_6	"	"	"
			148								"	$\overline{OE}$ to Q_7	"	"	"
10	Same tests, as subgroup 9 except T_C = +125°C, use limits from table I.														
11	Same tests as subgroup 10 except T_C = -55°C, use limits from table I.														

1/ Apply all voltages then apply 3 V, 0, 3 V to CP then make measurement.

2/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuit A
I_{IL1}	-.25/-.60

3/ A = 3.0 V minimum; B = 0.0 V or GND, H ≥ 2.5 V, L ≤ 0.5 V.

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

5/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 11 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	$\overline{OE}$	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	$\overline{Q}_7$
1 Tc = +25°C	V_{OH}	3006	1	0.8 V	0.8 V								GND	1/	
			2	"		0.8 V							"	"	
			3	"			0.8 V						"	"	
			4	"				0.8 V					"	"	
			5	"					0.8 V				"	"	
			6	"						0.8 V			"	"	
			7	"							0.8 V		"	"	
			8	"								0.8 V	"	"	-1.0 mA
	V_{OL}	3007	9	"	2.0 V								"	"	
			10	"		2.0 V							"	"	
			11	"			2.0 V						"	"	
			12	"				2.0 V					"	"	
			13	"					2.0 V				"	"	
			14	"						2.0 V			"	"	
			15	"							2.0 V		"	"	
			16	"								2.0 V	"	"	20 mA
	V_{IC}		17	-18 mA									"		
			18		-18 mA								"		
			19			-18 mA							"		
			20				-18 mA						"		
			21					-18 mA					"		
			22						-18 mA				"		
			23							-18 mA			"		
			24								-18 mA		"		
			25									-18 mA	"		
			26										"	-18 mA	
	I_{IH1}	3010	27	2.7 V									"		
			28		2.7 V								"		
			29			2.7 V							"		
			30				2.7 V						"		
			31					2.7 V					"		
			32						2.7 V				"		
			33							2.7 V			"		
			34								2.7 V		"		
			35									2.7 V	"		
			36										"	2.7 V	
	I_{IH2}	3010	37	7.0 V									"		
			38		7.0 V								"		
			39			7.0 V							"		
			40				7.0 V						"		
			41					7.0 V					"		
			42						7.0 V				"		
			43							7.0 V			"		
			44								7.0 V		"		
			45									7.0 V	"		
			46										"	7.0 V	
	I_{IL}	3009	47	0.5 V									"		
			48		0.5 V								"		
			49			0.5 V							"		
			50				0.5 V						"		
			51					0.5 V					"		
			52						0.5 V				"		
			53							0.5 V			"		
			54								0.5 V		"		
			55									0.5 V	"		
			56										"	0.5 V	

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X Test no.	13 $\overline{Q}_6$	14 $\overline{Q}_5$	15 $\overline{Q}_4$	16 $\overline{Q}_3$	17 $\overline{Q}_2$	18 $\overline{Q}_1$	19 $\overline{Q}_0$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 Tc = +25°C	V_{OH}	3006	1							-1.0 mA	4.5 V	$\overline{Q}_0$	2.5		V
			2						-1.0 mA		"	$\overline{Q}_1$	"		"
			3					-1.0 mA			"	$\overline{Q}_2$	"		"
			4				-1.0 mA				"	$\overline{Q}_3$	"		"
			5			-1.0 mA					"	$\overline{Q}_4$	"		"
			6		-1.0 mA						"	$\overline{Q}_5$	"		"
			7	-1.0 mA							"	$\overline{Q}_6$	"		"
			8								"	$\overline{Q}_7$	"		"
	V_{OL}	3007	9							20 mA	"	$\overline{Q}_0$		0.5	"
			10						20 mA		"	$\overline{Q}_1$		"	"
			11					20 mA			"	$\overline{Q}_2$		"	"
			12				20 mA				"	$\overline{Q}_3$		"	"
			13			20 mA					"	$\overline{Q}_4$		"	"
			14		20 mA						"	$\overline{Q}_5$		"	"
			15	20 mA							"	$\overline{Q}_6$		"	"
			16								"	$\overline{Q}_7$		"	"
	V_{IC}		17								"	$\overline{OE}$		-1.2	"
			18								"	D_0		"	"
			19								"	D_1		"	"
			20								"	D_2		"	"
			21								"	D_3		"	"
			22								"	D_4		"	"
			23								"	D_5		"	"
			24								"	D_6		"	"
			25								"	D_7		"	"
			26								"	CP		"	"
	I_{IH1}	3010	27								5.5 V	$\overline{OE}$		20	∝A
			28								"	D_0		"	"
			29								"	D_1		"	"
			30								"	D_2		"	"
			31								"	D_3		"	"
			32								"	D_4		"	"
			33								"	D_5		"	"
			34								"	D_6		"	"
			35								"	D_7		"	"
			36								"	CP		"	"
	I_{IH2}	3010	37								"	$\overline{OE}$		100	"
			38								"	D_0		"	"
			39								"	D_1		"	"
			40								"	D_2		"	"
			41								"	D_3		"	"
			42								"	D_4		"	"
			43								"	D_5		"	"
			44								"	D_6		"	"
			45								"	D_7		"	"
			46								"	CP		"	"

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	$\overline{Q}_6$	$\overline{Q}_5$	$\overline{Q}_4$	$\overline{Q}_3$	$\overline{Q}_2$	$\overline{Q}_1$	$\overline{Q}_0$	V_{CC}	terminal	Min	Max	Unit
1	I_{IL}	3009	47								4.5 V	$\overline{OE}$	2/	2/	mA
Tc = +25°C			48								"	D_0	"	"	"
			49								"	D_1	"	"	"
			50								"	D_2	"	"	"
			51								"	D_3	"	"	"
			52								"	D_4	"	"	"
			53								"	D_5	"	"	"
			54								"	D_6	"	"	"
			55								"	D_7	"	"	"
			56								"	CP	"	"	"

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TABLE III. Group A inspection for device type 11 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1 $\overline{OE}$	2 D_0	3 D_1	4 D_2	5 D_3	6 D_4	7 D_5	8 D_6	9 D_7	10 GND	11 CP	12 $\overline{Q}_7$
1 Tc = +25°C	I_{OS}	3011	57	0.0 V	0.0 V								"	1/	
			58	"		0.0 V							"	"	
			59	"			0.0 V						"	"	
			60	"				0.0 V					"	"	
			61	"					0.0 V				"	"	
			62	"						0.0 V			"	"	
			63	"							0.0 V		"	"	
			64	"								0.0 V	"	"	0.0 V
	I_{OZH}	3011	65	2.0 V	0.0 V								"	"	
			66	"		0.0 V							"	"	
			67	"			0.0 V						"	"	
			68	"				0.0 V					"	"	
			69	"					0.0 V				"	"	
			70	"						0.0 V			"	"	
			71	"							0.0 V		"	"	
			72	"								0.0 V	"	"	2.7 V
	I_{OZL}	3011	73	"	5.5 V								"	"	
			74	"		5.5 V							"	"	
			75	"			5.5 V						"	"	
			76	"				5.5 V					"	"	
			77	"					5.5 V				"	"	
			78	"						5.5 V			"	"	
			79	"							5.5 V		"	"	
			80	"								5.5 V	"	"	0.5 V
	I_{CCZ}	3011	81	5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"	"	
	I_{CCL}	3011	82	0.0 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.														
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.														
7 Tc = +25°C	Func-tional 3/	3014	83	B	B	A	B	A	B	A	B	A	GND	1/	L
			84	B	A	B	A	B	A	B	A	B	"	A	L
			85	B	A	B	A	B	A	B	A	B	"	B	L
			86	B	B	A	B	A	B	A	B	A	"	B	L
			87	B	A	B	A	B	A	B	A	B	"	B	L
			88	B	A	B	A	B	A	B	A	B	"	1/	H
			89	B	B	A	B	A	B	A	B	A	"	A	H
			90	B	B	A	B	A	B	A	B	A	"	B	H
			91	B	A	B	A	B	A	B	A	B	"	B	H
			92	B	B	A	B	A	B	A	B	A	"	B	H
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.														

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TABLE III. Group A inspection for device type 11 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X Test no.	13 $\overline{Q}_6$	14 $\overline{Q}_5$	15 $\overline{Q}_4$	16 $\overline{Q}_3$	17 $\overline{Q}_2$	18 $\overline{Q}_1$	19 $\overline{Q}_0$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 Tc = +25°C	I_{OS}	3011	57							0.0 V	5.5 V	$\overline{Q}_0$	-60	-150	mA
			58						0.0 V		"	$\overline{Q}_1$	"	"	"
			59					0.0 V			"	$\overline{Q}_2$	"	"	"
			60				0.0 V				"	$\overline{Q}_3$	"	"	"
			61			0.0 V					"	$\overline{Q}_4$	"	"	"
			62		0.0 V						"	$\overline{Q}_5$	"	"	"
			63	0.0 V							"	$\overline{Q}_6$	"	"	"
			64								"	$\overline{Q}_7$	"	"	"
	I_{OZH}	3011	65							2.7 V	"	$\overline{Q}_0$		50	∝A
			66						2.7 V		"	$\overline{Q}_1$		"	"
			67					2.7 V			"	$\overline{Q}_2$		"	"
			68				2.7 V				"	$\overline{Q}_3$		"	"
			69			2.7 V					"	$\overline{Q}_4$		"	"
			70		2.7 V						"	$\overline{Q}_5$		"	"
			71	2.7 V							"	$\overline{Q}_6$		"	"
			72								"	$\overline{Q}_7$		"	"
	I_{OZL}	3011	73							0.5	"	$\overline{Q}_0$		-50	"
			74						0.5		"	$\overline{Q}_1$		"	"
			75					0.5			"	$\overline{Q}_2$		"	"
			76				0.5				"	$\overline{Q}_3$		"	"
			77			0.5					"	$\overline{Q}_4$		"	"
			78		0.5						"	$\overline{Q}_5$		"	"
			79	0.5							"	$\overline{Q}_6$		"	"
			80								"	$\overline{Q}_7$		"	"
	I_{CCZ}	3011	81								"	V_{CC}		86	mA
	I_{CCL}	3011	82								"	V_{CC}		86	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.														
7 Tc = +25°C	Func-tional 3/	3014	83	H	L	H	L	H	L	H	4/	All outputs			
			84	H	L	H	L	H	L	H	"	"			
			85	H	L	H	L	H	L	H	"	"			
			86	H	L	H	L	H	L	H	"	"			
			87	H	L	H	L	H	L	H	"	"			
			88	L	H	L	H	L	H	L	"	"			
			89	L	H	L	H	L	H	L	"	"			
			90	L	H	L	H	L	H	L	"	"			
			91	L	H	L	H	L	H	L	"	"			
			92	L	H	L	H	L	H	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.														

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TABLE III. Group A inspection for device type 11 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X Test no.	1 $\overline{OE}$	2 D_0	3 D_1	4 D_2	5 D_3	6 D_4	7 D_5	8 D_6	9 D_7	10 GND	11 CP	12 $\overline{Q}_7$
9 Tc = +25°C	f_{MAX} 5/	3003 Fig. 4	93	0.0 V	IN								GND	IN	
			94	"		IN							"	"	
			95	"			IN						"	"	
			96	"				IN					"	"	
			97	"					IN				"	"	
			98	"						IN			"	"	
			99	"							IN		"	"	
			100	"								IN	"	"	OUT
	t_{PLH2}	3003	101	"	IN								"	"	
			102	"		IN							"	"	
			103	"			IN						"	"	
			104	"				IN					"	"	
			105	"					IN				"	"	
			106	"						IN			"	"	
			107	"							IN		"	"	
			108	"								IN	"	"	OUT
	t_{PHL2}	3003	109	"	IN								"	"	
			110	"		IN							"	"	
			111	"			IN						"	"	
			112	"				IN					"	"	
			113	"					IN				"	"	
			114	"						IN			"	"	
			115	"							IN		"	"	
			116	"								IN	"	"	OUT
	t_{PHZ2}	3003	117	IN	0.0 V								"	"	
			118	"		0.0 V							"	"	
			119	"			0.0 V						"	"	
			120	"				0.0 V					"	"	
			121	"					0.0 V				"	"	
			122	"						0.0 V			"	"	
			123	"							0.0 V		"	"	
			124	"								0.0 V	"	"	OUT
	t_{PLZ2}	3003	125	"	2.7 V								"	"	
			126	"		2.7 V							"	"	
			127	"			2.7 V						"	"	
			128	"				2.7 V					"	"	
			129	"					2.7 V				"	"	
			130	"						2.7 V			"	"	
			131	"							2.7 V		"	"	
			132	"								2.7 V	"	"	OUT
	t_{PZH2}	3003	133	"	0.0 V								"	"	
			134	"		0.0 V							"	"	
			135	"			0.0 V						"	"	
			136	"				0.0 V					"	"	
			137	"					0.0 V				"	"	
			138	"						0.0 V			"	"	
			139	"							0.0 V		"	"	
			140	"								0.0 V	"	"	OUT

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	$\overline{Q}_6$	$\overline{Q}_5$	$\overline{Q}_4$	$\overline{Q}_3$	$\overline{Q}_2$	$\overline{Q}_1$	$\overline{Q}_0$	V_{CC}	terminal	Min	Max	Unit
9 Tc = +25°C	f_{MAX} 5/	3003 Fig. 4	93							OUT	5.0 V	$\overline{Q}_0$	100		MHz
			94						OUT		"	$\overline{Q}_1$	"		"
			95					OUT			"	$\overline{Q}_2$	"		"
			96				OUT				"	$\overline{Q}_3$	"		"
			97			OUT					"	$\overline{Q}_4$	"		"
			98		OUT						"	$\overline{Q}_5$	"		"
			99	OUT							"	$\overline{Q}_6$	"		"
			100								"	$\overline{Q}_7$	"		"
	t_{PLH2}	3003 Fig. 4	101							OUT	"	CP to $\overline{Q}_0$	2.5	8.5	ns
			102						OUT		"	CP to $\overline{Q}_1$	"	"	"
			103					OUT			"	CP to $\overline{Q}_2$	"	"	"
			104				OUT				"	CP to $\overline{Q}_3$	"	"	"
			105			OUT					"	CP to $\overline{Q}_4$	"	"	"
			106		OUT						"	CP to $\overline{Q}_5$	"	"	"
			107	OUT							"	CP to $\overline{Q}_6$	"	"	"
			108								"	CP to $\overline{Q}_7$	"	"	"
	t_{PHL2}	3003 Fig. 4	109							OUT	"	CP to $\overline{Q}_0$	"	"	"
			110						OUT		"	CP to $\overline{Q}_1$	"	"	"
			111					OUT			"	CP to $\overline{Q}_2$	"	"	"
			112				OUT				"	CP to $\overline{Q}_3$	"	"	"
			113			OUT					"	CP to $\overline{Q}_4$	"	"	"
			114		OUT						"	CP to $\overline{Q}_5$	"	"	"
			115	OUT							"	CP to $\overline{Q}_6$	"	"	"
			116								"	CP to $\overline{Q}_7$	"	"	"
	t_{PHZ2}	3003 Fig. 4	117							OUT	"	$\overline{OE}$ to $\overline{Q}_0$	1.5	5.5	"
			118						OUT		"	$\overline{OE}$ to $\overline{Q}_1$	"	"	"
			119					OUT			"	$\overline{OE}$ to $\overline{Q}_2$	"	"	"
			120				OUT				"	$\overline{OE}$ to $\overline{Q}_3$	"	"	"
			121			OUT					"	$\overline{OE}$ to $\overline{Q}_4$	"	"	"
			122		OUT						"	$\overline{OE}$ to $\overline{Q}_5$	"	"	"
			123	OUT							"	$\overline{OE}$ to $\overline{Q}_6$	"	"	"
			124								"	$\overline{OE}$ to $\overline{Q}_7$	"	"	"

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	13	14	15	16	17	18	19	20	Measured	Limits		
			Test no.	$\overline{Q}_6$	$\overline{Q}_5$	$\overline{Q}_4$	$\overline{Q}_3$	$\overline{Q}_2$	$\overline{Q}_1$	$\overline{Q}_0$	V_{CC}	terminal	Min	Max	Unit
9 Tc = +25°C	t_{PLZ2}	3003 Fig. 4	125							OUT	5.0 V	$\overline{OE}$ to $\overline{Q}_0$	1.5	5.5	ns
			126						OUT		"	$\overline{OE}$ to $\overline{Q}_1$	"	"	"
			127					OUT			"	$\overline{OE}$ to $\overline{Q}_2$	"	"	"
			128				OUT				"	$\overline{OE}$ to $\overline{Q}_3$	"	"	"
			129			OUT					"	$\overline{OE}$ to $\overline{Q}_4$	"	"	"
			130		OUT						"	$\overline{OE}$ to $\overline{Q}_5$	"	"	"
			131	OUT							"	$\overline{OE}$ to $\overline{Q}_6$	"	"	"
			132								"	$\overline{OE}$ to $\overline{Q}_7$	"	"	"
	t_{PZH2}	3003 Fig. 4	133							OUT	"	$\overline{OE}$ to $\overline{Q}_0$	3.0	9.0	"
			134						OUT		"	$\overline{OE}$ to $\overline{Q}_1$	"	"	"
			135					OUT			"	$\overline{OE}$ to $\overline{Q}_2$	"	"	"
			136				OUT				"	$\overline{OE}$ to $\overline{Q}_3$	"	"	"
			137			OUT					"	$\overline{OE}$ to $\overline{Q}_4$	"	"	"
			138		OUT						"	$\overline{OE}$ to $\overline{Q}_5$	"	"	"
			139	OUT							"	$\overline{OE}$ to $\overline{Q}_6$	"	"	"
			140								"	$\overline{OE}$ to $\overline{Q}_7$	"	"	"

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TABLE III. Group A inspection for device type 11 – Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2,R, S, and X	1	2	3	4	5	6	7	8	9	10	11	12
			Test no.	$\overline{OE}$	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	GND	CP	$\overline{Q}_7$
9 Tc = +25°C	t_{PZL2}	3003	141	IN	2.7 V								GND	IN	
			142	"		2.7 V							"	"	
			143	"			2.7 V						"	"	
			144	"				2.7 V					"	"	
			145	"					2.7 V				"	"	
			146	"						2.7 V			"	"	
			147	"							2.7 V		"	"	
			148	"								2.7 V	"	"	OUT
10	Same tests as subgroup 9 except T_C = +125°C, use limits from table I.														
11	Same tests as subgroup 10 except T_C = -55°C, use limits from table I.														

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.8 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, R, S, and X	13	14	15	16	17	18	19	20	Measured	Limits		Unit
			Test no.	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0	V_{CC}	terminal	Min	Max	
9 Tc = +25°C	t_{PZL2}	3003	141							OUT	5.0 V	$\overline{OE}$ to $\overline{Q}_0$	3.0	9.0	ns
			142						OUT		"	$\overline{OE}$ to $\overline{Q}_1$	"	"	"
			143					OUT			"	$\overline{OE}$ to $\overline{Q}_2$	"	"	"
			144				OUT				"	$\overline{OE}$ to $\overline{Q}_3$	"	"	"
			145			OUT					"	$\overline{OE}$ to $\overline{Q}_4$	"	"	"
			146		OUT						"	$\overline{OE}$ to $\overline{Q}_5$	"	"	"
			147	OUT							"	$\overline{OE}$ to $\overline{Q}_6$	"	"	"
			148								"	$\overline{OE}$ to $\overline{Q}_7$	"	"	"
10	Same tests as subgroup 9 except T_C = +125°C, and use limits from table I.														
11	Same tests as subgroup 10 except T_C = -55°C, and use limits from table I.														

1/ Apply all voltages then apply 3 V, 0, 3 V to CP then make measurement.

2/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Circuit A
I_{IL1}	-.25/-.60

3/ A = 3.0 V minimum; B = 0.0 V or GND, H ≥ 2.5 V, L ≤ 0.5 V.

4/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

5/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

I Requirements for "JAN" marking.

j. Packaging Requirements (see 5.1)

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

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6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in *MIL-PRF-38535*, MIL-HDBK-1331, and as follows:

GND ..... Ground zero voltage potential
I_{IN} ..... Current flowing into an input terminal
V_{IN} ..... Voltage level at an input terminal

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	54F074
02	54F109
03	54F112
04	54F175
05	54F374
06	54F534
07	54F174
08	54F378
09	54F379
10	54F574
11	54F564

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6.8 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Circuit A	Circuit B	Circuit C	Circuit D
	National Semiconductor	Motorola	Signetics	Texas Instruments
01	X	X	X	X
02	X	X	X	X
03	X		X	X
04	X	X	X	X
05	X	X	X	X
06	X	X		
07	X	X		
08	X	X		
09	X	X		
10	X			
11	X			

6.9 Change from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – SM, MI
Navy - AS, CG, MC, SH TD
Air Force – 03, 19, 99

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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